

ELLIOTT WAVE / NEOWAVE

100% SOURCE-LOCKED MASTER DEVELOPER SPECIFICATION

For a Pine Script / Python developer with zero prior Elliott Wave knowledge. Includes a START HERE programmer onboarding section, full supplied-note coverage, and the audited current Pine Script update map.

Version 3.1 - Full Notes Coverage + START HERE Developer Onboarding + Acceptance Pack - 21 Aug 2026

Purpose

This manual explains the Elliott concepts, exact coding states, formulas, invalidation logic, correction classifiers, Fibonacci/time/MACD rules, controlled recount behavior and acceptance tests required to upgrade the current 1,007-line Elliott overlay into a state-based automatic Elliott Wave engine.

Important

Where the supplied source notes disagree with each other, this manual does NOT silently invent a rule. Those items are marked OWNER DECISION / DO NOT GUESS and must remain configurable or blocked until approved.

V3 SOURCE-LOCK NOTICE: This version supersedes the earlier developer handoff. It adds a complete source register, cross-source conflict matrix, full page-by-page source ledger, binding implementation scope, and developer sign-off conditions. Where supplied notes conflict, the developer must preserve the conflict as configurable/TBD rather than silently choosing a textbook rule.

START HERE

ELLIOTT WAVE FOR A PINE SCRIPT DEVELOPER

Assumption: you are an expert programmer, but you do not need prior Elliott Wave knowledge. Read this section first; then use the full source-locked specification as the detailed rule reference.

NON-NEGOTIABLE IMPLEMENTATION PRINCIPLE

A wave label is the OUTPUT of validation, never the INPUT. Do not label the first swing as Wave 1, the second as Wave 2, etc. Raw pivots create candidates. Structure, degree, internal subdivision, Fibonacci, time, and supporting evidence determine whether the candidate becomes a confirmed Elliott label.

This methodology is source-specific. Do not replace a source rule with a generic textbook Elliott rule. If the source is contradictory or blank, keep the item configurable or mark it TBD-BLOCKED until the owner confirms the intended version.

SH-1. The Mental Model in 5 Minutes

Elliott analysis in this project is a hierarchical pattern-classification problem. Market price produces many swings. Only some swings belong to the active Elliott degree. A large wave can contain smaller waves, and a correction can evolve into several different structures before the final type is known.

Programming object	Meaning	What NOT to assume
RawPivot	Confirmed local swing high/low from the pivot engine.	Every RawPivot is an Elliott wave endpoint.
DegreePivot	A pivot large enough to matter at the active Elliott degree.	The latest extreme automatically resets the full count.
WaveCandidate	Possible 1/2/3/4/5/A/B/C/W/X/Y/XX/Z with calculated rules.	The candidate label is final.
PatternCandidate	Possible Impulse, Zig-Zag, Flat, Double, Triple, Triangle, LD, ED, etc.	Only one correction type exists while the structure is still forming.
ConfirmedWave	Candidate that passed required hard rules and confirmation policy.	It may move every time a minor pivot appears.
AlternateCount	A still-valid secondary interpretation.	Ambiguity must be forced into one label immediately.

- Impulse phase: normally five main waves labeled 1-2-3-4-5.
- Correction phase: can be A-B-C, W-X-Y, W-X-Y-XX-Z, or A-B-C-D-E depending on the candidate structure.
- The same early corrective path may still fit several correction types. Keep those candidates in parallel until rules eliminate them.
- MACD, RSI, gaps, candles, trendlines and channels support/rank the count; they do not replace price structure unless a specific source rule explicitly makes them mandatory.

SH-2. State Machine - How the Engine Must Think

17. Recommended indicator state engine (implementation translation)

This section is recommended software architecture, not a new trading rule. It translates the source into a deterministic implementation that can handle incomplete patterns without repainting historical labels arbitrarily.

17.1 Candidate lifecycle

State	Meaning
SEARCHING	No valid base/current candidate yet.
FORMING	Enough structure exists to propose a wave/pattern, but all confirmation conditions are not met.
CONFIRMED	Source-defined structural, fib/time and required confirmation conditions are met.
ALTERNATE	A secondary count remains possible because correction rules are guideline based.
INVALID	A hard source rule has been violated. Store the exact reason.
TBD-BLOCKED	A decision depends on a source blank/conflict; show it rather than guessing.

17.2 Evaluation order

- **Step 1:** determine analysis timeframe / degree and locate Important High-Low plus candidate base point.
- **Step 2:** create impulse candidate and subordinate pivots; apply hard price-structure rules before supporting indicators.
- **Step 3:** calculate Fib retracement/extension and time relationships at every confirmed pivot.
- **Step 4:** classify Trending vs Terminal impulse, diagonal subtype and extension subtype if applicable.
- **Step 5:** when correction begins, maintain Zig-Zag / Flat / Double / Triple / Triangle candidates in parallel.
- **Step 6:** use MACD, gap, candlestick, channel and trendline events as the source describes - primarily as confirmation/ranking, not as replacements for wave structure.
- **Step 7:** confirm/invalidate labels only after the relevant source rule is satisfied/violated; retain reason codes for debugging.

17.3 Suggested internal record

```
WaveCandidate { degree, timeframe, direction, label, patternType, startBar, endBar,
startPrice, endPrice, durationBars, fibRetrace, fibExtension, internalCount, macdState,
divergenceState, gapState, candlestickState, trendlineState, channelState, status,
invalidReason, sourceRuleId }
```

18. Chart display requirements (recommended handoff)

Element	Required display behavior
Wave labels	Show 0/Base,1,2,3,4,5,A,B,C,W,X,Y,XX,Z and A-B-C-D-E triangle labels at confirmed pivots; optional dotted label for forming candidates.
Pattern name	Display active structure: Impulse, Terminal, Leading Diagonal, Ending Diagonal, Zig-Zag, Flat subtype, Double, Triple, Triangle subtype, Extension subtype.
Fib levels	Draw only the levels relevant to the current candidate and source rule; label the anchor pair (e.g., 0-1-2,0-A-B,0-W-X,0-Y-XX).
Trendlines/channels	Draw source 0-2,1-3,2-4,0-B and triangle boundaries with anchor labels and break state.
MACD support	Display compact status (lowest/highest/top/bottom, +div/-div) rather than replacing wave labels.
Gap / candle support	Small labels at the source-defined context.
Status panel	Show current degree, current wave, candidate pattern, confirmed/forming/alternate/invalid, next source condition and conflict/TBD if applicable.
Debug mode	Developer-only mode should print rule IDs passed/failed and numeric ratios for backtesting/validation.

19. Inputs / settings required for implementation

Source-derived state-engine concept: do not finalize a label while required conditions remain unresolved.

State	Developer meaning	Chart behavior
SEARCHING	No valid base/current wave candidate.	No final Elliott label.

State	Developer meaning	Chart behavior
FORMING	Enough evidence to propose a candidate, but confirmation is incomplete.	Optional DEV / dotted label.
CONFIRMED	All required hard rules and chosen confirmation rules pass.	Lock the confirmed pivot/label.
ALTERNATE	Secondary interpretation is still valid.	Show only in dashboard/debug or subtle alternate label.
INVALID	A hard rule is violated.	Reject the candidate; keep reason code; controlled recount.
TBD-BLOCKED	Decision depends on a source conflict/blank.	Do not guess; display pending/owner decision.

```
enum EwStatus
    SEARCHING
    FORMING
    CONFIRMED
    ALTERNATE
    INVALID
    TBD_BLOCKED
```

```
type WaveCandidate
    int degree
    int direction
    string label
    string patternType
    int startBar
    int endBar
    float startPrice
    float endPrice
    int durationBars
    float fibRetrace
    float fibExtension
    int internalCount
    string macdState
    string divergenceState
    EwStatus status
    string invalidReason
    string sourceRuleId
```

SH-3. End-to-End Processing Order

1. Select the analysis timeframe/degree and load enough source context (including the approved Important High/Low window).
2. Generate confirmed RAW pivots. Do not attach Elliott labels yet.
3. Build Base / Point-0 candidates using Important High/Low + MACD/divergence + approved structural context.
4. Lock a base only after the degree-development rule is satisfied. The base remains locked until a valid recount condition occurs.
5. Build Wave-1 from subordinate structure; evaluate its degree threshold and special Leading Diagonal possibility.
6. After W1, open a Wave-2 correction container. Maintain Zig-Zag / Flat / W-X-Y candidates as appropriate.
7. After W2 is confirmed, build W3. Classify Trending / Terminal / Extended; do not treat <161.8% as automatically wrong.
8. After W3, open a Wave-4 correction container. Do NOT start Wave-5 merely because price turns up/down.
9. Confirm W4 only when a valid correction candidate completes. Then build W5 and evaluate normal / truncated / extended / Ending Diagonal paths.
10. After the five-wave impulse, open the larger correction classifier and maintain A-B-C / W-X-Y / W-X-Y-XX-Z / Triangle candidates until resolved.
11. At every transition, calculate price/Fib/time/internal-structure first, then supporting MACD/RSI/gap/candle/trendline evidence.

12. On hard invalidation, reject only the affected candidate and perform a controlled recount. Never restart because a small raw swing appeared.

RAW OHLC -> RawPivot[]

- > Degree Filter / Important H-L
- > BaseCandidate -> BASE_LOCKED
- > W1 Candidate(s) -> W1 CONFIRMED
- > W2 Correction Container -> W2 CONFIRMED
- > W3 Trending/Terminal/Extension -> W3 CONFIRMED
- > W4 Correction Container -> W4 CONFIRMED
- > W5 Normal/Truncated/Extended/ED -> W5 CONFIRMED
- > Larger Correction Candidates
- > New degree / controlled recount

SH-4. Worked Bullish Example - From Point 0 to Wave 5

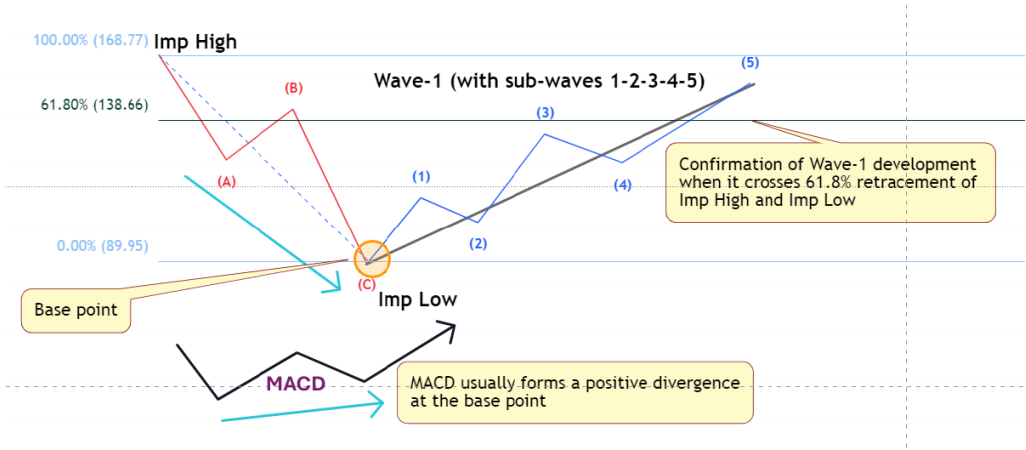
The numbers below are a coding example only. They show how to calculate candidate ratios. The full specification later in this document determines the exact source mode when multiple note versions conflict.

Point	Example price	Meaning	Calculation / gate	State example
P0	100.00	Locked bullish base	Important Low + base evidence; degree context active	BASE_LOCKED
P1	120.00	W1 end candidate	W1 length = 20.00; subordinate 5-move structure required	W1_CONFIRMED
P2	110.00	W2 end candidate	Retrace = $(120-110)/20 = 50\%$	W2_CONFIRMED if correction completes
P3	142.36	W3 candidate	0-1-2 extension: $110 + 1.618*(120-100) = 142.36$	W3_TRENDING candidate
P4	130.00	W4 end candidate	Must be a valid W4 correction; use approved Base->W3 retracement/time/alternation rules	W4_CONFIRMED only after correction completion
P5	145.70+	W5 candidate	3-4 min projection example: $130 + 1.27*(142.36-130) \approx 145.70$	W5_FORMING / CONFIRMED after full gates

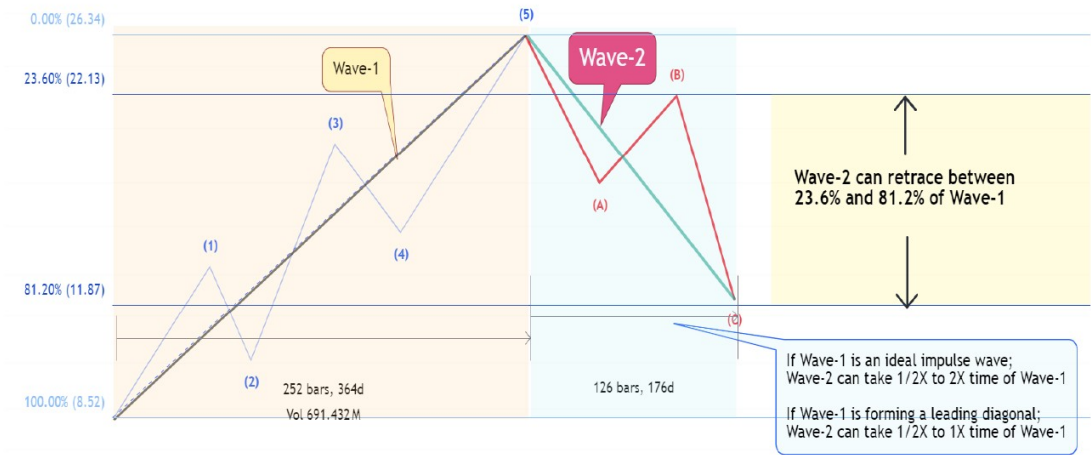
Critical: P4 is NOT "the next low".

After P3, the engine enters a W4 correction container. It must classify and complete a permitted correction. Only after W4 is structurally confirmed can the next impulse candidate be promoted to Wave 5. This is the structural fix for the earlier premature-Wave-5 problem.

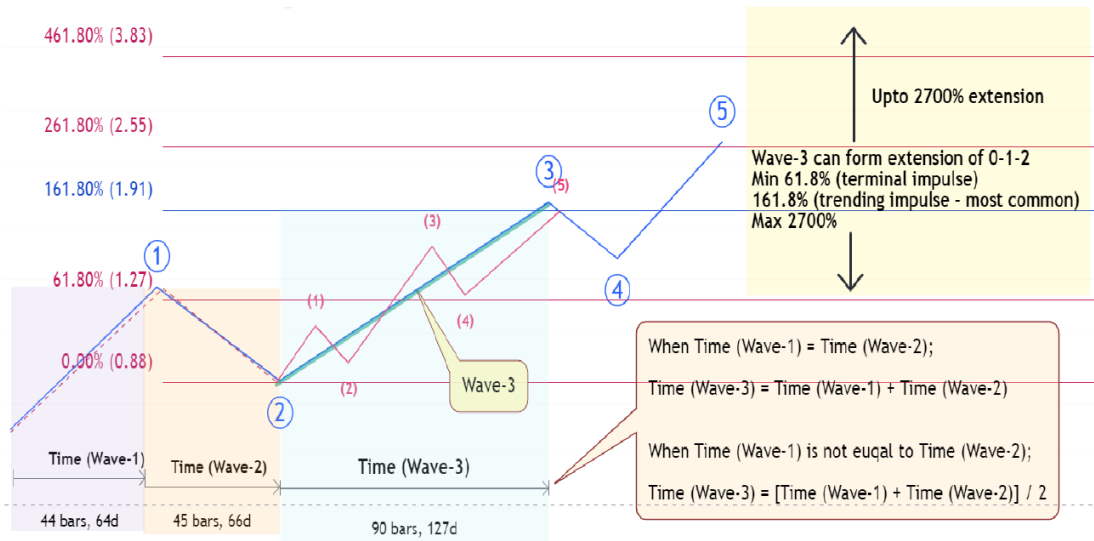
Development of Wave-1



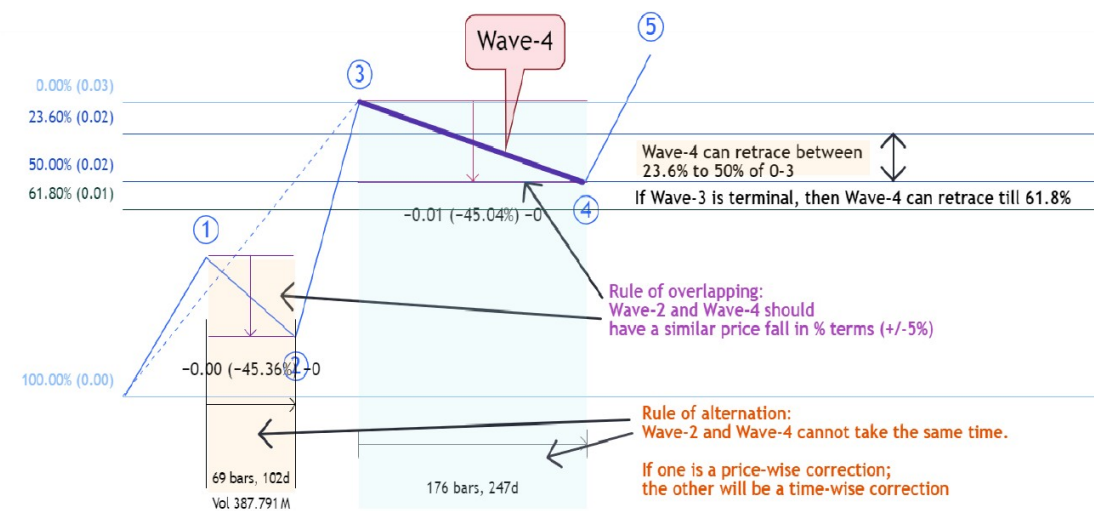
Wave-1 source snapshot - Important High/Low, base, internal sub-waves and 61.8% degree-development concept.



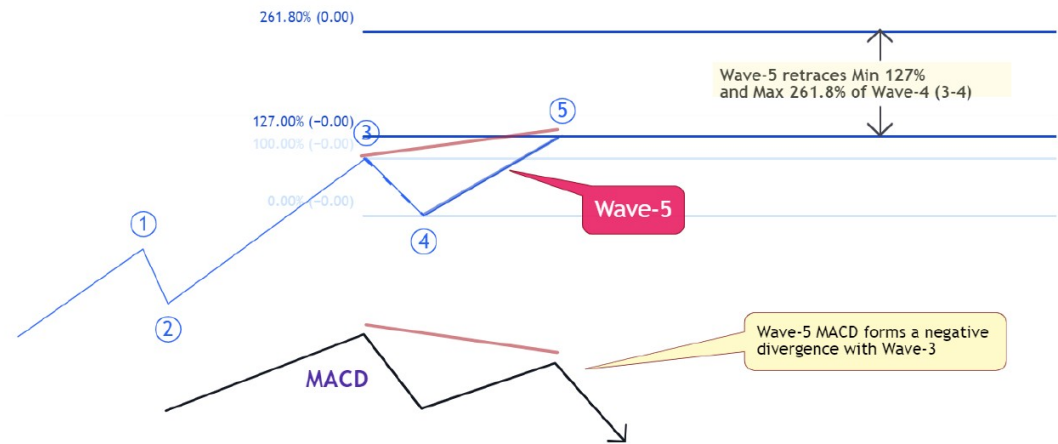
Wave-2 source snapshot - corrective structure and retracement context.



Wave-3 source snapshot - 0-1-2 extension, Trending vs Terminal/extension context.



Wave-4 source snapshot - complex-correction container, retracement, alternation and overlap context.



Wave-5 source snapshot - 3-4 projection and MACD divergence context.

SH-5. Wave 2 Is a Correction Engine, Not One Pivot

After W1 is confirmed, the next falling move in a bullish count is only a W2 candidate. W2 itself normally contains corrective structure. At minimum the engine must be able to represent A-B-C or W-X-Y paths described in the supplied notes.

```
w1Range = abs(P1 - P0)
// bullish example
w2Retrace = (P1 - candidateLow) / w1Range

if candidateLow <= P0
  W2 = INVALID // hard structural failure
else
  evaluate normal/deep/microscopic source mode
  evaluate corrective internal candidates
  evaluate W2 time mode
  if correctionCompleted and required gates pass
    W2 = CONFIRMED
  else
    W2 = FORMING
```

Do not hard-code 14% as the only normal W2 minimum.

The supplied notes contain different presentations: the detailed Chartking rule table uses 23.6%-81.2% as the normal range and treats 14% as a microscopic special clue; other notes show 14% as a broad minimum. This is a SOURCE CONFLICT and must use the document's conflict-mode/owner-decision handling, not a silent developer choice.

W2 situation	What developer should do	Typical status	Important consequence
Normal retracement context	Evaluate approved normal W2 range + internal correction.	FORMING/CONFIRMED	W3 candidate may begin only after W2 completion.
Deep 61.8%-81.2% context	Evaluate project HP BUY/SELL overlay only if W2 structure is valid.	FORMING + HP WATCH	Fib touch alone is not an HP signal.
Microscopic ~14% clue	Use separate microscopic flag according to selected source mode.	SPECIAL/FORMING	Prepare W3 extension candidate.

Break of Point 0	Reject bullish W1/W2 interpretation.	INVALID	Controlled recount from prior valid structure.
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SH-6. Correction Classifier - Why B=111% Is Not a Universal Rule

Correction rules are guideline-based in the supplied methodology. The engine must keep more than one candidate while the structure is unresolved. A single B retracement threshold cannot define every correction.

Candidate	Outer labels	Internal idea	Key early discriminator	Do not do
Zig-Zag	A-B-C	5-3-5	B is below the Flat B zone in the detailed source; C is impulsive.	Do not force B=61.8-111.
Flat	A-B-C	3-3-5	B 61.8%-111% in the source; C subtype depends on B/C relationships.	Do not call every ABC a Zig-Zag.
Double	W-X-Y	Combination	X can be small or large; Y has 0-W-X extension/time rules.	Do not just rename 3 pivots W-X-Y.
Triple	W-X-Y-XX-Z	Combination	XX and Z have their own price/time relationships.	Do not use a second X without XX logic.
Triangle	A-B-C-D-E	Five corrective legs	Geometry, relative leg sizes, boundaries and thrust rules.	Do not detect Triangle in W2 when source excludes it.

6. Correction classifier - simple correction

Correction recognition must be candidate-based because the source explicitly calls corrections guideline based. The same early A-B path may later resolve to Zig-Zag, Flat, Double, Triple or Triangle.

6.1 Zig-Zag Correction (A-B-C)

Field	Source requirement
Type / occurrence	Single, price-wise correction; mostly Wave-2, sometimes Wave-4 and inside Double/Triple.
Count	3 moves A-B-C; internal 5-3-5.
Speed	A fast, B slow, C fast.
Wave-A	Min 61.8% retracement of previous degree Wave-5.
Wave-B time	$B \geq A$, up to 5X A.
Wave-B price	Less than 61.8%; source says if $>61.8\%$ it is Wave-2.
Wave-C projection	$C = A$ length-wise; min 61.8%, source table max 261.8% extension of 0-A-B.
Wave-C retracement	Min 127% of Wave-B.
C time	If $A=B$, $C=A+B$; if $A < B$, $C=(A+B)/2$.
Entry	On crossing the 3rd Fib ratio in Wave-C.
SL	Previous Fib level of entry.
Targets / reversal support	Bullish candle at 61.8%, 100%, 161.8%, 261.8%; 0-B trendline break in half C time; point B crossed in same C time.
Pattern end	Complete correction minimum 38.2% of entire impulse.
C types	Normal 61.8%-161.8%; Elongated source also shows up to 461.8%; Truncated reverses before 61.8% and uses an 81.2% confirmation.

Zig-Zag Correction (A-B-C)

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Correction Type	• Single correction	Entry	• On crossing 3 rd Fib Ratio in Wave-C
Occurrence	• Mostly found in Wave-2/ sometimes in Wave-4 and in double or triple correction	SL	• Previous Fib Level of entry
Category	• Price-wise correction	Target	• Bullish Candle formation at Fib levels 61.8%, 100%, 161.8%, 261.8%
No. of moves	• 3 (A-B-C)	Pattern confirmation-1	• When price crosses 0-B trendline in half the time (Wave-C)
Internal count	• 5-3-5	Pattern confirmation-2	• When price crosses point B at the same time (Wave-C)
Speed	• A Fast, B Slow, C Fast	Pattern End	• Complete correction should retrace Min 38.2% of the entire impulse move
Rule for Wave-A	• Min 61.8% retracement of previous degree Wave-5		
Wave-B Time period	• $B \geq A$ (up to 5 times of A)		
Fib Ext Wave-C (0-A-B)	• $C = A$ (length wise), Min 61.8%, Max 261.8% Fib extension (0-A-B)		
Wave-C retracement	• Wave-C should be Fib retracement 127% of Wave-B		
Wave-B retracement	• Less than 61.8%, if $>61.8\%$ then it is Wave-2		
Time taken for Wave-C : 1	• If $A = B$, $C = A + B$	Types of C:	Normal C
Time taken for Wave-C : 2	• If $A < B$, $C = (A + B) / 2$	Trend based Fib extension on 0-A-B	Elongated C
			Truncated C
			61.8% to 161.8%
			161.8% to 461.8%
			1. Reverses before 61.8%
			2. Confirmation: 81.2% retracement of A-B-C (0-C)

Source snapshot - page 37: complete Zig-Zag rule table.

Zig-Zag source rule summary - use structure + Fib + time + 0-B trendline, not three adjacent pivots only.

Flat Correction (A-B-C)

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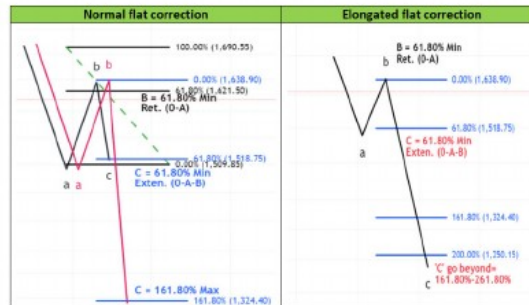
Correction Type	• Single correction	Point of Entry	• Start of Wave-X or Wave-1
Occurrence	• Mostly found in Wave-2	Confirmation:1	• Bullish Candle formation at Fib levels 61.8%, 100%, 161.8%, 261.8%
Category	• Price-wise correction	Confirmation:2	• On crossing 3 rd Fib Ratio
No. of moves	• 3 (A-B-C)	SL	• 61.8%, 100%, 161.8%, 261.8%
Internal count	• 3-3-5	Exit (time period)	• Half the time of Wave-C
Speed	• A Slow, B Fast, C Fast	Reversal criteria	• When price crosses 0-B trendline in half the time (Wave-C)
Rule for Wave-A	• Min 38.2% retracement of previous degree Wave-5	Pattern confirmation	• When price crosses point B at the same time (Wave-C)
Rule for Wave-B	• B >= A (up to 2 times of A)	Pattern End	• Complete correction should retrace Min 38.2% of the entire impulse move
Wave-B retracement	• Min 61.8% , Max 111% , (SL 127%)		
Time taken for Wave-C : 1	• If time (A) = time (B), time (C) = time (A) + time (B)		
Time taken for Wave-C : 2	• If time (A) < time (B), time (C) = [time (A) + time (B)]/2		

	Normal C (B : Min 61.8%, Max 61.2%)	61.8% to 161.8%
Types of C : Trend based Fib extension on 0-A-B	Elongated C (B: Min 61.8%, Max 61.2%)	161.8% to 261.8%
	Strong Irregular (B: Min 61.8%, Max 111%)	61.8% to 161.8%
	Running (B: Min 61.8%, Max 111%)	161.8% to 261.8%

Source snapshot - page 43: Flat correction rule table.

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Normal flat correction	Elongated flat correction	Truncated flat correction	Irregular flat correction (Strong-B)	Running flat correction
A= max61.80%	A= max61.80%	A= max61.80%	A= max61.80%	A= max61.80%
B=61.80% min = 81.20% max	B=61.80% min = 81.20% max	B=61.80% min = 111% max	B=Between 100% to 111%	B=Between 100% to 111%
C=61.20% min 161.80% max	C=Go beyond 161.80%-261.80%	C=does not go beyond 61.80%	C=61.20% min 161.80% max	C=above 161.80% -261.80%



Source snapshot - page 47: Normal, Elongated, Truncated, Irregular and Running Flat comparison.

Flat source rule summary - B 61.8%-111% belongs to Flat context; subtype determines C behavior.

```
// Simplified candidate concept after A and B exist
zigzagPossible = rulesZigZagAB_pass
flatPossible = rulesFlatAB_pass
doublePossible = contextAllowsDouble
trianglePossible = contextAllowsTriangle and enoughLegs
```

```
// Keep candidates in parallel.  
// Eliminate only when a source hard rule fails or a later structure confirms another path.
```

SH-7. The Premature Wave-5 Bug - Exact Structural Fix

The current/earlier implementation problem occurred because the engine advanced its wave phase after a certain swing count. That approach can label a rebound after W3 as W5 even while W4 is still building A-B-C or a complex correction.

Required replacement

After W3 is confirmed, the next state is not W5. The next state is W4_CORRECTION_CONTAINER. Inside it, run the permitted correction candidates. Only a completed valid correction may publish W4. Only then may W5_FORMING begin.

```
if W3.status == CONFIRMED  
    engineState := W4_CORRECTION_CONTAINER  
  
if engineState == W4_CORRECTION_CONTAINER  
    updateZigZagCandidate()  
    updateFlatCandidate()  
    updateDoubleCandidate()  
    updateTripleCandidate()  
    updateTriangleCandidate()  
  
    if oneCandidateConfirmed()  
        W4 := confirmedCorrectionEndPivot  
        engineState := W5_SEARCHING  
  
// Never: phase = (pivotIndex - anchor) % cycleLength
```

SH-8. Diagonal and Extension Exceptions - Do Not Apply Normal Rules Universally

A programmer who does not know Elliott Wave can easily make one dangerous mistake: applying normal impulse overlap/Fib rules to every impulse. The supplied notes define terminal/diagonal and extension subtypes with different behavior. Classify the subtype before applying subtype-specific invalidations.

Subtype	Where it appears	Internal structure / clue	Coding consequence
Leading Diagonal	Wave 1 / Wave A context	5-3-5-3-5 source model; terminal W3; overlap can occur	Normal overlap rejection cannot be used blindly.
Ending Diagonal	Wave 5 / Wave C	3-3-3-3-3; W3/W5 momentum relationship	Enable ED candidate and post-pattern correction logic.
W3 Extension	Usually W3	Move count can expand 5->9->13->17->21	Keep base fixed until degree-fall; count internal moves.
W5 Truncation	Special double-extension context	W5 can fail normal 127% target under source subtype	Do not mark invalid only for missing 127% if truncation context passes.
W5 Extension	Source has instrument restrictions/conflicts	Megaphone/equality context	Use instrument/source conflict mode; do not generalize to all markets.

Leading Diagonal (5-3-5-3-5)

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Nature	• Terminal impulse	Feature-5	• Wave-5 is Min 161.8% and Max 261.8% , Wave-5 and Wave-1 may be equal
Sub-moves	• Sub-waves (1,3,5) should have 5 moves 1-2-3-4-5, all are slow	Time : W2	• Subordinate W2 = ¼ or ½ Subordinate W1
Occurrence	• Within Wave-1 and Wave-A (Zig-Zag correction)	Time:W3	• If W1 not equal to W2; then W3 = [W1+ W2]/2 • If W1 = W2 [+/- 5]; then W3 = W1 + W2
Retracement	• Wave-4 can retrace Min 23.2% and Max till 61.8% of 0-3	Next pattern	• Wave-2 (degree) after the entire leading diagonal should take ¼X, ½X or X
Feature-1	• Wave-4 may enter Wave-1's territory and will not go below end of Wave-2	Wave-3	• Wave-3 (degree) after Wave-1 is an LD, would mostly be extended
Feature-2	• Wave-3 Fib extension will be terminal Min 61.8% and Max <161.8%	Special feature	• Fall of Wave-2 (degree) after completion of the leading diagonal can go till sub-Wave-4, sub-Wave-2 or Max till 81.2%
Feature-4	• Wave-3 will be terminal and is slow in nature		

Leading Diagonal source snapshot - subtype-specific overlap, Fib and time rules.

Ending Diagonal (3-3-3-3-3)

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Nature	• Terminal impulse	Feature-4	• Wave-3 is slow in speed
Sub-moves	• 3-3-3-3-3 (Triple Zig-Zag)	Feature-5	• Wave-5 is Min 127% and Max 261.8%, Wave-5 and Wave-1 may be equal in length
Occurrence	• Wave-5 and Wave-C (If Wave-3 is in extension, Wave-4 is a sharp move, then Wave-5 will be an ED)	MACD	• MACD forms negative divergence from Wave-3 (degree)
Retracement	• Wave-4 can retrace Min 23.2% and Max till 61.8% of 0-3	Time : W2	• Subordinate W2 = ¼ or ½ Subordinate W1
Feature-1	• Wave-4 enters Wave-1's territory and will not break Wave-2 low	Time:W3	• If W1 not equal to W2; then W3 = [W1+ W2]/2 • If W1 = W2 [+/- 5]; then W3 = W1 + W2
Feature-2	• Wave-3 Fib extension may be terminal Min 61.8% and Max <161.8%	Pattern End	• Correction after the entire ending diagonal should take 1/4 th or ½ its time
Feature-3	• Wave-2 and 4 won't retrace more than 50%	Criteria	• LD and ED will not form in the same impulse move (If W1 = LD, W5 = ED)

Ending Diagonal source snapshot - Wave-5/C location, 3-3-3-3-3 and divergence context.

SH-9. Price Structure First; Time, MACD and Trendlines Next

The supplied files contain several time-analysis variants. Treat time as a dedicated engine with selectable/owner-approved source mode. Time must not silently redefine a price pivot. Also note that time anchors can differ from Elliott base anchors in Flat corrections; the separate Time Analysis note explicitly warns not to mix those concepts.

Time Analysis

Confidence in trading can be achieved using study of market structure (Elliott Wave theory + Fib ratios) + Time analysis + Price Action (Candlestick patterns, indicators)

Market remains in correction 80% of the time and in impulse only 20% of the time.

Time period is a different concept than Elliott waves. While Elliot wave (base point in case of flat correction) may begin from a lower high than the price top (where top is 111%), time period is to be considered from market high and lows only (not to be misunderstood/mixed with base point 0 in case of flat correction)

Time analysis

1. Elliot Wave time periods (tp)
2. Fib Wave proportions (wp)
3. Fib sequence Numbers (fn)

Few important points about time period

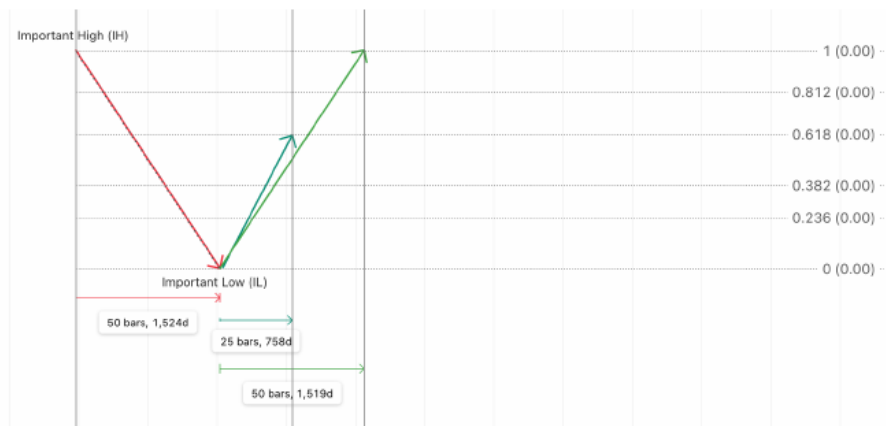
- a. Reversal zones: Points in time where all these time periods come together to form clusters and act as probable reversal zones
- b. High/Low price levels of 1/4th time period are most important for trading purposes

Elliott Wave Time Periods

Wave 1 time period: Wave 1 often starts as impulse after a correction.

What qualifies a good impulse:

1. When price recovers at least 61.8% of previous move (usually, downfall in stock price) from Important high (IH) to Important low (IL) in about ½ of the period taken by previous move (IH to IL)
2. When price recovers 100% of the fall in about the same time (time from IH to IL)



Impulse if **confirmed** only after formation of wave 3 (sharp upmove). Before that it is just an assumption/possibility. Fast up moves after sharp fall (correction) can also be X wave, after which the price will most likely enter into double / triple correction.

Time-analysis source overview - time period is a separate engine and uses market highs/lows according to the note.

TRENDLINES

- There are three types of trendline namely:-
 1. BASE TRENDLINE-Drawn between 0 point to 2 point and extended.
 2. ACTION TRENDLINE-Base trendline is copy paste and then put to wave 1 is called action line.
 3. MATURITY TRENDLINE-This trendline is drawn between wave 2 and wave 4.

Source trendline concept - Base, Action and Maturity lines provide wave-strength/reversal evidence.

Support engine	Example use	Rule of precedence
MACD	Base lowest low/+div, W3 top, W5 negative divergence, C/Y/Z completion clues.	Supports/ranks valid structure; does not rescue a hard-invalid wave.
RSI divergence	Project support at important pivots where approved.	Use corresponding Elliott pivots, not random local oscillator swings.
Trendlines/channels	0-2, 1-3, 2-4, 0-B, correction boundaries.	Draw from confirmed/active anchors and track break time/state.
Gaps/candles	Wave-specific supporting evidence.	Context-gated; not standalone wave-number generators.
Time	Wave duration relationships and reversal clusters.	Run selected source mode; conflicts remain explicit.

SH-10. Live-Chart Behavior: Forming, Locking, Recount and Non-Repaint

Elliott counts can evolve while a wave is forming. The goal is not to pretend every live candidate is permanent; the goal is to make the lifecycle explicit so confirmed history is stable and only genuinely invalidated structures are recounted.

- A latest raw swing may move to a more extreme price while its wave candidate is FORMING.
- A CONFIRMED wave pivot should be locked and should not migrate because a new minor pivot appears.
- If a hard Elliott rule later becomes impossible under the approved methodology, mark the candidate INVALID and log the first failing rule.
- Recount from the nearest structurally justified checkpoint; do not clear the entire history automatically.
- FBD/FBO is a project-specific recount CONTEXT, not an unconditional reset command.
- If two correction interpretations remain valid, keep a Primary and an ALTERNATE instead of forcing one final label.

```
onNewRawPivot(p):
  updateFormingCandidates(p)

  for candidate in activeCandidates
    evaluateHardStructure(candidate)
    if candidate.hardFail
      candidate.status := INVALID
      candidate.invalidReason := firstFailedRule
      promoteAlternateOrControlledRecount()
```

```

else if requiredConfirmationsPass(candidate)
    candidate.status := CONFIRMED
    lockPivot(candidate.endPivot)
else
    candidate.status := FORMING

```

SH-11. What to Do With the Existing Pine Script First

Do not rewrite useful infrastructure unnecessarily. Reuse the raw-pivot, oscillator-at-pivot, drawing-array and formatting utilities where they are sound. Replace the phase-assignment architecture that turns accepted pivots into wave numbers by cycle position.

Action	Current component type	Developer instruction
KEEP / REUSE	Confirmed raw pivot collection, ATR/range filtering, oscillator values at pivots, drawing cleanup	These are infrastructure; feed them into the new candidate engine.
REFACTOR	Important H/L and base scanner	Make degree/timeframe-aware; fix 144-day vs 144-chart-bar issue; add base lock lifecycle.
REPLACE	Modulo/fixed cycle phase assignment	Remove as the main wave-number decision engine.
REPLACE	"assign label then warn" hard-invalid logic	Hard invalidation must reject candidate and trigger alternate/recount logic.
ADD	Wave/Pattern candidate records and states	SEARCHING/FORMING/CONFIRMED/ALTERNATE/INVALID/TBD-BLOCKED.
ADD	Correction classifier	Zig-Zag, Flat, Double, Triple, Triangle candidates in parallel.
ADD	Subtype engines	Trending/Terminal, LD/ED, extensions, truncation.
ADD	Debug reason codes	Show numeric ratios, active candidates, next condition and first failed rule.

SH-12. Suggested Pine v6 Module Layout

```

// 1) DATA ACQUISITION
f_collectRawPivots()
f_collectOscillatorAtPivot()
f_degreeContext()

// 2) BASE ENGINE
f_baseCandidates()
f_confirmAndLockBase()
f_baseInvalidation()

// 3) IMPULSE ENGINE
f_wave1Candidate()
f_wave2CorrectionContainer()
f_wave3Classifier()
f_wave4CorrectionContainer()
f_wave5Classifier()

```

```
// 4) SUBTYPE ENGINES
f_leadingDiagonal()
f_endingDiagonal()
f_extensionEngine()
f_truncationEngine()

// 5) CORRECTION ENGINE
f_zigzagCandidate()
f_flatCandidate()
f_doubleCandidate()
f_tripleCandidate()
f_triangleCandidate()

// 6) SUPPORT ENGINES
f_waveSpecificFib()
f_timeEngine()
f_macdSupport()
f_divergenceSupport()
f_trendlineChannel()
f_gapCandleSupport()

// 7) LIFECYCLE
f_confirmCandidate()
f_invalidateCandidate()
f_promoteAlternate()
f_controlledRecount()

// 8) OUTPUT
f_drawConfirmedWaves()
f_drawFormingWave()
f_debugDashboard()
```

Pine resource limits

If Pine object/token/runtime limits prevent a literal one-to-one implementation, document the limitation BEFORE removing a source rule. Prefer compact records, bounded arrays, active-candidate windows and debug toggles. Do not silently simplify Elliott logic just to make the script compile.

SH-13. First Sprint - Minimum Demonstration Before Continuing

Before implementing every correction subtype, demonstrate that the new architecture is real. The following must work on both bullish and bearish examples.

13. Raw pivots are generated without automatic Elliott numbers.
14. A Point-0 candidate is selected, confirmed and locked with correct degree context.
15. Wave 1 remains FORMING until its required development criteria pass.
16. Wave 2 is a correction container and Point-0 violation rejects the impulse candidate.
17. Wave 3 is classified as Trending or Terminal rather than only PASS/WARN around 161.8%.
18. After W3, the script enters W4_CORRECTION_CONTAINER and cannot print W5 until W4 is confirmed.
19. A hard-invalid candidate is removed/reclassified, not just painted red.
20. A confirmed historical wave remains locked while the current forming wave can update.
21. The debug panel shows candidate, status, Fib/time/internal values, and exact invalidation reason.
22. The same OHLC sample yields equivalent confirmed states in Pine and Python for this core flow.

STOP CONDITION FOR DEVELOPMENT REVIEW

Do not approve later modules if the developer is still assigning wave numbers by pivot index/cycle position. Fix the core state/candidate architecture first. Adding more Elliott rules on top of a sequential labeler will make the code larger without solving the counting problem.

SH-14. Quick Glossary for the Programmer

Term	Plain developer-language meaning
Point 0 / Base	Origin of the active degree candidate. A locked checkpoint, not simply the latest low/high.
Degree	Scale of the Elliott structure. A lower-degree wave can exist inside a higher-degree wave.
Subordinate wave	Internal smaller-degree wave used to validate the parent wave.
Impulse	Directional five-wave structure 1-2-3-4-5, with subtype rules.
Correction	Counter-trend/complex structure that can resolve to several candidate types.
Retracement	Price movement measured back against a prior wave/range.
Extension	Projection beyond a prior range using a specified anchor pair.
Terminal	Impulse subtype where W3/overlap behavior differs from normal trending impulse.
Leading Diagonal	Terminal/diagonal structure permitted in W1/A context under source rules.
Ending Diagonal	Terminal/diagonal structure permitted in W5/C context under source rules.
Alternation	W2 and W4 should differ in corrective character/time according to source rules.
Rule of similarity	W2/W4 price-fall percentages can be similar within the approved tolerance.
Controlled recount	Re-evaluate only when structural invalidation/superior base/approved event justifies it.
Alternate count	Second still-valid interpretation kept until later evidence eliminates it.

SH-15. Instructions to the Developer Before Reading the Full Specification

- You are not expected to know Elliott Wave Theory before starting this project.
- Use this START HERE section to understand the software model, then use the remaining source-locked document for exact rules, exceptions, source variants and acceptance tests.
- Do not use a generic Elliott library or textbook rule to override a supplied source rule.
- Do not interpret a manual label option as an automatic detector. A pattern name must be produced by its classifier.
- Do not choose a source conflict silently. Expose it as a configuration/owner decision.
- When a qualitative source phrase such as fast/slow/microscopic lacks a numeric threshold, use the approved configurable tolerance or keep it supporting/TBD as defined later in the document.
- For each implemented module, provide at least one accepted example and one rejected example with debug reason codes.

Read next

Continue with Executive Summary, How to Read This Manual, Source Hierarchy and Conflict Policy, then Part I onward. The remainder of V3.1 is the 100% source-locked rule/reference pack.

Executive Summary

The current Pine Script contains useful components: confirmed pivot detection, ATR/range swing filtering, Important High/Low context, MACD/RSI values at pivots, a 61.8% degree concept, several Fibonacci levels, partial Wave-2/Wave-3/Wave-4/Wave-5 checks and a time-rule table. These should be reused.

The central problem is that the script still assigns Elliott labels mainly by swing sequence / cycle position and then checks rules afterward. A true Elliott engine must do the reverse: build a candidate, validate price structure first, then Fibonacci/time/internal structure, then use MACD/RSI/trendlines as supporting evidence, and only then confirm the wave label.

One sentence for the developer

Do not ask “what label comes next in the sequence?” Ask “which Elliott candidate(s) can this new confirmed pivot legally complete?”

Required Processing Order



Priority Summary

Priority	Required change	Reason
P0	Replace fixed/modulo wave cycle with candidate state machine	Without this, later rules only create warnings; they do not control the count.
P0	Separate raw pivots from Elliott pivots	Small internal swings must not advance the main-degree count.
P0	Lock Point 0 / Base after confirmation	Prevents anchor drift and uncontrolled recount.
P0	Make hard invalidation reject/recount	An invalid wave must not remain as a red but accepted label.

Priority	Required change	Reason
P0	Fix 144-day/degree Important H/L handling	144 chart bars is not 144 days on intraday charts.
P1	Implement Wave 1-5 structural engines	Core impulse logic.
P1	Automatic correction classifier	Zig-Zag, Flat, Double, Triple and Triangle cannot be manual label choices.
P1	Leading / Ending Diagonal + extensions	Needed for legitimate overlap and special Wave-5/C behavior.
P2	Wave-specific Fib/time/MACD/trendlines	Required for confirmation and debugging.
P2	Controlled recount / non-repaint / alternate counts	Required for live trading stability.
P3	Gaps, candlestick support, production debug report	Final supporting modules and validation.

How to Read This Manual

- **SOURCE RULE** = rule taken from the supplied Elliott/NeoWave notes/specification.
- **PROJECT REQUIREMENT** = additional behavior required for this project, such as HP BUY/SELL and FBD/FBO controlled recount.
- **HARD RULE** = violation invalidates that candidate.
- **GUIDELINE / SUPPORT** = affects rank or confidence, but should not override valid price structure unless the approved source explicitly makes it mandatory.
- **OWNER DECISION** = the supplied source contains conflicting or incomplete instructions; do not guess.

Source Hierarchy and Conflict Policy

The coding rules in this manual are grounded primarily in the supplied “EW Notes / Chartking” material and the derived “Elliott Wave Indicator - Developer Requirement Specification”. Some additional time-analysis material contains alternative values. When two source sections disagree, preserve the conflict instead of silently reconciling it.

OWNER DECISION / DO NOT GUESS

Examples of unresolved source conflicts include Wave-2 time, some Wave-4 retracement limits during extensions, precise terminal/trending Wave-3 boundaries, Wave-C elongated maximum, Double Correction confirmation, selected X/XX/Z rules, LD/ED coexistence wording, exact MACD parameters and the exact pivot definition. Implement a visible conflict mode or keep the affected candidate FORMING/TBD-BLOCKED until the owner chooses the rule.

PART I - Elliott Wave Concepts for a Programmer

1. Core Terminology

Term	Meaning for code
Point 0 / Base	Starting pivot of the current degree impulse/correction. It must be selected and then locked.
Impulse	Directional sequence normally described as Wave 1-2-3-4-5.
Correction	Counter-trend structure after/inside impulse: ABC, WXY, WXY-XX-Z, triangle, etc.
Degree	Scale of the wave. Main-degree pivots must be separated from subordinate/internal pivots.

Term	Meaning for code
Internal structure	The smaller waves that compose a larger wave. A large wave cannot be confirmed only from one start and one end pivot.
Candidate	A possible Elliott interpretation that is not yet confirmed.
Confirmed pivot	A pivot that has passed the required confirmation rules and should normally remain locked.
Alternate count	Secondary still-valid Elliott interpretation retained while the market is ambiguous.
Hard invalidation	A price/structure event that makes a specific candidate impossible.
Supporting evidence	MACD/RSI divergence, gaps, candlesticks, trendline/channel behavior, etc., used after structure.

2. State Model

State	Meaning	Drawing behavior
SEARCHING	No valid base/current candidate	No Elliott label or only debug marker.
FORMING	Enough structure exists for a possible wave, but required confirmation is incomplete	Optional DEV/FORMING label; may move/recount.
CONFIRMED	Required structural/Fib/time/confirmation conditions met	Lock main label unless later hard structural invalidation permits controlled recount.
ALTERNATE	Secondary valid interpretation remains possible	Show only in debug/alternate mode.
INVALID	Hard source rule broken	Reject candidate; store reason; do not keep as accepted red wave label.
TBD-BLOCKED	Source itself is blank/conflicting	Do not invent a rule; keep visible in debug.

3. Required Data Model

```
WaveCandidate {
    degree, timeframe, direction, label, patternType, subtype,
    startBar, endBar, startTime, endTime, startPrice, endPrice,
    durationBars, priceLength, fibRetrace, fibExtension,
    internalCount, internalPattern,
    macdLine, macdHist, rsi, divergenceState,
    trendlineState, channelState, gapState, candleState,
    status, hardInvalid, invalidReason, sourceRuleId,
    parentWaveId, childWaveIds[]
}
```

Also maintain raw pivot records separately from Elliott candidates. The same raw pivot may be used as an internal sub-wave without becoming a main-degree Elliott pivot.

PART II - Pivot, Degree and Base Engine

4. Raw Pivot Engine

The current pivot/ATR filtering infrastructure can be retained as a raw swing provider. However, raw pivot acceptance must not directly assign a main Elliott label.

Required rule

A raw pivot is market-structure data. An Elliott pivot is a validated semantic event. Keep these as different objects/arrays.

```
onNewConfirmedRawPivot(p):
    updateRawSwingBuffer(p)
    updateCurrentFormingLeg(p)
    evaluateExistingWaveCandidates(p)
    proposeNewCandidatesOnlyWhereContextAllows(p)
    neverAdvanceMainWaveNumberOnlyBecausePivotCountIncreased
```

5. Important High / Low and 144-Day Context

The current script uses a numeric lookback of 144 chart bars. On a 15-minute chart that is only 36 hours, so it cannot represent a 144-day source context. The Important H/L scanner must be degree/timeframe-aware.

```
importantHigh = highest(sourceDegreeHigh, lookbackEquivalentTo144Days)
importantLow  = lowest(sourceDegreeLow,  lookbackEquivalentTo144Days)
```

Recommended implementation: calculate the Important H/L on the approved source/degree timeframe using request.security(), and store the source timeframe, timestamp and price with the base candidate.

6. Base / Point 0 Candidate

A bullish base is an important low from which a degree-up impulse may begin. A bearish base is the mirrored important high.

Source basis: Wave-1 source framework: Important H/L context, MACD lowest low or positive divergence, and minimum 61.8% degree movement before confirming the degree Wave-1.

```
BullishBaseCandidate = importantLowContext
                      AND directionCanMoveUp
                      AND degreePotential
                      AND oscillatorEvidenceConfigured

BearishBaseCandidate = mirrored logic
```

Do not make “MACD extreme” and “MACD/RSI divergence” sound identical. The current script allows MACD extreme OR divergence. Make this user-selectable and visible in debug output.

7. Base Lock and Controlled Release

```
BASE_CANDIDATE -> BASE_CONFIRMED -> BASE_LOCKED
```

Release BASE_LOCKED only when:

- hard Elliott invalidation occurs, OR
- higher-degree completion / degree fall requires a new origin, OR
- approved FBD/FBO structural recount candidate proves superior, OR
- existing count becomes structurally impossible.

PROJECT REQUIREMENT

FBD/FBO may nominate a new Elliott origin/recount candidate, but must never blindly reset the Elliott count. The existing confirmed count remains until the new origin passes degree/structure tests or the old count is invalidated.

PART III - Impulse Engine: Wave 1 to Wave 5

8. Wave 1 Coding Algorithm

Source basis: Wave 1: impulsive, five subordinate moves / 5-3-5-3-5 internal framework; MACD lowest low or positive divergence at the base; degree confirmation after minimum 61.8% Important H/L movement; Leading Diagonal may occur in Wave 1.

```
W1 candidate:
  P0 = locked base
  build subordinate impulse candidate from P0
  track degree progress = abs(currentPrice - P0) / importantRange

if degreeProgress < 0.618:
  status = FORMING
else if internalImpulseValid and hardRulesPass:
  status = CONFIRMED
else:
  remain FORMING / INVALID depending on failed rule
```

Do not do this

Do not call the first raw pivot after P0 "Wave 1" simply because it is the first swing.

9. Wave 2 Coding Algorithm

Source basis: Normal Wave-2 retracement: 23.6%-81.2% of W1. Special microscopic clue around 14% is treated separately and is associated with an extended W3 possibility. W2 may form ABC or WXY; MACD generally falls slowly and should not violate the approved base-MACD condition.

```
For bullish impulse:
  W1Length = P1 - P0
  W2Retrace = (P1 - currentLow) / W1Length

For bearish impulse:
  W1Length = P0 - P1
  W2Retrace = (currentHigh - P1) / W1Length
```

Retracement	Classification	Action
~14% special case	W2_MICROSCOPIC candidate	Do not use as normal minimum; prepare W3-extension context.
23.6%-61.8%	Normal W2 range	Keep W2 candidate if internal correction valid.
61.8%-81.2%	Deep W2 / project HP zone	Still requires correction completion and confirmation.
>81.2% but Point 0 not broken	Outside normal range	Warn / apply owner-selected source rule; do not automatically call normal W2.
Crosses Point 0	Hard invalid	Reject bullish/bearish impulse candidate and controlled recount.

10. Wave 2 Internal Structure

Wave 2 is a correction, not merely the next opposite pivot. Maintain correction candidates underneath W2: Zig-Zag, Flat and W-X-Y where context allows.

```
W2_FORMING
|- ZigZag candidate (A-B-C)
|- Flat candidate (A-B-C)
`- WXY candidate
```

Confirm W2 only when one correction candidate completes and the Point 0 invalidation is preserved.

11. HP BUY / HP SELL - Project Trading Signal

PROJECT REQUIREMENT

The 61.8%-81.2% Wave-2 zone is a candidate zone, not an automatic trade. HP BUY/SELL requires Wave-2 corrective completion plus confirmation. Base/Point 0 is the structural invalidation reference. The project may also use support/AVWAP/Ripple confirmation and a confirmation candle; these should be implemented as configurable project filters rather than changing the Elliott count itself.

```
HP BUY candidate:
    bullish W1 CONFIRMED
    AND W2 is valid correction
    AND W2 retracement in 61.8%-81.2% zone
    AND Point 0 remains valid
    AND correction completion / reversal confirmation
    AND project confirmation filters pass
```

HP SELL = exact bearish mirror

Chart should show W1, W2, 61.8%, 81.2%, HP zone, confirmation state and invalidation. Do not print HP BUY/SELL just because price touches a Fib line.

12. Wave 3 Coding Algorithm

Source basis: Wave 3 is impulsive; source main degree rule uses minimum 161.8% of the 0-1-2 extension, while terminal subordinate W3 may exist from the terminal minimum below 161.8%. W3 normally has the strongest/top MACD context and can extend to 9/13/17/21 moves.

```
W3Projection = extensionFrom(P0, P1, P2, currentPrice)

if W3Projection >= 1.618 and internalImpulseValid:
    subtypeCandidate = TRENDING_W3
elif terminalRulesPass and W3Projection >= terminalMin and W3Projection < 1.618:
    subtypeCandidate = TERMINAL_W3
else:
    do not confirm W3 yet
```

The current script treats below-161.8% mainly as a warning. Replace this with an explicit trending-versus-terminal classifier.

13. Wave 3 Time and MACD

Keep the existing W3 time relationships, but make the selected source version visible. One source version uses the average of W1/W2 when their times differ and the sum when they are approximately equal; other notes present a simpler minimum-time relationship. This is a conflict-sensitive area.

OWNER DECISION / DO NOT GUESS

Do not hard-code one Wave-3 time formula as universally correct. Expose the selected rule set in the debug panel and use relative tolerance consistently.

MACD W3 support:

- compare degree MACD behavior at W3 candidate
- record top/highest behavior
- if price advances but MACD structure strongly contradicts W3 source clue, reduce rank / keep alternate correction candidate
- never let MACD create W3 without valid price structure

14. Wave 4 Coding Algorithm

Source basis: Wave 4 may be Zig-Zag, Flat, Triangle, Double or Triple correction. Normal source range is approximately 23.6%-50% using the approved Base-to-W3 convention; terminal cases may allow 61.8%. W4 time is related to W2 and normal impulse overlap rules differ from diagonal/terminal cases.

W4_FORMING:

- open correction candidates = {ZigZag, Flat, Double, Triple, Triangle}
- calculate approved Base->W3 retracement
- evaluate W2/W4 alternation and time
- apply overlap rule only after impulse subtype is known

The current script uses a 14% normal W4 minimum. Replace that with the approved 23.6% normal minimum; reserve special extension/microscopic handling for the contexts where the notes actually use it.

15. Wave 4 Overlap

Context	Wave-4 / Wave-1 overlap
Normal impulse	Normally invalid according to the normal impulse rule.
Leading Diagonal / terminal context	May be permitted according to diagonal rules.
Ending Diagonal internal structure	Overlap is expected/allowed within the ED rules.
Unknown subtype	Do not hard-invalidate until subtype tests are evaluated.

16. Wave 5 Coding Algorithm

Source basis: Normal W5 uses an impulsive internal structure, 127%-261.8% projection from 3-4 in the source summary, negative MACD divergence vs W3, possible Ending Diagonal, and special truncation/extension cases.

W5Projection = $\text{abs}(P5 - P4) / \text{abs}(P3 - P4)$

Normal W5 candidate:

- 1.27 <= W5Projection <= 2.618
- AND internal impulse valid
- AND W3-not-shortest hard rule passes
- AND W5/W3 divergence context passes configured rule

Also evaluate:

- W5_TRUNCATED
- W5_EXTENDED
- W5_ENDING_DIAGONAL

17. Wave 5 Negative Divergence

Compare the oscillator values at the actual degree Wave-3 and Wave-5 pivots. Do not compare random local pivots.

```
Bullish example:
  price(P5) > price(P3)
  AND MACD(P5) < MACD(P3)
  -> negative divergence support

Bearish example:
  price(P5) < price(P3)
  AND MACD(P5) > MACD(P3)
  -> positive/mirrored divergence support
```

18. Truncated Wave 5

Source basis: When W1 and W3 are both in extension, source notes include a special truncated W5 context with maximum around 81.2% of the 3-4 reference while still containing five sub-moves. Treat as separate subtype, not as a failure of the normal 127% rule.

PART IV - Correction Classifier

19. Why Correction Classification Must Be Candidate-Based

Early A-B price action can legitimately fit more than one correction. Therefore the engine must retain parallel candidates and eliminate them only as rules fail. A manual dropdown that simply changes labels is not a classifier.

```
After impulse/corrective start:
  candidates = {ZIGZAG, FLAT, DOUBLE, TRIPLE, TRIANGLE}

for each new confirmed subordinate pivot:
  evaluate each candidate
  invalidate impossible candidates
  update Fib/time/internal state
  if exactly one candidate meets confirmation -> CONFIRMED
  if more than one remains -> PRIMARY + ALTERNATE
```

20. Zig-Zag A-B-C

Source basis: Zig-Zag: A-B-C, internal 5-3-5; A fast, B slow, C fast; A minimum around 61.8% of previous-degree W5; B below the Flat-style range; C uses 0-A-B projection and minimum relationship to B; 0-B trendline and correction-end rules support confirmation.

Component	Coding requirement
A	Five-move/impulsive behavior for Zig-Zag context; minimum source retracement.
B	Corrective; generally below 61.8% in the Zig-Zag rule set; detailed developer summary gives roughly 1%-50%.
C	Impulsive five-move candidate; 0-A-B projection; minimum 127% retracement relationship to B in source summary.
Time	B >= A up to selected source limit; C time derives from A/B relationship.
Confirmation	0-B trendline / point-B reversal timing rules where applicable.
End	Whole correction minimum retracement requirement must be evaluated.

21. Flat A-B-C

Source basis: Flat: internal 3-3-5; A minimum around 38.2%; B 61.8%-111%; C behavior determines subtype. Source describes Normal, Elongated, Strong Irregular and Running variants.

If B retracement is 61.8%-111% of A:
 Flat candidates become eligible
else:
 do not force Flat just because the label sequence is A-B-C

Flat subtype	B context	C context (source summary)
Normal	61.8%-81.2%	C roughly 61.8%-161.8%
Elongated	61.8%-81.2%	C above normal range; source contains max-range conflict
Strong Irregular	61.8%-111%	C roughly 61.8%-161.8%
Running	61.8%-111%	C roughly 161.8%-261.8%

OWNER DECISION / DO NOT GUESS

Wave-C elongated maximum conflicts across the supplied notes (some tables stop at 261.8%, while other pages/graphics extend to 461.8%). Implement the selected conflict mode; do not silently choose.

22. Wave B - Critical Misclassification Rule

Do not use 61.8%-111% as a universal B-wave definition. It belongs to the Flat candidate. Zig-Zag B has a different range. This is the direct coding fix needed to prevent the “B becomes Wave 5 too early” problem from being solved only by a forced label sequence.

```
if B < 0.618 of A:  
    keep ZigZag candidate where internal structure supports it  
  
if 0.618 <= B <= 1.11 of A:  
    keep Flat candidate  
  
if B > 1.11 of A:  
    invalidate/reclassify Flat according to approved rule  
  
Never: “after W3, 2 more pivots occurred, therefore next pivot = W5”
```

23. Wave A

Wave A cannot be defined only as the first pivot after Wave 5. In Zig-Zag context it behaves as a five-move impulsive A; in Flat context it can have a three-move form. It can also form a Leading Diagonal. Use the correction candidate to decide the expected internal structure.

24. Wave C - Completion Must Be Real

Source basis: Wave C: impulsive five moves / 5-3-5-3-5, positive divergence in the source correction rules, 0-A-B extension, minimum 127% retracement of B, normal/truncated/elongated types, A/B-based time rule, and possible Ending Diagonal.

C_FORMING:
 maintain subordinate impulse/ED candidate
 calculate 0-A-B projections
 calculate C-vs-B retracement
 track C time expectation
 track divergence

C_CONFIRMED only when completion conditions pass.
Place C on the actual completion pivot/candle, not simply the third corrective pivot.

PROJECT REQUIREMENT

For this project, the C completion event should be attached to the actual completion candle/pivot when the required Fib objective and completion conditions are achieved. C/Z target events should not be shifted to a later arbitrary swing.

25. Double Correction W-X-Y

Source basis: Double Correction: complex/time-wise, mostly Wave 4; W and X can be Zig-Zag or Flat; Y may be Zig-Zag, Flat or Triangle; small X max 50%, large X 61.8%-111%; Y 61.8%-161.8% of 0-W-X; X time < W; Y time = W, W+X or (W+X)/2; positive MACD divergence at Y is supporting evidence. One source confirmation field is literally unresolved.

DOUBLE_FORMING:

```

classify W structure
classify X as small or large
validate X time < W
classify Y
validate 0-W-X Y projection 61.8%-161.8%
track MACD divergence near Y completion

```

If unresolved source confirmation is required:

```

status = TBD_BLOCKED or FORMING
do not invent a rule

```

26. Triple Correction W-X-Y-XX-Z

Use `XX` internally for the second connecting wave in the Triple Correction. If the chart display must retain the earlier W-X-Y-X-Z wording for familiarity, treat that as a display alias only; the internal data model should distinguish X and XX.

Source basis: Master notes identify Triple W-X-Y-XX-Z and explicitly warn that several source cells are incomplete; later pages supply some XX/Z detail but not every blank field is resolved.

TRIPLE_FORMING:

```

W -> X -> Y -> XX -> Z
store independent durations and Fib relationships
require X != XX semantic identity even if both are connecting corrections
evaluate MACD/time toward Y/Z completion
leave unresolved fields visible as TBD

```

27. Triangle A-B-C-D-E

Do not label every five-pivot sideways sequence as a triangle. The engine needs geometry, size sequence and boundary checks.

Source basis: Horizontal Contracting Triangle: trendlines travel in opposite price directions; $b < a$, $c < b$, $d < c$, $e < d$; A is widest; thrust after completion is normally 75%-125% of widest wave. Irregular Contracting Triangle has its own B-size condition and c/d/e contraction rules.

Contracting triangle candidate:

```

legs = {A,B,C,D,E}
confirm alternating direction
confirm decreasing leg sizes for subtype
build upper/lower boundary lines
validate convergence / source-specific apex rule
confirm E completion before thrust projection
thrustTarget = 0.75 to 1.25 * widestLeg (normal source range)

```

Implement expanding triangle separately using its source-specific size sequence. Do not reuse the contracting inequalities in reverse without checking the supplied rule table.

PART V - Diagonals and Extensions

28. Leading Diagonal

Source basis: *Leading Diagonal occurs in Wave 1 and Wave A; source specifies 5-3-5-3-5 framework, special overlap, W3 terminal range, W4 retracement, W5 projection and time behavior.*

```
LD candidate only if parentPosition in {W1, A}
AND internalPattern == 5-3-5-3-5
AND diagonal overlap/range rules pass
AND geometry/trendline behavior passes

Do not reject W4/W1 overlap using normal impulse rules until LD is ruled out.
```

29. Ending Diagonal

Source basis: *Ending Diagonal occurs in Wave 5 and Wave C; source uses 3-3-3-3-3, W4 overlap, W2/W4 restrictions, W5 127%-261.8%, negative divergence from W3 MACD and a post-pattern time expectation.*

```
ED candidate only if parentPosition in {W5, C}
AND internalPattern == 3-3-3-3-3
AND overlap rules pass
AND W2/W4 restrictions pass
AND Fib + divergence support pass
```

30. Extension Engine

Source basis: *Extension move counts use 5, 9, 13, 17, 21; maximum 21. Every extension subtype changes follow-on retracement/structure expectations. W1, W3 and W5 may extend, with W3 being the common case.*

```
extensionMoveCount in {5,9,13,17,21}

if internalCount > 5:
    classify parent as extension candidate
    apply subtype-specific rules
    do not let every internal pivot advance parent wave number
```

This module is one of the main reasons raw pivots and main-degree pivots must be separate.

PART VI - Fibonacci, Time, MACD, Trendlines and Project Context

31. Wave-Specific Fibonacci Anchors

Candidate	Required anchor idea
Wave 2	Point 0 -> Wave 1 retracement
Wave 3	0-1-2 extension
Wave 4	Approved Base -> W3 source convention
Wave 5	3-4 projection
Wave C	0-A-B extension / B retracement relationship
Wave Y	0-W-X extension
Wave Z	Triple-correction source anchors; expose selected formula in debug

Candidate	Required anchor idea
Degree W1	Important High/Low range

Draw only levels relevant to the active candidate. Label the anchor pair. A generic Important H/L ladder is useful as a guide but is not a substitute for wave-specific Fibonacci logic.

32. Source Fibonacci Universe

The supplied notes use levels including 14/14.2, 23.2/23.6, 38.2, 50, 61.8, 81.2, 100, 111, 127/127.2, 161.8, 200, 261.8, 361.8 and 461.8, with larger W3 extension values in special cases. Do not display all levels at once; activate by candidate.

33. Time Engine

Time rules must be part of candidate evaluation, not only a dashboard note. Use duration from actual confirmed pivot times/bars. Preserve source conflicts.

Structure	Time relationship to implement
Wave 2	Source variants include 1/4, 1/2, 1X, 2X of W1; exact allowed set is conflict-sensitive.
Wave 3	Source variants derive from W1/W2; implement selected rule and show it.
Wave 4	Related to W2; source versions differ (e.g., 1/2, 2X, 3X in one summary; other notes show 5X). W4 should not simply equal W2 in the relevant rule set.
Wave B	$B > A$; subtype-dependent upper timing context.
Wave C	If $A=B$, $C=A+B$; if $A < B$ /different, source uses an average formula in the developer summary.
Wave X	$X < W$.
Wave Y	$Y = W$, $W+X$ or $(W+X)/2$.
Wave XX/Z	Use source-specific Triple Correction relationships; do not invent missing rows.

```
relativeTimeError = abs(actual - expected) / expected
timePass = relativeTimeError <= configuredTolerance
```

Use relative tolerance consistently. The current script mixes relative and absolute ratio tolerance in different places.

34. Wave-Specific MACD / RSI

Stage	MACD / oscillator role
Base / W1	Lowest low or positive divergence support.
W2	Falls slowly; should respect approved base-MACD behavior.
W3	Top/highest momentum clue.
W4	Can fall sharply / form low reading in price-wise correction.
W5	Negative divergence vs W3.
Simple C	Positive divergence can support completion.

Stage	MACD / oscillator role
B / X	MACD can move with speed; advanced classifier uses high MACD without matching price extension as clue.
Y / Z	Positive divergence may support complex-correction completion.
RSI	Project supporting divergence; use corresponding Elliott pivots, not random local swings.

Keep MACD fast/slow/signal and divergence pivot sensitivity configurable because the source does not specify exact numerical parameters. Also expose whether divergence is computed from MACD line, histogram or both.

35. Trendline / Channel Engine

Support the Elliott-specific lines requested by the source: 0-2, 1-3, 2-4, 0-B and triangle boundaries. Store line anchors and break time/state. These lines are part of confirmation and should not be decorative only.

36. FBD / FBO Controlled Recount

PROJECT REQUIREMENT

FBD/FBO is a structural context trigger for this project. It should nominate a possible new Elliott origin, not directly assign a new wave count.

```
onConfirmedFBDorFBO(event):
    candidateOrigin = event.structuralPivot
    if candidateOrigin passes important/degree tests:
        compare with current locked base
        if current count hard-invalid OR candidate clearly superior by approved rule:
            start controlled alternate/recount
        else:
            keep current count; store event as supporting context
```

37. Degree / Multi-Timeframe Handling

Every candidate must store its degree and source timeframe. The source documents contain a default Daily / 75m / 15m degree framework, while the broader project may use a richer timeframe ladder. Implement degree mapping as a configurable routing layer; do not let a 5-minute internal pivot and a Daily major pivot live in the same wave array without degree metadata.

PROJECT REQUIREMENT

The project requires multi-timeframe Elliott degrees to remain distinct. A lower-timeframe sub-wave can explain the internal structure of a higher-degree wave, but must not automatically overwrite the higher-degree count.

PART VII - Live Behavior, Recount and Non-Repaint

38. Developing vs Confirmed Pivots

The most recent raw pivot may continue to move to a more extreme price before its reversal is confirmed. That is acceptable for a FORMING/DEV wave. It is not acceptable to visually present that pivot as permanently confirmed and later move it without explanation.

```
FORMING label -> may update
CONFIRMED label -> locked
```

A confirmed label changes only after an approved hard structural invalidation / controlled recount event.

39. Controlled Recount Rules

Trigger	Allowed behavior
Minor new raw swing	Do not restart main count.
Hard wave invalidation	Reject candidate and promote valid alternate / recount.
New valid Important H/L base	Open alternate origin; do not instantly replace locked base.
Confirmed FBD/FBO important reversal	Open structural recount candidate.
Correction resolves differently	Promote surviving correction candidate.
Higher-degree completion / degree fall	Start new degree candidate according to routing rules.

40. Alternate Count

Primary: W4 FORMING - Flat candidate
 Alternate: Wave B FORMING - larger correction candidate

Keep both until:
 one violates a hard rule, OR
 one reaches its confirmation conditions.

Do not force Wave 5 simply because it is the next expected label after a fixed number of pivots.

PART VIII - Exact Updates Required in the Current Pine Script

Current element	Keep / replace	Required update
ta.pivohigh / ta.pivotlow infrastructure	KEEP	Use as raw confirmed pivot provider; preserve confirmation delay semantics.
ATR / minimum swing filters	KEEP	Use for raw swing quality only, not Elliott semantics.
swing arrays storing MACD/RSI	KEEP	Extend with degree, status, parent/child and candidate IDs.
f_phase / cycleLength / modulo logic	REPLACE	Remove as primary wave-number engine.
f_valid_anchor_index rescanning	REPLACE	Use persistent base candidate / locked base lifecycle.
f_rule_state warnings	REFACTOR	Convert hard rules into candidate rejection; soft rules into ranking/debug evidence.
Wave 2 min 14%	FIX	Normal W2 min 23.6%; 14% separate microscopic subtype.
Wave 2 no explicit 81.2 normal max	FIX	Implement normal max/context and owner-selected invalidation behavior.
Flat B 61.8-111 applied generically	FIX	Apply only inside Flat candidate.
showWave4Internal forced A/B/C sequence	REPLACE	Use actual correction candidates; do not force positional A/B/C before W5.
Manual correctionPattern dropdown	KEEP AS DISPLAY FILTER ONLY	Detection must be automatic.

Current element	Keep / replace	Required update
Wave 4 min 14%	FIX	Use approved normal 23.6% source range.
W4 overlap universal	FIX	Make subtype-aware; diagonals/terminal exceptions.
W5 127-261.8 target infrastructure	KEEP	Integrate into proper W5 classifier plus truncation/extension/ED.
Generic Important H/L fib ladder	KEEP AS GUIDE	Add wave-specific Fib engines.
Time table	KEEP	Refactor into actual candidate constraints; show selected conflict mode.
Drawing cleanup arrays	KEEP	Reuse for labels/lines, but attach objects to candidate IDs/status.
No Leading/Ending Diagonal engine	ADD	Implement dedicated subtype detectors.
No extension move-count engine	ADD	5/9/13/17/21 internal count classifier.
No WXY/Triple structural engine	ADD	Implement pattern rules instead of text labels.
No controlled FBD/FBO recount	ADD	Project requirement.
No HP BUY/SELL	ADD	Project Wave-2 signal with confirmation.
No Pine/Python parity output	ADD	Same OHLC must produce materially same confirmed count.

PART IX - Suggested Engine Pseudocode

41. Main Event Loop

```

onBarClose():
    updateIndicators()
    newPivot = detectConfirmedRawPivot()

    if newPivot:
        rawSwings.upsert(newPivot)
        updateImportantHLAndDegreeContext()
        updateBaseCandidates(newPivot)

        if lockedBase exists:
            updateImpulseCandidates(newPivot)
            updateCorrectionCandidates(newPivot)
            updateAlternateCounts(newPivot)

        applyHardInvalidations()
        promoteConfirmedCandidates()
        performControlledRecountIfRequired()
        updateDrawingsFromCandidateState()
        updateDebugPanel()

```

42. Candidate Validation Order

```
validate(candidate):
    if not parentContextAllows(candidate): INVALID
    if hardPriceRuleFails(candidate): INVALID
    if internalStructureImpossible(candidate): INVALID

    calculateFib(candidate)
    calculateTime(candidate)
    calculateMomentumSupport(candidate)
    calculateTrendlineSupport(candidate)

    if mandatorySourceConditionsComplete:
        CONFIRMED
    else:
        FORMING

    return status + exact reason codes
```

43. Example Reason Codes

Code	Meaning
W2_ORIGIN_BREAK	Wave 2 crossed Point 0.
W2_RETRACE_OUTSIDE_NORMAL	W2 outside normal source retracement range.
W3_BELOW_TRENDING_MIN	W3 below 161.8%; terminal classifier must decide.
W3_INTERNAL_FAIL	Expected impulsive internal structure failed.
W4_NORMAL_OVERLAP	Normal impulse W4 entered W1 territory.
W5_W3_SHORTEST	Proposed W5 makes W3 the shortest impulse wave.
FLAT_B_GT_111	Flat B exceeded approved maximum.
C_INTERNAL_INCOMPLETE	C Fib reached but internal completion missing.
DOUBLE_CONFIRM_TBD	Source confirmation field unresolved.
BASE_REPLACEMENT_NOT_PROVEN	New origin candidate is not strong enough to replace locked base.

PART X - Acceptance Test Matrix

ID	Module	Test condition	Expected result
T01	Base/W1	Important H/L + base oscillator evidence + 61.8% degree progress	W1 FORMING -> CONFIRMED; base locked.
T02	Normal W2	23.6%-81.2% retracement with valid correction	W2 remains valid; Point 0 preserved.
T03	Microscopic W2	Special ~14% case	W2_MICROSCOPIC; W3 extension candidate.
T04	Invalid W2	W2 crosses Point 0	Reject W2/impulse; controlled recount.

ID	Module	Test condition	Expected result
T05	HP BUY	Bull W1 + W2 61.8%-81.2% + correction completion + confirmation	HP BUY only after confirmation.
T06	HP SELL	Bearish mirror	HP SELL only after confirmation.
T07	Trending W3	Internal impulse + $\geq 161.8\%$ + MACD support	W3_TRENDING CONFIRMED.
T08	Terminal W3	Valid terminal structure below 161.8%	W3_TERMINAL; not generic warning.
T09	Extended W3	9/13/17/21 internal moves	Extension subtype; main count remains W3.
T10	Normal W4	Valid 23.6%-50% correction	W4 classification and time/alternation visible.
T11	Complex W4	Double/Triple/Triangle example	No premature W5.
T12	Normal W5	127%-261.8% 3-4 + internal + divergence	W5 CONFIRMED.
T13	Truncated W5	Valid double-extension context	W5_TRUNCATED, not rejected by normal min.
T14	Leading Diagonal	Valid W1/A LD	LD subtype; overlap allowed by LD rules.
T15	Ending Diagonal	Valid W5/C ED	ED subtype + divergence/post-pattern logic.
T16	Zig-Zag	5-3-5 + correct A/B/C	Automatic Zig-Zag classification.
T17	Flat	3-3-5 + B 61.8%-111%	Correct Flat subtype.
T18	B below Flat range	B < 61.8 with valid Zig-Zag context	Do not force Flat.
T19	B > 111	Flat invalidation case	Flat candidate rejected/reclassified.
T20	Wave C	C internal impulse + Fib/time/divergence	C on actual completion pivot/candle.
T21	Double WXY	Small/large X and valid Y projection/time	Automatic Double candidate; unresolved confirmation marked TBD.
T22	Triple	W-X-Y-XX-Z	All components and time/Fib relationships tracked.
T23	Contracting Triangle	ABCDE with correct geometry/size sequence	Triangle subtype + thrust projection.
T24	Expanding Triangle	Source-valid expanding case	Separate expanding subtype.
T25	False small swings	Many internal pivots inside W3	Main count remains W3.
T26	FBD/FBO recount	Important confirmed reversal event	Open recount candidate; no blind reset.

ID	Module	Test condition	Expected result
T27	Alternate count	Ambiguous W4 vs B / Flat vs Zig-Zag	Primary + alternate until resolution.
T28	Non-repaint	Live sequence with developing last leg	Only FORMING pivot moves; confirmed labels stay locked.
T29	144-day test	Daily, 1H, 15m	Same intended Important H/L source context, not 144 local bars.
T30	Pine/Python parity	Same OHLC dataset	Materially same confirmed count; differences documented.

44. Mandatory Developer Evidence for Each Test

- Chart screenshot with Elliott labels and candidate status.
- Exact raw pivots used for the main-degree count.
- Numeric Fib ratios and anchors.
- Wave durations and selected time rule.
- MACD/RSI values at compared pivots where divergence is used.
- First hard invalidation reason if a candidate is rejected.
- Primary vs alternate candidate where applicable.
- Pine output and Python output for parity cases.

PART XI - Source Conflict Register / Owner Decisions

ID	Topic	Conflict / uncertainty	Developer action
C01	Wave-2 time	Source pages give different allowed time relationships.	Configurable conflict mode; do not silently choose.
C02	W4 max during W1 extension	38.2% vs 50% appears in different sections/examples.	Keep context-specific setting until owner locks.
C03	W3 terminal/trending boundary	Main degree min 161.8%; terminal subordinate may begin near 61.8% and stay below 161.8%.	Explicit classifier; define equality/tolerance.
C04	LD/ED coexistence wording	Source wording is ambiguous about coexistence in same impulse.	Owner decision / visible mode.
C05	Zig-Zag / Wave-C max	261.8% vs 461.8% appears across tables/graphics.	Do not auto-reconcile.
C06	Double correction confirmation	Source cell is "???".	TBD-BLOCKED / FORMING; do not invent.
C07	Double/Triple move totals	Summary totals differ by pattern combination.	Implement per-pattern tables and log combination.
C08	Small vs large X	Different general/specific ranges appear.	Classify small vs large X explicitly.
C09	Triple X/XX time	Different X-time constraints appear.	Context-specific rule + owner choice.
C10	Z failure / failure percentage	Different nearby notes reference 31.8 vs 38.2 and different wave labels.	Do not hard-code until confirmed.

ID	Topic	Conflict / uncertainty	Developer action
C11	MACD parameters	No exact fast/slow/signal or divergence algorithm in source.	Inputs + visible settings.
C12	Pivot definition	No single exact wave endpoint algorithm in source.	Owner-approved pivot engine; do not let implementation silently redefine theory.
C13	Qualitative terms	Fast/slow/steep/microscopic/not long are not fully quantified.	Keep non-blocking or expose thresholds until approved.

PART XII - Required Implementation Sequence

Phase	Deliverable	Acceptance gate
Phase 1 - P0	State engine, raw/main pivot separation, locked base, hard invalidation, 144-day fix, controlled recount skeleton	T01, T04, T25, T28, T29 must pass before proceeding.
Phase 2 - Core impulse	W1/W2/W3/W4/W5 + HP + normal Fib/time	T02-T13.
Phase 3 - Corrections	Zig-Zag, Flat, A/B/C, WXY, Triple, Triangle	T16-T24.
Phase 4 - Special impulse	Leading/Ending Diagonal + extensions/truncation	T09, T13-T15.
Phase 5 - Supporting engines	MACD/RSI, trendlines/channels, FBD/FBO recount, degree routing	T26-T27 plus debug evidence.
Phase 6 - Parity/production	Python parity, dashboard/debug, performance optimization	T30 plus full regression suite.

45. What the Developer Must Return

23. Updated Pine Script source.
24. Updated Python implementation using the same rule/state model.
25. Implementation checklist: DONE / PARTIAL / TBD for every numbered requirement in this manual.
26. Visible list of all owner decisions / source conflicts still unresolved.
27. Screenshots and debug records for the acceptance tests.
28. Pine-vs-Python comparison on identical OHLC data.
29. Non-repaint statement explaining what is mutable (FORMING) and what is locked (CONFIRMED).
30. Performance note: maximum raw pivots, candidate limits, request.security usage and drawing-object limits.

46. Definition of Done

The Elliott module is complete only when...

The engine can explain why Point 0 was selected; why W1/W2/W3/W4/W5 are structurally valid; which correction type is active; why a B move is not prematurely W5; where C actually completed; which Fib anchors and time rule were used; what invalidates the count; why a recount occurred; which labels are locked; and whether Pine and Python agree.

Do not accept this shortcut

"The chart displays 1-2-3-4-5-A-B-C" is NOT proof that Elliott Wave logic is implemented. The labels must be the output of the rules, not the input assumption.

Appendix A - Quick Formula Reference

Formula / concept	Implementation
Bull W2 retracement	$(P1 - P2) / (P1 - P0)$
Bear W2 retracement	$(P2 - P1) / (P0 - P1)$
W5 projection base	$\text{abs}(P3-P4)$; W5 ratio = $\text{abs}(P5-P4)/\text{abs}(P3-P4)$
Flat B retracement	$\text{abs}(B-A) / \text{abs}(A-\text{startOfA})$, using exact candidate anchors
Relative time tolerance	$\text{abs}(\text{actual-expected})/\text{expected} \leq \text{tolerance}$
Bull W5 negative divergence	$\text{price5} > \text{price3}$ AND $\text{oscillator5} < \text{oscillator3}$
Bear W5 mirrored divergence	$\text{price5} < \text{price3}$ AND $\text{oscillator5} > \text{oscillator3}$
Contracting triangle basic size order	$A > B > C > D > E$ for horizontal contracting subtype
Normal HP zone	W2 retracement 61.8%-81.2% + correction completion + project confirmation

Appendix B - Developer Checklist Before Coding

- [] Have you removed the modulo/fixed-cycle count as the primary wave engine?
- [] Are raw swings and Elliott pivots stored separately?
- [] Can a hard-invalid candidate be rejected instead of merely colored red?
- [] Is the confirmed base persistent/locked?
- [] Is Important H/L truly degree/timeframe-aware rather than 144 local bars?
- [] Does W2 use 23.6%-81.2% as normal range and keep 14% as a special microscopic case?
- [] Does the Flat B 61.8%-111% rule apply only to Flat candidates?
- [] Can the engine keep Zig-Zag and Flat as simultaneous early candidates?
- [] Does C wait for actual completion instead of fixed pivot position?
- [] Can W3 be Trending, Terminal or Extended?
- [] Can W5 be Normal, Truncated, Extended or Ending Diagonal?
- [] Are Leading/Ending Diagonal overlap rules separated from normal impulse rules?
- [] Does extension counting use internal 5/9/13/17/21 logic?
- [] Can W4 automatically become Zig-Zag/Flat/Double/Triple/Triangle?
- [] Is Triple Correction stored as W-X-Y-XX-Z internally?
- [] Are Fib anchors wave-specific and shown in debug?
- [] Are time-rule source conflicts visible/configurable?
- [] Are MACD parameters/divergence settings configurable?
- [] Can FBD/FBO nominate a recount without blindly resetting?
- [] Are FORMING and CONFIRMED labels visually/state-wise different?
- [] Can the engine maintain an ALTERNATE count?
- [] Do Pine and Python return materially the same confirmed count?

Appendix C - Source Basis Used for This Manual

- Current Pine Script supplied for audit: "Elliott Wave Notes Overlay" (1,007-line pasted Pine Script).
- EW Notes / Chartking v2 reference material: impulse Waves 1-5, diagonals, extensions, Zig-Zag, Flat, Double, Triple, Triangle, MACD, trendlines/channels and time rules.
- Elliott Wave Indicator - Developer Requirement Specification: normalized rule tables, candidate-state architecture, conflict register and acceptance matrix.
- Elliott Wave Master Notes / Pivot Table: consolidated pivot-by-pivot quick reference and unresolved-source warnings.
- Project-specific requirements carried into this handoff: HP BUY/SELL at deep Wave-2 completion with confirmation; FBD/FBO controlled recount; actual C/Z completion-event marking; degree-aware multi-timeframe separation; Pine/Python parity.

END OF DEVELOPER CODING MANUAL

DETAILED IMPLEMENTATION ADDENDUM - FOR A PINE SCRIPT EXPERT WITH ZERO ELLIOTT WAVE BACKGROUND

Why this addendum exists

The earlier manual gives the architecture and missing modules. This addendum goes one level deeper: it explains how a programmer should think about each Elliott object, which prices become anchors, how to calculate ratios, when a candidate is only FORMING, what confirms it, what invalidates it, how to preserve source conflicts, and exactly how the current Pine overlay should be refactored.

Developer mindset

Treat Elliott Wave as a constrained parser over confirmed price pivots, not as a label sequence. The algorithm should parse raw pivots into several possible structures, reject impossible structures, rank the remaining structures using Fib/time/MACD/trendline evidence, and only then print confirmed labels.

A. What You Must Understand Before Writing Any More Pine Code

A1. There Are Three Different Things Called a “Swing”

Object	What it is	Can it advance main Elliott count?	Can it move later?
Raw pivot	A confirmed local high/low from pivot/ATR logic.	No. It is only input data.	The newest same-type raw pivot may be replaced until its reversal is confirmed.
Internal/subordinate pivot	A raw pivot selected as part of a larger wave internal structure.	No, unless it also qualifies at the parent degree.	Only while parent/internal leg is FORMING.
Main-degree Elliott pivot	A pivot that completed the required structure/Fib/time conditions for the selected degree.	Yes.	Normally locked once CONFIRMED; only controlled recount can change it.

IMPLEMENTATION WARNING

The current script does not make this separation strongly enough. Its accepted swing sequence directly drives the wave phase. That is the root cause of small swings becoming Elliott labels too early.

A2. The Correct Question at Every New Pivot

WRONG QUESTION:

“What label comes after the last label?”

CORRECT QUESTIONS:

1. Which currently FORMING candidates can this pivot legally complete?
2. Which candidates become impossible because of this pivot?
3. Does the pivot belong to the current degree or only to an internal degree?
4. Do Fib/time/internal rules confirm anything now?
5. If multiple candidates remain, which is PRIMARY and which is ALTERNATE?

A3. Impulse Rules vs Correction Guidelines

The source explicitly distinguishes impulse analysis as rule-based and correction analysis as guideline-based. In software terms, this means impulse has stronger hard-invalid conditions, while corrections need parallel candidates because the same early A-B path can later become Zig-Zag, Flat, Double, Triple or Triangle.

Source basis: *Developer Requirement Specification p.2; Chartking EW Notes p.3.*

B. Exact Audit of the Current 1,007-Line Pine Script

The following line ranges refer to the supplied file "Elliott Wave Notes Overlay". The aim is to tell the developer what can be reused and what must be replaced.

Current line range / function	What it currently does	Decision	Exact update
147-197: f_correction_count / f_cycle_length / f_phase / f_phase_text	Builds a fixed repeating sequence of 0,1,2,3,A,B,4/C,5 and correction labels.	REPLACE AS COUNT ENGINE	Keep only optional text-format helpers. Wave identity must come from candidate validation, never modulo phase.
257-321: f_start_confirmation_note	Important H/L, ATR, degree, MACD extreme and RSI/MACD divergence checks.	KEEP + REFACTOR	Split into BaseCandidate scoring and BaseConfirmation. Make oscillator mode explicit: extreme OR divergence is not the same as mandatory divergence.
323-353: f_valid_anchor_index / f_wave_phase	Rescans and lets the latest passing anchor determine cycle phase.	REPLACE	Persist BASE_CANDIDATE / BASE_LOCKED. A new passing base opens an alternate origin; it does not automatically replace the locked base.
355-483: swing filters / f_upsert_swing	Maintains filtered alternating pivots and replaces same-type pivots with more extreme ones.	KEEP AS RAW PIVOT ENGINE	Rename conceptually to rawSwings. Add pivot confirmation state and do not attach Elliott labels directly.
485-689: f_rule_state	Calculates warning/invalidation booleans after phase label is already known.	REFACTOR	Each wave validator should return CandidateStatus. Hard invalid must reject the candidate; soft warning should only reduce rank/show debug.
509-537: W2	Origin invalidation + minimum retrace/time warning.	FIX	Normal W2 range depends on selected source mode; implement explicit microscopic subtype and max-range handling.
539-579: W3	161.8% ratio + one time formula.	EXPAND	Add internal 5-move/extension count, Trending vs Terminal subtype, MACD top support, source-selected time rule.
581-657: W4 A/B/C and W4	Forces internal A/B/C by pivot position and uses Flat B range.	REPLACE	Create real correction candidates. Do not force every W4 to be A-B-C or every B to use 61.8%-111%.
659-687: W5	Normal W5 127%-261.8 + W3-shortest invalidation.	KEEP + EXPAND	Add W5 divergence, truncation, W5 extension, Ending Diagonal, internal move count.
691-797: time table	Displays selected time relationships.	KEEP UI + REWRITE LOGIC	Make time rules part of candidate evaluation and show which source mode is active.
819-887: Fib guide + W5 targets	Important H/L Fib ladder and W5 projection.	KEEP DRAWING INFRA	Add wave-specific anchor sets: 0-1,0-1-2,0-3,3-4,0-A-B,0-W-X,0-Y-XX.
891-947: redraw	Draws every phase label even if hardInvalid.	REPLACE DRAW DECISION	FORMING candidates may be optional/dotted; CONFIRMED labels locked; INVALID candidates should not look like accepted waves.
949-1007: pivot calculation	Confirmed pivots with right-bar delay.	KEEP	Document delay as confirmation latency, not repainting; attach pivot to raw degree buffer only.

C. Foundation Math Helpers the Developer Should Build First

Before implementing any pattern, create a small library of direction-safe math functions. This reduces long/short asymmetry bugs.

C1. Direction Convention

```
direction = +1 for bullish candidate
direction = -1 for bearish candidate

moveSigned(a, b, direction) = direction * (b - a)
moveAbs(a, b) = abs(b - a)
```

C2. Retracement and Extension Functions

Generic retracement	$\text{retr} = \text{abs}(\text{retraceEnd} - \text{retraceStart}) / \text{abs}(\text{referenceEnd} - \text{referenceStart})$
Bull W2	$(P1 - P2) / (P1 - P0)$
Bear W2	$(P2 - P1) / (P0 - P1)$
W3 0-1-2 extension ratio	$\text{direction} * (P3 - P2) / \text{abs}(P1 - P0)$
W4 Base-to-W3 retracement	$\text{abs}(P3 - P4) / \text{abs}(P3 - P0)$ [use only if this is the owner-approved anchor convention]
W5 3-4 projection ratio	$\text{direction} * (P5 - P4) / \text{abs}(P3 - P4)$
ABC 0-A-B C extension	$\text{directionOfA} * (C - B) / \text{abs}(A - \text{originA})$
C retracement of B	$\text{abs}(C - B) / \text{abs}(B - A)$
WXY Y extension	$\text{directionOfW} * (Y - X) / \text{abs}(W - \text{originW})$
Triple Z extension	$\text{directionOfY} * (Z - XX) / \text{abs}(Y - \text{originTriple})$

Important

The formulas above are coding translations of the source Fib anchor descriptions. The debug panel must always print the anchor prices used. If the owner chooses a different source convention for a conflicted anchor, change the helper input - not scattered pattern code.

C3. Time Helpers

Duration	$\text{endBar} - \text{startBar}$ (or exact timestamp difference when required)
----------	---

Relative tolerance	$\text{abs}(\text{actual} - \text{expected}) / \text{max}(\text{expected}, \text{eps}) \leq \text{tolerancePct}$
--------------------	--

Absolute bar tolerance	$\text{abs}(\text{actualBars} - \text{expectedBars}) \leq \text{toleranceBars}$
------------------------	---

IMPLEMENTATION WARNING

Do not compare 0.5x time using a raw absolute tolerance while comparing another rule using percentage tolerance. The current script mixes these styles. Standardize the calculation and expose the tolerance source.

C4. Divergence Helper

Bullish positive divergence:
price2 < price1 AND oscillator2 > oscillator1

Bearish negative divergence:
price2 > price1 AND oscillator2 < oscillator1

For W5 vs W3, compare only the actual degree W3 and W5 pivots.
For base divergence, compare same-type important lows/highs at the selected degree.

D. Base Point and Degree Formation - Step-by-Step

D1. Timeframe and Degree Are Configuration, Not One Universal Mapping

The supplied notes contain more than one timeframe mapping. Chartking uses a Daily/75m/15m framework for Major/Intermediate/Minor in its main notes, while the Hardik material changes degree timeframes depending on investment, swing or day-trading horizon. Do not silently combine them.

Source	Major	Intermediate	Minor / lower	Implementation
Chartking main framework	Daily	75-minute	15-minute	Use as one selectable preset.
Hardik - 1 year holding	Monthly	Weekly	Daily	Selectable long-horizon preset.
Hardik - swing 1-3 months	Weekly	Daily	75m / 288m / 1h commodity	Selectable swing preset.
Hardik - day trading	Daily	75m	15m	Essentially matches Chartking day-trading mapping.

SOURCE CONFLICT - DO NOT GUESS

Timeframe/degree mapping differs across the supplied notes. Add a "Degree Mapping Preset" input and show the active mapping in the dashboard. Never let timeframe routing change silently because current price crosses an arbitrary threshold unless that source mode is explicitly enabled.

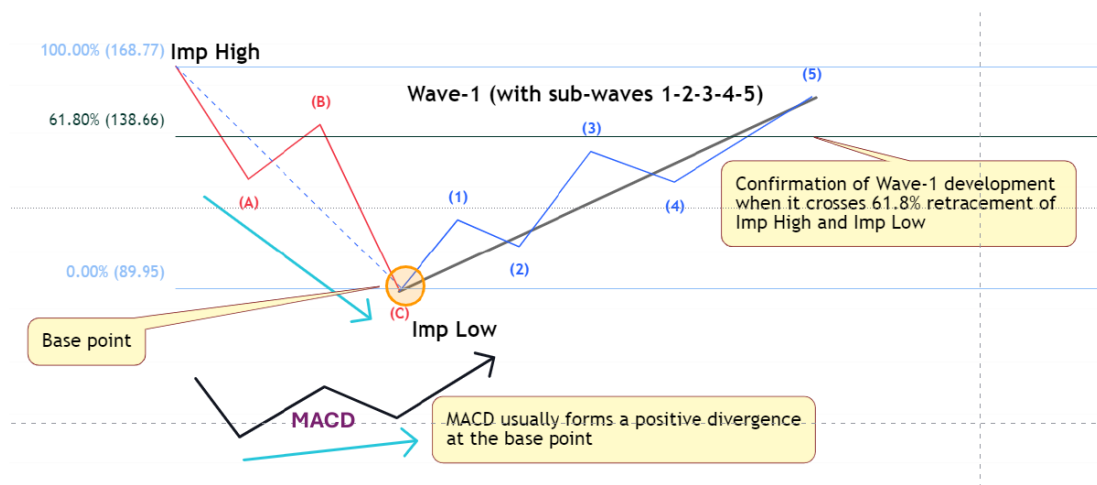
D2. Base Search Workflow

- Choose the analysis/degree timeframe preset.
- Build Important High and Important Low for the approved lookback. Chartking says max 144 days for the Daily start search; Hardik also requires significant high/low context.
- Find candidate Point 0 near a MACD lowest point and/or positive divergence according to the selected oscillator mode.
- Store both bullish and bearish base candidates until structure chooses one; do not assume direction too early.
- Do not CONFIRM Wave 1 merely because Point 0 exists. Point 0 is only the origin candidate.

Source basis: Chartking EW Notes pp.3-6; Hardik pp.8-10; Developer Specification pp.3-5.

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Development of Wave-1



D3. 61.8% Degree Confirmation

The source uses the move from Important Low toward Important High (or the bearish mirror) to determine whether the developing move has reached degree scale. The developer must separate “five internal moves completed” from “degree Wave 1 confirmed”. If five moves finish before the degree threshold, the notes may treat them as subordinate rather than the full degree wave.

```
if fiveInternalMovesCompleted AND degreeProgress < selectedDegreeThreshold:  
    classify structure as SUBORDINATE candidate  
    do not advance parent degree to W1 confirmed  
  
if degreeProgress >= selectedDegreeThreshold AND internalStructureValid:  
    W1 can move from FORMING -> CONFIRMED
```

D4. 144 Days Is Not 144 Bars

```
// WRONG on an intraday chart  
importantHigh = ta.highest(high, 144)  
  
// CORRECT concept  
importantHigh = request.security(symbol, approvedDegreeTf, highest(high, barsEquivalentToSourceWindow))  
// or use a timestamp-based 144-day window if the source says calendar days.
```

IMPLEMENTATION WARNING

The current script input “Important H/L lookback bars = 144” changes meaning with chart timeframe. This must be fixed before the wave-count logic can be trusted across timeframes.

E. Wave 1 - Detailed Theory-to-Code Translation

E1. What Wave 1 Means

Wave 1 is not simply “the first higher high after a low.” It is the first directional degree candidate after the validated base. The source describes it as impulsive with five subordinate movements, with MACD base evidence and degree confirmation from the Important H/L range.

Requirement	What the developer must store/check
Origin	Locked base Point 0, price/time/degree.
Internal movement	At least a valid five-move impulse candidate, unless a specifically allowed diagonal subtype is being parsed.
Degree progress	Progress from Important Low/High to the source 61.8% degree threshold.
MACD support	Lowest low or positive divergence at the base according to selected oscillator configuration.
Channel	After the source threshold, channel/trendline module can activate.
Status	FORMING before all requirements; CONFIRMED only after source conditions.

E2. Bullish Wave 1 Pseudocode

```
evaluateBullW1(base, rawPivots):
    candidate = buildFiveMoveImpulseFrom(base, rawPivots)
    candidate.degreeProgress = (candidate.endPrice - importantLow) / (importantHigh - importantLow)

    if candidate.breaksBase:
        return INVALID("W1_BASE_BREAK")

    if not candidate.internalFiveMovePossible:
        return FORMING or INVALID depending on structure

    if candidate.degreeProgress >= selectedW1DegreeThreshold:
        return CONFIRMED

    return FORMING("SUBORDINATE_OR_DEGREE_NOT_REACHED")
```

E3. Wave 1 Time - Source Variants

The Hardik document gives an additional time method: measure bars from significant high to significant low, divide by 50 for a maximum Wave-1 time reference, and also describes a 50% price recovery within that time as sufficient when five moves exist. The separate Time Analysis document instead describes 61.8% recovery in about half the previous decline time or 100% recovery in about the same time. These are not identical rules.

Oct 02, 2025 15:36 UTC+5:30

3.31 -0.38 (-1.32%)
10.50 0

Time Analysis

➤ Wave 1

- Time of wave 1 can be measured by putting date and time measurement tool from significant high to significant low.
- Now divide no. of bars by 50.
- And then again put tool from base point to the no. of bars which we have derived from dividing with 50.
- This is maximum time taken by wave 1 to complete. Remember it can also take less time than max time.
- If price reaches fib retracement till 50% in a given time period then also we can consider as wave 1 (only if we get 5 moves)



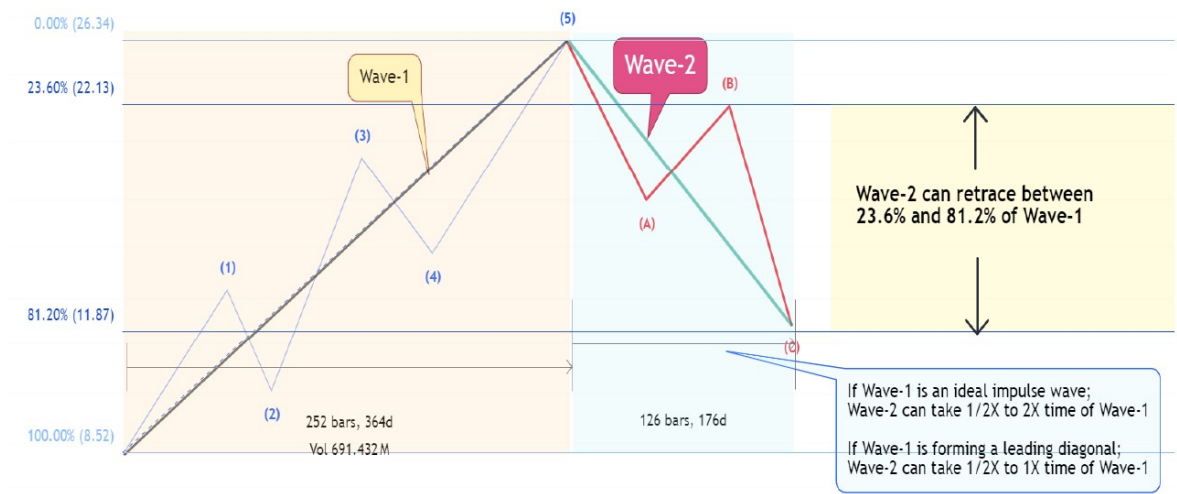
Source visual: Hardik Wave-1 time method using significant High-Low measurement (Hardik p.14).

SOURCE CONFLICT - DO NOT GUESS

Wave-1 time has multiple source methodologies. Implement them as separate Time Rule presets (e.g., CHARTKING/HARDIK/TIME_ANALYSIS) rather than averaging them into a new rule.

F. Wave 2 - Detailed Correction and Trade Logic

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Source visual: Wave-2 retracement zone and time context (Chartking EW Notes p.8).

F1. Why the Current 14% Rule Is Dangerous

One source table says normal W2 minimum 23.6% with a separate microscopic 14% clue. Other supplied impulse/Hardik notes describe the minimum as 14% or 14%-23.6%. Therefore 14% cannot be silently treated as a single universal meaning.

Source reading	14% meaning	23.6% meaning	Implementation
Chartking primary table	Microscopic special case that often implies W3 extension	Normal W2 minimum	Recommended "Chartking" mode.
Hardik degree notes	Minimum can begin 14%-23.6%	Also an important minimum zone	Alternative source preset.
Impulse Notes	W2 minimum stated as 14%	Included as part of Fib ladder	Alternative legacy preset.

SOURCE CONFLICT - DO NOT GUESS

Add W2MinimumMode. Do not "fix" the source by choosing one rule invisibly. The dashboard/debug log must show whether the active W2 candidate is NORMAL, MICROSCOPIC, or LEGACY_14_MIN.

F2. W2 Structural Sequence

After W1 confirmation:
open W2 correction container
allowed child candidates = ZigZag, Flat, WXY, approved single-leg case

A raw low is NOT W2 simply because price retraced.
W2 becomes CONFIRMED only after one child correction candidate completes
and Point 0/base is still structurally valid.

F3. W2 Range and Invalidation

Event	State effect
Retracement below first source threshold	W2 FORMING / microscopic possibility, depending mode.
Retracement inside selected normal range	Valid W2 price context, but not yet confirmed without internal completion.
Retracement 61.8%-81.2%	Deep W2 context; project HP zone becomes eligible.
Retracement beyond 81.2%	Flag according to selected source mode; keep base-origin hard rule separate.
Price crosses Point 0	Hard invalidation of the proposed impulse; reject W1-W2 interpretation and recount.

F4. W2 Time - Preserve the Contradictions

Source	W2 time guidance
Chartking table	One section says W2 cannot take more time than W1; another graphic permits broader relationships.
Hardik	W1/4, W1/2, W1, or 2*W1.
Separate Time Analysis	W2 should be W1/4 or W1/2; if W2 time equals W1, it may actually be Wave B.
Impulse Notes	W2 time described as same as W1 or double W1.

SOURCE CONFLICT - DO NOT GUESS

This is a major conflict. Do not make time a hard invalidation until the owner selects a time preset. Recommended software behavior: calculate all source variants, show which pass, and keep W2 FORMING/TBD if time is the only unresolved item.

F5. HP BUY / HP SELL - Separate the Trade Signal from the Elliott Count

PROJECT-SPECIFIC REQUIREMENT

HP BUY/SELL is a project trading signal built on a valid W2 interpretation. It must never change the Elliott count just because a trade filter passes or fails.

```

HP_BUY = w1_CONFIRMED
        AND w2_candidate.status in {FORMING_NEAR_COMPLETION, CONFIRMED}
        AND w2Retrace in 0.618..0.812
        AND Point0Valid
        AND correctionCompletionEvidence
        AND ownerSelectedTradeFilters

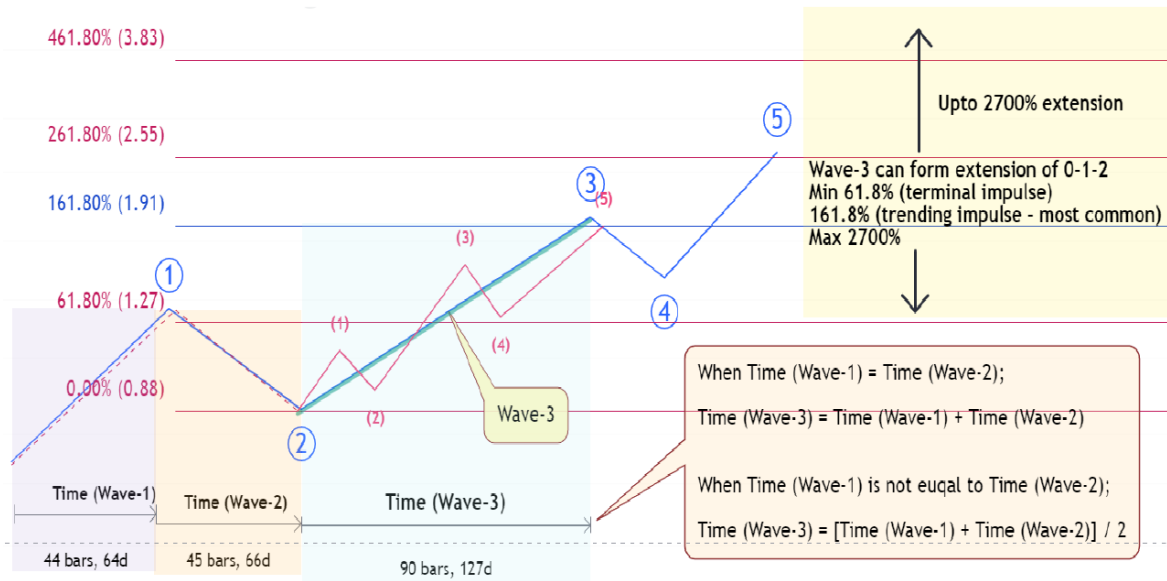
```

```

HP_SELL = mirror

```

G. Wave 3 - Trending, Terminal, and Extended

[Index](#)


Source visual: Wave-3 0-1-2 extension and time logic (Chartking EW Notes p.11).

G1. W3 Must Never Be Determined Only by the Next High

Wave 3 needs an internal impulsive structure and a subtype classification. The source distinguishes a normal/trending degree Wave 3 (commonly at or beyond 161.8%) from terminal/subordinate behavior below 161.8%, and also allows extensions with 9/13/17/21 moves.

G2. W3 Projection

```
bullW3Ratio = (P3 - P2) / abs(P1 - P0)
bearW3Ratio = (P2 - P3) / abs(P0 - P1)

if ratio >= 1.618:
    TRENDING_W3 candidate
elif ratio >= terminalMin AND ratio < 1.618 AND terminalStructureValid:
    TERMINAL_W3 candidate
else:
    do not confirm degree W3
```

G3. W3 Is Not the Shortest

The "Wave 3 is never the shortest among 1,3,5" rule cannot be fully evaluated until Wave 5 exists. Therefore store it as a deferred hard rule. When W5 candidate forms, test W1, W3 and W5 lengths and reject the complete impulse if W3 becomes the shortest under the selected source interpretation.

G4. MACD and Gap Evidence

Evidence	Use
MACD top/highest	Support/rank the W3 candidate. Do not create W3 without price structure.
Breakaway gap above W1 high in bullish case	Source clue for W3; gap persistence rule is market/session dependent.
Price passes source extension but MACD does not show expected W3 behavior	Keep alternate correction/terminal interpretation alive rather than force W3.

G5. W3 Maximum Conflict

SOURCE CONFLICT - DO NOT GUESS

Chartking normalized specification states degree W3 can extend up to 2700%, while Impulse Notes show 2100%. Store the active maximum as a source-mode parameter. Never silently replace one with the other.

G6. W3 Time Conflict

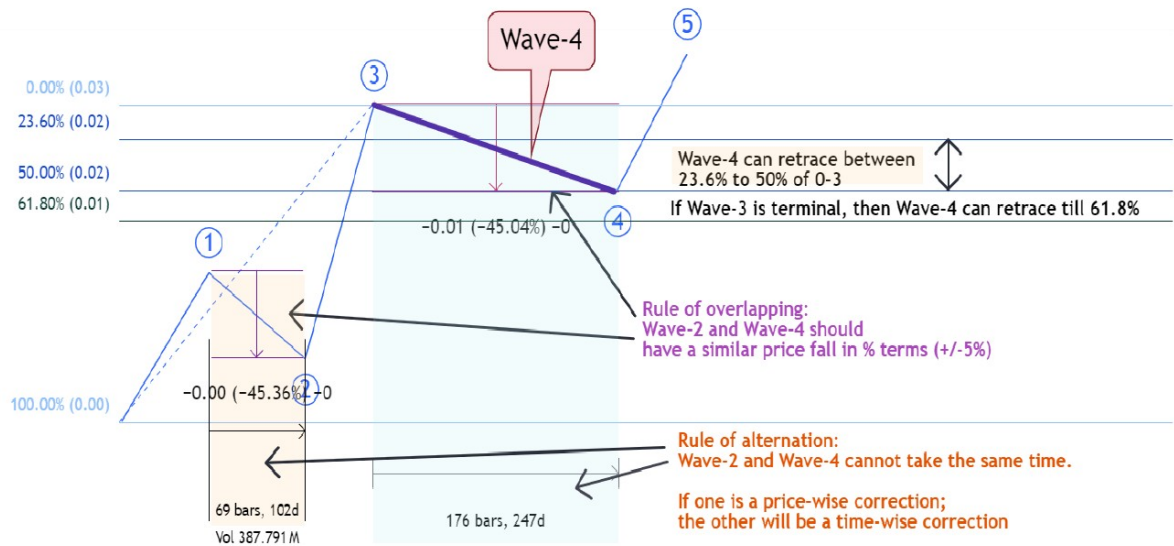
Source	Time rule
Chartking	If $W1 \neq W2$, $W3 = (W1 + W2) / 2$; if approximately equal, $W3 = W1 + W2$.
Hardik	W3 minimum time described as $W1 + W2$.
Separate Time Analysis	W3 time = $W1 + W2$; also combines price equality / 127% W2 retracement confirmation.

SOURCE CONFLICT - DO NOT GUESS

Keep time-rule preset separate from price subtype. A valid trending W3 price structure should not be relabeled solely because an unresolved time source disagrees.

H. Wave 4 - The Main Source of Premature Wave 5 Errors

Index



Source visual: Wave-4 Base-to-W3 retracement, alternation and similarity (Chartking EW Notes p.13).

H1. W4 Is a Correction Container, Not One Leg

Wave 4 is where complex corrections are common. The engine should open multiple child correction candidates: Zig-Zag, Flat, Double, Triple and Triangle. Until one completes, the parent state is W4_FORMING. This is why the current forced A-B-C sequence is not enough.

H2. W4 Price Rules

Source mode	Normal W4 minimum	Normal max	Special/terminal
Chartking primary	23.6%	50%	May allow 61.8% in terminal context.
Hardik / Impulse Notes	14% appears as minimum in some tables	50%	Rare 61.8%.
W3 extension context	Context-specific	Often 38.2%, rare 50% in some notes	Complex correction expected.

SOURCE CONFLICT - DO NOT GUESS

As with W2, preserve a W4MinimumMode. Do not use one 14% constant everywhere.

H3. W4 Overlap Depends on Impulse Subtype

```
if impulseSubtype == TRENDING_NORMAL:
    apply normal overlap prohibition
elif impulseSubtype in {LEADING_DIAGONAL, ENDING_DIAGONAL, TERMINAL}:
    apply subtype-specific overlap rules
else:
    delay hard overlap decision until subtype can be evaluated
```

H4. Rule of Alternation and Similarity

The notes repeatedly use Wave 2 vs Wave 4 comparison. Price fall percentages may be similar within about +/-5% in the Chartking source, while time/structure should differ. This should be an explicit comparison object in code, not a visual-only note.

```
w2FallPct = abs(P1-P2) / referenceForW2
w4FallPct = abs(P3-P4) / referenceForW4
similarityDiff = abs(w2FallPct - w4FallPct)

priceSimilarityPass = similarityDiff <= ownerTolerance
timeAlternationPass = not timeApproximatelyEqual(w2time, w4time)
structureAlternation = type(W2) != type(W4) where source expects it
```

H5. W4 Time Conflicts

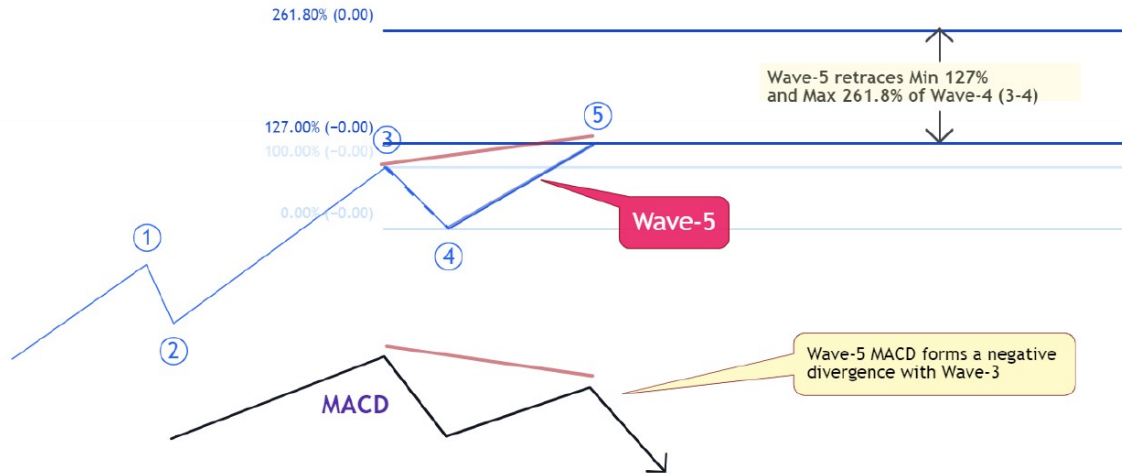
Source	W4 time options
Chartking summary	1/2, 2x or 3x W2; not same as W2.
Hardik	W2/2, 2x W2 or 5x W2.
Separate Time Analysis	W2/2; (W2+W3)/4; (W2+W3)/2; W2+W3.
W3 extension notes	W4 may be 5-10x / 10x W2 when W1/W2 microscopic.

SOURCE CONFLICT - DO NOT GUESS

Do not compress these into one arbitrary range. Calculate a list of expected windows per active source/subtype and show the nearest matching window in debug.

I. Wave 5 - Completion, Divergence, Truncation and Extension

[Index](#)



Source visual: Wave-5 127%-261.8% 3-4 projection and W3/W5 divergence (Charting EW Notes p.15).

I1. Normal W5

```
normalW5PricePass = 1.27 <= W5Ratio <= 2.618
internalPass = fiveMoveImpulseCandidate
w3ShortestPass = not makesWave3Shortest(W1,W3,W5)
divergenceSupport = compareDegreeW3W5MACD()

if hard rules pass and required completion conditions pass:
    W5_CONFIRMED
```

I2. Do Not Use One W5 Branch

Subtype	When to evaluate	Key distinction
Normal W5	Normal/trending impulse context	127%-261.8% 3-4 projection.
Truncated W5	Source double-extension context	Can terminate below normal 127 rule; source shows max around 81.2% of 3-4 while retaining five submoves.
W5 Extension / Megaphone	Specific source context, instrument restrictions/conflicts	Can extend toward source maximum; separate from normal W5.
Ending Diagonal W5	Terminal/late impulse context	3-3-3-3-3 and overlap/slow/divergence rules.

I3. W5 Time Is Also Source-Dependent

Hardik says W5 usually takes W1 time and rarely W4 time. The separate Time Analysis document provides a wider set: W1, (W1+W3)/4, (W1+W3)/2, W1+W3 and an additional (W3+W4)/2 case. Keep these as selectable/diagnostic windows.

J. Trending vs Terminal Impulse, Leading Diagonal and Ending Diagonal

Leading Diagonal (5-3-5-3-5)

[Index](#)

Nature	• Terminal impulse	Feature-5	• Wave-5 is Min 161.8% and Max 261.8% , Wave-5 and Wave-1 may be equal
Sub-moves	• Sub-waves (1,3,5) should have 5 moves 1-2-3-4-5, all are slow	Time : W2	• Subordinate W2 = ¼ or ½ Subordinate W1
Occurrence	• Within Wave-1 and Wave-A (Zig-Zag correction)	Time:W3	• If W1 not equal to W2; then W3 = [W1 + W2]/2 • If W1 = W2 [+/- 5]; then W3 = W1 + W2
Retracement	• Wave-4 can retrace Min 23.2% and Max till 61.8% of 0-3	Next pattern	• Wave-2 (degree) after the entire leading diagonal should take ¼X, ½X or X
Feature-1	• Wave-4 may enter Wave-1's territory and will not go below end of Wave-2	Wave-3	• Wave-3 (degree) after Wave-1 is an LD, would mostly be extended
Feature-2	• Wave-3 Fib extension will be terminal Min 61.8% and Max <161.8%	Special feature	• Fall of Wave-2 (degree) after completion of the leading diagonal can go till sub-Wave-4, sub-Wave-2 or Max till 81.2%
Feature-4	• Wave-3 will be terminal and is slow in nature		

Source visual: Leading Diagonal rules and overlap behavior (Chartking EW Notes p.18).

Ending Diagonal (3-3-3-3-3)

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Nature	• Terminal impulse	Feature-4	• Wave-3 is slow in speed
Sub-moves	• 3-3-3-3-3 (Triple Zig-Zag)	Feature-5	• Wave-5 is Min 127% and Max 261.8%, Wave-5 and Wave-1 may be equal in length
Occurrence	• Wave-5 and Wave-C (If Wave-3 is in extension, Wave-4 is a sharp move, then Wave-5 will be an ED)	MACD	• MACD forms negative divergence from Wave-3 (degree)
Retracement	• Wave-4 can retrace Min 23.2% and Max till 61.8% of 0-3	Time : W2	• Subordinate W2 = ¼ or ½ Subordinate W1
Feature-1	• Wave-4 enters Wave-1's territory and will not break Wave-2 low	Time:W3	• If W1 not equal to W2; then W3 = [W1 + W2]/2 • If W1 = W2 [+/- 5]; then W3 = W1 + W2
Feature-2	• Wave-3 Fib extension may be terminal Min 61.8% and Max <161.8%	Pattern End	• Correction after the entire ending diagonal should take 1/4 th or ½ its time
Feature-3	• Wave-2 and 4 won't retrace more than 50%	Criteria	• LD and ED will not form in the same impulse move (If W1 = LD, W5 = ED)

Source visual: Ending Diagonal rules and W5/C context (Chartking EW Notes p.22).

J1. Trending vs Terminal Gate

Feature	Trending / normal	Terminal / diagonal family
W3 extension	Normally $\geq 161.8\%$ in the source main degree rule	May be $\geq 61.8\%$ but below 161.8% under terminal rules.
W4 overlap	Normal overlap prohibition	May enter W1 territory according to diagonal/terminal subtype rules.

Feature	Trending / normal	Terminal / diagonal family
W3 speed / channel	Usually stronger trending behavior	Often slower / repeated trendline interaction.
W5	Normal impulsive or extension	Can be diagonal depending on position/subtype.

J2. Leading Diagonal Parser

LD allowed only in source-approved positions (W1 and Wave A in the Chartking spec).

Expected high-level structure: 5-3-5-3-5.

Evaluate:

- W3 terminal range
- W4 overlap allowed but cannot violate LD-specific W2 boundary
- W4 retracement source range
- W5 source projection
- wedge / trendline behavior
- subordinate W2 time

Do not identify LD just because two trendlines converge.

J3. Ending Diagonal Parser

ED allowed only in W5 or Wave C.

Expected structure: 3-3-3-3-3.

Evaluate:

- valid position
- W2/W4 retracement limits
- overlap behavior
- W5 projection
- W3-vs-W5 MACD divergence
- post-pattern correction timing

ED_CONFIRMED only after all hard subtype conditions are met.

J4. LD/ED Coexistence Conflict

SOURCE CONFLICT - DO NOT GUESS

The source wording about whether Leading and Ending Diagonals can occur in the same impulse is internally ambiguous. Add a configuration or keep this criterion non-blocking/TBD until the owner selects the intended rule.

K. Extension Engine - 5, 9, 13, 17, 21 Moves

K1. Why “More Pivots” Does Not Mean “Next Main Wave”

An extended Wave 3 can contain 9, 13, 17 or 21 moves while the main degree is still Wave 3. This is exactly why a main count cannot be tied to consecutive raw pivots.

Move count	Meaning for engine
5	Normal internal impulse candidate.
9	First extension count in +4 sequence.
13	Further extension; still same parent wave.
17	Further extension; still same parent wave.
21	Maximum extension count described in source.

Source basis: *Chartking Developer Specification p.11; Hardik pp.20-27; Impulse Notes pp.19-21.*

K2. Generic Extension Rules

```
while parentWave.status == FORMING_EXTENDED:
    count internal confirmed moves
    require source minimum internal swing retracement/projection
    keep extension base fixed until degree-fall rule triggers
    if internalCount in {5,9,13,17,21}:
        evaluate completion candidate
    if degreeFallDetected:
        close parent extension / promote next-degree correction candidate
```

K3. Microscopic Extension

Hardik says microscopic W1/W2 may be hard to see, W1+W2 candle count can be small (e.g. <=13 in that note), W3 then tends to extend, and W4 may stay shallow. Treat “microscopic” as a subtype with configurable quantitative thresholds because the source uses qualitative wording in other places.

K4. Megaphone / W5 Extension

Some supplied notes describe a W5 extension / megaphone condition when W1 and W3 are similar. The Developer Specification also carries instrument restrictions (source says not indices, only stocks) and equality wording conflicts. This must be its own subtype, not folded into normal W5.

SOURCE CONFLICT - DO NOT GUESS
W5 extension applicability and W1/W3 equality criteria vary by source. Keep instrument gating and equality tolerance configurable.

L. Correction Classifier - The Most Important Refactor After P0

L1. Parallel Candidate Model

```
onCorrectionStart(parentWave):
    candidates = []
    if positionAllowsZigZag: candidates += ZIGZAG
    if positionAllowsFlat:   candidates += FLAT
    if positionAllowsDouble: candidates += DOUBLE
    if positionAllowsTriple: candidates += TRIPLE
    if positionAllowsTriangle: candidates += TRIANGLE

onSubPivot(p):
    for c in candidates:
        c.evaluate(p)
        if c.hardInvalid: remove(c)

    rankRemainingCandidates()
    if one candidate fully confirmed: promote it
    else show PRIMARY + ALTERNATE in debug
```

IMPLEMENTATION WARNING

Do not ask the user to choose "A-B-C / W-X-Y / W-X-Y-X-Z / A-B-C-D-E" so the script knows what to label. That dropdown can remain only as a display filter or manual debug override.

M. Zig-Zag - Exact Coding Checklist

6. Correction classifier - simple correction

Correction recognition must be candidate-based because the source explicitly calls corrections guideline based. The same early A-B path may later resolve to Zig-Zag, Flat, Double, Triple or Triangle.

6.1 Zig-Zag Correction (A-B-C)

Field	Source requirement
Type / occurrence	Single, price-wise correction; mostly Wave-2, sometimes Wave-4 and inside Double/Triple.
Count	3 moves A-B-C; internal 5-3-5.
Speed	A fast, B slow, C fast.
Wave-A	Min 61.8% retracement of previous degree Wave-5.
Wave-B time	$B \geq A$, up to 5X A.
Wave-B price	Less than 61.8%; source says if >61.8% it is Wave-2.
Wave-C projection	$C = A$ length-wise; min 61.8%, source table max 261.8% extension of 0-A-B.
Wave-C retracement	Min 127% of Wave-B.
C time	If $A=B$, $C=A+B$; if $A < B$, $C=(A+B)/2$.
Entry	On crossing the 3rd Fib ratio in Wave-C.
SL	Previous Fib level of entry.
Targets / reversal support	Bullish candle at 61.8%, 100%, 161.8%, 261.8%; 0-B trendline break in half C time; point B crossed in same C time.
Pattern end	Complete correction minimum 38.2% of entire impulse.
C types	Normal 61.8%-161.8%; Elongated source also shows up to 461.8%; Truncated reverses before 61.8% and uses an 81.2% confirmation.

Zig-Zag Correction (A-B-C)

[Index](#)

Correction Type	•Single correction	Entry	• On crossing 3 rd Fib Ratio in Wave-C	
Occurrence	•Mostly found in Wave-2/ sometimes in Wave-4 and in double or triple correction	SL	• Previous Fib Level of entry	
Category	• Price-wise correction	Target	• Bullish Candle formation at Fib levels 61.8%, 100%, 161.8%, 261.8%	
No. of moves	• 3 (A-B-C)	Pattern confirmation-1	• When price crosses 0-B trendline in half the time (Wave-C)	
Internal count	• 5-3-5	Pattern confirmation-2	• When price crosses point B at the same time (Wave-C)	
Speed	• A Fast, B Slow, C Fast	Pattern End	• Complete correction should retrace Min 38.2% of the entire Impulse move	
Rule for Wave-A	• Min 61.8% retracement of previous degree Wave-5	Types of C : Trend based Fib extension on 0-A-B	Normal C	61.8% to 161.8%
Wave-B Time period	• B ≥ A (up to 5 times of A)		Elongated C	161.8% to 461.8%
Fib Ext Wave-C (0-A-B)	• C = A (length wise), Min 61.8%, Max 261.8% Fib extension (0-A-B)		Truncated C	1. Reverses before 61.8% 2. Confirmation: 81.2% retracement of A-B-C (0-C)
Wave-C retracement	• Wave-C should be Fib retracement 127% of Wave-B			
Wave-B retracement	• Less than 61.8%, if ≥61.8% then it is Wave-2			
Time taken for Wave-C : 1	• If A = B, C = A + B			
Time taken for Wave-C : 2	• If A < B, C = (A + B) / 2			

Source snapshot - page 37: complete Zig-Zag rule table.

Rule	Code check	State effect
Structure	A-B-C; internal 5-3-5.	Hard structural requirement for confirmed Zig-Zag.
A	Source min 61.8% of previous-degree W5 in this pattern context.	Below threshold -> remain FORMING/alternate according to source.
B price	Below Flat-style 61.8%; detailed summary gives 1%-50%.	If B enters Flat range, Flat candidate becomes stronger.
B time	B >= A; source up to 5A for Zig-Zag.	Time support / conflict mode.
C projection	0-A-B extension; normal/elongated/truncated subtypes.	Determines C subtype and target zone.
C vs B	Source min 127% retracement of B.	Required where active source says mandatory.
Trendline	0-B break in source timing.	Completion/reversal support.
Whole correction	Min 38.2% of entire impulse in source table.	Pattern-end check.

N. Flat - Exact Coding Checklist

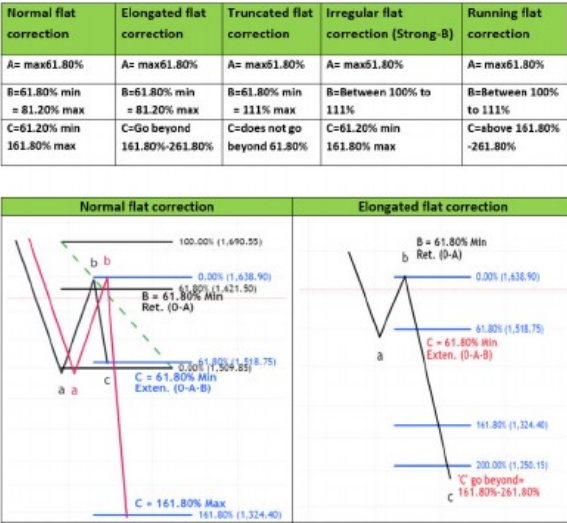
Flat Correction (A-B-C)

Index

Correction Type	• Single correction	Point of Entry	• Start of Wave-X or Wave-1	
Occurrence	• Mostly found in Wave-2	Confirmation:1	• Bullish Candle formation at Fib levels 61.8%, 100%, 161.8%, 261.8%	
Category	• Price-wise correction	Confirmation:2	• On crossing 3 rd Fib Ratio	
No. of moves	• 3 (A-B-C)	SL	• 61.8%, 100%, 161.8%, 261.8%	
Internal count	• 3-3-5	Exit (time period)	• Half the time of Wave-C	
Speed	• A Slow, B Fast, C Fast	Reversal criteria	• When price crosses O-B trendline in half the time (Wave-C)	
Rule for Wave-A	• Min 38.2% retracement of previous degree Wave-5	Pattern confirmation	• When price crosses point B at the same time (Wave-C)	
Rule for Wave-B	• B >= A (up to 2 times of A)	Pattern End	• Complete correction should retrace Min 38.2% of the entire impulse move	
Wave-B retracement	• Min 61.8% , Max 111%, (SL 127%)			
Time taken for Wave-C : 1	• If time (A) = time (B), time (C) = time (A) + time (B)	Types of C : Trend based Fib extension on O-A-B	Normal C (B : Min 61.8%, Max 81.2%)	61.8% to 161.8%
Time taken for Wave-C : 2	• If time (A) < time (B), time (C) = [time (A) + time (B)]/2		Elongated C (B: Min 61.8%, Max 81.2%)	161.8% to 261.8%
			Strong Irregular (B: Min 61.8%, Max 111%)	61.8% to 161.8%
			Running (B: Min 61.8%, Max 111%)	161.8% to 261.8%

Source snapshot - page 43: Flat correction rule table.

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Source snapshot - page 47: Normal, Elongated, Truncated, Irregular and Running Flat comparison.

```
Flat candidate gates:
  structure = 3-3-5
  A >= selected source minimum (38.2% in Chartking Flat table)
  B retracement in 61.8%-111%
  classify C subtype from B range + C projection
  evaluate time and 0-B reversal rules
```

Subtype	B	C	Developer note
Normal Flat	61.8%-81.2%	61.8%-161.8%	Normal C candidate.
Elongated Flat	61.8%-81.2%	161.8%-261.8% in one table	Do not silently extend to 461.8 without owner rule.
Strong Irregular	61.8%-111%	61.8%-161.8%	B can exceed A while remaining below source max.
Running Flat	61.8%-111%	161.8%-261.8% in summary	Preserve structural requirements, not only ratios.

O. Wave B - How to Prevent the Old Premature Wave 5 Bug

The old bug happened because the engine expected a fixed sequence. The correct fix is not merely “allow B to 111%”. The correct fix is to keep correction candidates alive until their structure resolves.

```
After a candidate A:
  if B < 0.618 of A:
    keep ZigZag candidate (if internal rules allow)
  if 0.618 <= B <= 1.11:
    keep Flat candidate
  if correction is already complex:
    keep Double/Triple/Triangle candidates according to position

Only after W4 correction completion can the next parent impulse leg be evaluated as W5.
Never infer W5 from “next pivot number”.
```

P. Wave C - Completion Engine

Wave C is not simply the third pivot in A-B-C. It is an impulsive child wave (or ED where allowed), has source-specific Fib/time/MACD behavior, and its completion is important for both correction confirmation and project signals.

```

evaluateC(corr):
  build C internal impulse / ED candidate
  cExt = extension012(originA, A, B, currentC)
  cVsB = abs(C-B) / abs(B-A)
  cTime = duration(B,C)

  evaluate normal / elongated / truncated subtype
  evaluate positive divergence support
  evaluate 0-B trendline / point-B timing support

  if completion conditions pass:
    corr.C = CONFIRMED at actual completion pivot
  else:
    keep C FORMING

```

PROJECT-SPECIFIC REQUIREMENT

C (and later Z) should be marked on the actual completion pivot/candle when the pattern completes. Do not shift the label to a later unrelated swing just to make the count visually neat.

Q. Double Correction W-X-Y

8. Double Correction (W-X-Y)

Field	Source requirement
Type / occurrence	Complex, time-wise correction; mostly Wave-4.
Internal structure	W and X: Zig-Zag or Flat; Y: Zig-Zag, Flat or Triangle.
Move count	Source summary gives max 39 with W(5-3-5), X up to 11, Y(3-3-3-3-3); later examples provide specific totals by combination.
Wave-X small	Max 50% retracement of W.
Wave-X large	Min 61.8%, max 111% retracement of W.
Wave-Y fib	0-W-X extension: min 61.8%, max 161.8%.
Wave-Y time	$\text{Time}(Y) = \text{Time}(W)$ OR $\text{Time}(W+X)$ OR $\text{Time}(W+X)/2$.
Wave-X time	$\text{Time}(X) < \text{Time}(W)$.
Entry evidence	Positive MACD divergence at Y.
Confirmation	Source cell is literally "???" - must remain TBD.
Entry	Close above 38.2% retracement of 0-Y.
SL	Trailing SL of 2 Fib levels; if price crosses 81.2%, source says SL at 50%.
Reversal	0-X TL crossed in half $\text{Time}(Y)$; Point B crossed in less than $\text{Time}(Y)$.

Double Correction (W-X-Y)

Index

Correction Type	• Complex correction	Time taken for Wave-Y	• $\text{Time}(Y) = \text{Time}(W)$ OR $\text{Time}(W+X)$ OR $\text{Time}(W+X)/2$
Occurrence	• Mostly found in Wave-4	Time taken for Wave-X	• $\text{Time}(X) < \text{Time}(W)$
Category	• Time-wise correction	Point of Entry	• Positive divergence in MACD at Y
No. of moves	• Total Max 39 -> W(5-3-5), X(11), Y(3-3-3-3-3)	Confirmation	• ???
Internal structure	• W & X: Zig-Zag or Flat ; Y: Zig-Zag, Flat or Triangle	Entry	• Close above 38.2% retracement of 0-Y
Rule for Wave-W	• If 1 st Wave takes 1/2, 1/3 or 1/4 Time of 2 nd Wave, then 1 st Wave is "W" and 2 nd Wave is "X"	SL	• Trailing SL of 2 Fib levels (SL will be at 50% if price crosses 81.2% Fib level)
Rule for Wave-X (small)	• Max 50% retracement of Wave-W	Exit (time period)	
Rule for Wave-X (Large)	• Min 61.8% and Max 111% retracement of Wave-W	Reversal criteria:1	• When price crosses 0-X TL in 1/2 Time (Wave-Y)
Rule for Wave-Y	• 0-W-X Extension: Min 61.8%, Max 161.8%	Reversal criteria:2	• When price crosses Pt. B in < Time (Wave-Y)

Source snapshot - page 56: Double Correction rule table including unresolved confirmation field.

Q1. Structure

W = simple correction candidate (ZigZag or Flat)
X = connector correction
Y = second correction candidate (ZigZag, Flat or Triangle)

Do not represent W, X and Y as three arbitrary raw pivots. Each component is itself a structure/container.

Q2. X and Y Rules

Component	Source rule	Coding action
Small X	Max 50% of W.	Classify X_SMALL.
Large X	61.8%-111% of W.	Classify X_LARGE.
X time	$X < W$.	Store pass/fail and selected tolerance.
Y Fib	0-W-X extension 61.8%-161.8%.	Project from X using W length.
Y time	W, W+X, or $(W+X)/2$.	Generate expected windows, not one continuous max.
Y MACD	Positive divergence at end in source.	Supporting completion evidence.
Confirmation	Source cell contains "???".	TBD-BLOCKED; do not invent textbook confirmation.

R. Triple Correction W-X-Y-XX-Z

10. Triple Correction (W-X-Y-XX-Z)

Field	Source requirement
Type / occurrence	Complex, time-wise; mostly Wave-4.
Initial structure	W & X: Zig-Zag or Flat; Y: Zig-Zag, Flat or Triangle.
Small X	Max 50% of W (another page: min 14.2%-max50%).
Large X	61.8%-111% of W.
Y	0-W-X extension min61.8%-max161.8%.
XX	50%-61.8% retracement of Y; cannot go above Point X.
Z	61.8%-161.8% extension of 0-Y-XX; source says failure may reverse from 31.8%.
Time X	$X < W$; another diagram says $X < 1/2 W$.
Time Y	$Y = W$ OR $W+X$ OR $(W+X)/2$.
Time XX	$XX = X$ and $< Y$.
Time Z	$Z = Y$ OR $Y+XX$ OR $(Y+XX)/2$.
Point of Entry	Start of Wave-Y in the source summary.
Blank source fields	Rule for W, Rule for XX/Z in summary, Confirmation, Entry, SL and Exit cells are blank. Use detailed later pages only where they explicitly provide values.

Triple Correction (W-X-Y-XX-Z)

Index

Correction Type	• Complex correction	Time taken for Wave-Y	•Time (Y) = Time (W) OR Time (W+X) OR Time (W+X)/2
Occurrence	• Mostly found in Wave-4	Time taken for Wave-X	•Time (X) < Time (W)
Category	• Time-wise correction	Point of Entry	•Start of Wave-Y
No. of moves	• Total Max 37 -> W(5-3-5), X(11), Y(5-3-5)	Confirmation	
Internal structure	• W & X: Zig-Zag or Flat ; Y: Zig-Zag, Flat or Triangle	Entry	
Rule for Wave-W		SL	
Rule for Wave-X (small)	• Max 50% retracement of Wave-W	Exit (time period)	
Rule for Wave-X (Large)	• Min 61.8% and Max 111% retracement of Wave-W	Reversal criteria:1	•When price crosses 0-X TL in 1/2 Time (Wave-Y)
Rule for Wave-Y	• 0-W-X Extension: Min 61.8%, Max 161.8%	Reversal criteria:2	•When price crosses Pt. B in < Time (Wave-Y)
Rule for Wave-XX			
Rule for Wave-Z			

Source snapshot - page 73: Triple Correction summary with blank fields.

R1. Exact Component Model

TRIPLE = W + X + Y + XX + Z

W = first correction
X = first connector
Y = second correction
XX = second connector (do NOT call it another generic X internally)
Z = final correction

Component	Source rule / range
Small X	Max 50% of W; another page gives min14.2%-max50%.
Large X	61.8%-111% of W.
Y	0-W-X extension 61.8%-161.8%.
XX	50%-61.8% retracement of Y; cannot go above Point X.
Z	0-Y-XX extension 61.8%-161.8%; failure percentage has source conflict.
Time X	X<W; another diagram says X<1/2 W.
Time Y	Y=W OR W+X OR (W+X)/2.
Time XX	XX=X and <Y.
Time Z	Z=Y OR Y+XX OR (Y+XX)/2.

IMPLEMENTATION WARNING

The current display option “W-X-Y-X-Z” should be changed internally to W-X-Y-XX-Z so the two connectors have distinct identities, histories and time/Fib rules.

S. Triangle Engine - Six Families, Not One ABCDE Label

11. Triangle Correction engine

11.1 Triangle families and common rules

Family	Types
Contracting	Horizontal Contracting, Irregular Contracting, Running Contracting.
Expanding	Horizontal Expanding, Irregular Expanding, Running Expanding.

Common contracting rule	Source requirement
Retracement condition	Three waves must retrace more than 50% of the preceding wave length.
Structure	A-B-C-D-E; each wave is corrective.
Trendlines	Always draw on candle chart; both trendlines max 5 touch points.
Wave-E guideline	Typically around 38.2% retracement.
Where triangles appear	Mostly Wave-4, B, E, X, Y, XX, Z.
Exclusion	Not found in Wave-2.

Types of Triangles

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Contracting Triangle:

- Horizontal Contracting Triangle.
- Irregular Contracting Triangle.
- Running Contracting Triangle

Expanding Triangle:

- Horizontal Expanding Triangle.
- Irregular Expanding Triangle.
- Running Expanding Triangle.

Source snapshot - page 87: six triangle types.

S1. Common Rules

Rule	Developer interpretation
Structure	A-B-C-D-E; every leg is corrective.
Location	Mostly W4, B, E, X, Y, XX, Z in the source; not W2.
Trendlines	Triangle boundaries must be built from the correct leg pivots; source limits touch-point behavior.
Retracement relation	Common contracting rule requires multiple legs to retrace >50% of preceding leg.
E guideline	Often near a source Fib guideline; exact subtype rules differ.
Thrust	Calculate only after confirmed triangle completion; subtype determines expected thrust.

S2. Horizontal Contracting

```

size(A) > size(B) > size(C) > size(D) > size(E)
trendlines converge / move in opposite price directions
A = widest
source thrust ~75%-125% of widest wave
apex relation must be evaluated where source mode requires it

```

S3. Irregular Contracting

The source permits B to be the widest segment, often below 161.8% of A and not above the larger maximum described in the notes. C, D and E then contract sequentially. The thrust is again tied to the widest wave in the common rule.

S4. Running Contracting

Running Contracting has different trendline slopes and size ordering. The supplied source also contains conflicting thrust statements (normal 75%-125% vs exceptional much larger thrust). Preserve this as a source conflict.

S5. Expanding Family

Expanding triangles reverse the size concept: A or B is among the smallest, E is usually the largest, and E often breaks the A-C boundary. The Hardik notes also give a 61.8% of E thrust guideline for expanding triangles. Do not apply the contracting thrust formula to expanding patterns automatically.

Source basis: *Developer Specification pp.24-27; Hardik triangle section; Chartking triangle pages.*

T. MACD Engine - Support the Structure, Never Replace It

Stage	Source clue	How code should use it
Base / W1	MACD lowest low or positive divergence.	Base score / start evidence.
W2	MACD falls slowly; mostly positive divergence; selected source may protect base MACD.	Support W2, not a standalone label.
W3	MACD highest / forms top.	Support trending W3.
W4	MACD may fall rapidly / become very low in price-wise correction.	Degree-fall / W4 support.
W5	MACD lower than W3; negative divergence.	Completion support / ED clue.
A	MACD lower than prior impulse W2/W4 in source correction rules.	Correction classification support.
B / X	MACD moves with speed.	Connector/B support.
C	Positive divergence can indicate simple correction completion.	C completion support.
Y	Positive divergence; lowest-low behavior may signal triple continuation in source.	Double vs Triple clue.
XX	MACD higher high.	Triple connector support.
Z	Positive divergence.	Triple completion support.

Source basis: *Charting MACD Rules - Impulse/Correction; Developer Specification pp.28-30.*

SOURCE CONFLICT - DO NOT GUESS

The source does not specify exact MACD fast/slow/signal parameters or exact divergence pivot algorithm. Keep these inputs configurable. The current code should also expose whether it is comparing MACD line, histogram, or both.

U. Gaps - Supporting Context Only

The supplied Impulse Notes and Developer Specification place different gap classes in specific wave contexts. Treat gaps as supporting evidence and be careful in 24-hour markets where "opening gap" behavior differs.

Gap	Source contexts	Implementation
Common	W1/W2/W4, Flat/Triangle/B/X contexts in supplied notes.	Small support tag; track fill window if applicable to instrument.
Breakaway	Primarily W3 after W1 breakout / news.	Support strong W3, do not force W3.
Runaway	W3, sometimes Zig-Zag A / X in supplied notes.	Trend-strength support.
Exhaustion	W5, C, Y, Z near maturity.	Completion/reversal support.

V. Trendline and Channel Engine

TRENDLINES

- There are three types of trendline namely:-
 1. BASE TRENDLINE-Drawn between 0 point to 2 point and extended.
 2. ACTION TRENDLINE-Base trendline is copy paste and then put to wave 1 is called action line.
 3. MATURITY TRENDLINE-This trendline is drawn between wave 2 and wave 4.

Hardik source page showing Base, Action and Maturity trendline definitions (p.39).

14. Trendline and channel engine

14.1 Impulse channel

Rule	Source requirement
Activation	When price crosses 61.8% of Important High-Low, draw channel to check W1 trend continuity.
Use	Impulse only.
Construction	Join base point with 2; ensure price does not fall outside the channel.
Exit meaning	After channel exit, price can fall 38.2%-81.2%, signifying end of trend.
Re-entry note	Entry after fall with reversal candle; source says price does not re-enter channel after exit.

Channel

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Whenever price crosses 61.8% of the Imp High – Low, draw a channel to check trend continuity of Wave-1

To be used only in Impulse waves. Price won't reverse till the time it does not exit the channel

Make it by joining the base point with 2 and make sure that price does not fall anywhere outside the channel

After exiting from the channel, price can fall till 38.2% to 81.2% levels and signifies end of trend

Entry can be done after the fall with a reversal candle pattern. Price does not enter the channel after exiting it

Source snapshot - page 108: Channel rules.

14.2 Trendline construction and projections

Stage	Source requirement
Base channel	Connect 0-2 and parallel through 1. No part of Wave-3 should break 0-2 TL.
Wave-4 projection	Connect 1-3 and parallel through 2; W4 can occur where line through 2 matches Fib levels.
Maturity line	While price remains above 2-4 TL, source expects Wave-5 to reach 127%.
Wave-C projection	Connect origin of A with B and place parallel line through A to project end of C.
After W3	Connect 1-3 and parallel through 2 to estimate W4 boundary.
After W4	Connect 2-4 and parallel through 3 to forecast W5 end.
Throw-over	If W4 moves out of channel, expect W5 throw-over.
Alternative W5 method	If W3 very strong/vertical and W4 sideways, connect 2-4 and parallel through W1 extreme.

V1. Objects to Store

```
TrendlineRecord {
  type: BASE_0_2 | ACTION_PARALLEL_1 | MATURITY_2_4 | CORRECTION_0_B | TRIANGLE_BOUNDARY,
  anchor1Bar, anchor1Price, anchor2Bar, anchor2Price,
  slope, projectedPrice, breakBar, breakPrice, breakConfirmed, sourceRuleId
}
```

V2. Impulse Lines

Line	Construction	Use
Base / 0-2	Connect 0 to2.	Trend continuity / W3 weakness clue.
Action	Parallel copy through W1.	Strong W3 behavior when price travels above/near action line in source.
1-3 / projection	Connect1-3 and parallel through2.	W4 projection/channel boundary.
Maturity / 2-4	Connect2-4.	W5 projection and post-W5 reversal confirmation.
0-B correction line	Connect correction origin to B.	ABC completion/reversal timing support.

V3. Reversal Timing after W5

Hardik material says to measure the time of the 4-to-5 leg; a maturity-line break within about half that time is a reversal confirmation clue, while reaching Wave 4 within an equal time is a pattern confirmation clue. Keep these as source-specific post-W5 confirmation rules, not universal Elliott textbook assumptions.

W. Time Analysis - Keep It as a Separate Engine

Time Analysis

Confidence in trading can be achieved using study of market structure (Elliott Wave theory + Fib ratios) + Time analysis + Price Action (Candlestick patterns, indicators)

Market remains in correction 80% of the time and in impulse only 20% of the time.

Time period is a different concept than Elliott waves. While Elliott wave (base point in case of flat correction) may begin from a lower high than the price top (where top is 111%), time period is to be considered from market high and lows only (not to be misunderstood/mixed with base point 0 in case of flat correction)

Time analysis

1. Elliot Wave time periods (tp)
2. Fib Wave proportions (wp)
3. Fib sequence Numbers (fn)

Few important points about time period

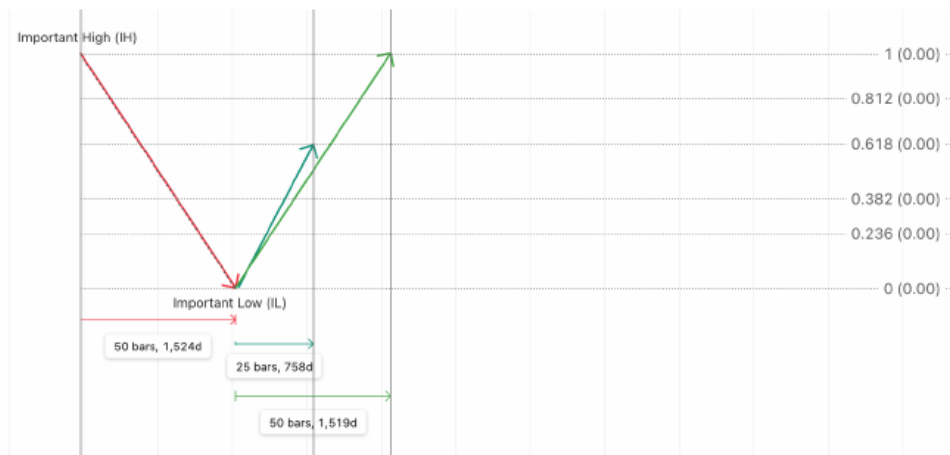
- a. Reversal zones: Points in time where all these time periods come together to form clusters and act as probable reversal zones
- b. High/Low price levels of 1/4th time period are most important for trading purposes

Elliott Wave Time Periods

Wave 1 time period: Wave 1 often starts as impulse after a correction.

What qualifies a good impulse:

1. When price recovers at least 61.8% of previous move (usually, downfall in stock price) from Important high (IH) to Important low (IL) in about $\frac{1}{2}$ of the period taken by previous move (IH to IL)
2. When price recovers 100% of the fall in about the same time (time from IH to IL)



Impulse is **confirmed** only after formation of wave 3 (sharp upmove). Before that it is just an assumption/possibility. Fast up moves after sharp fall (correction) can also be X wave, after which the price will most likely enter into double / triple correction.

Separate Time Analysis source: time origin and Wave-1 concept (p.1).

Wave 2 time period (W2 tp):

1. Wave 2 time period (tp) can never be equal to wave 1 time period (tp).
2. It will be either $\frac{1}{2}$ or $\frac{1}{4}$ of the wave 1 time period
3. If wave 2 tp is equal to wave 1 tp, it may not be wave 2 but wave B which (tp) is equal to wave A
4. Wave 2 is sharp most of the time as it is zig zag correction (price correction more often than time correction). This happens as market is generally not very bullish at this point after fall from important high to important low (correction)
5. Wave 2 can retrace between 23.6% to 81.2% in about $\frac{1}{2}$ or $\frac{1}{4}$ of the wave 1 time period.

Summary:

- a. $\frac{1}{2}$ of wave 1 time period
- b. $\frac{1}{4}$ of wave 1 time period

Wave 3 time period (W3 tp):

1. Wave 3 formation can only be confirmed by two ways
 - a. Following rule of equality where wave 3 is equal to price range of wave 1. This can be checked by applying three point fib extension tool from base point 0 to high/end of wave 1 and low/end of wave 2. It should be 100% i.e. same price range as wave 1
 - b. As retracement of wave 2 price, being minimum 127%. This can be verified using fib retracement from high/end of wave 1 to low/end of wave 2 towards wave 3.
2. Most of the time fib extension of 100% and 127 retracement will be at similar level creating confluence of price level.
3. ## Wave 3 is generally sharp move supported by price action indicators such as upper bollinger band (std. 2) turning upwards, RSI>60 and ADX above 12-15 and rising
4. Time period for wave 3
Wave 1 time period + Wave 2 time period : (W1 tp + W2tp)

Wave 4 time period (W4 tp)

1. Wave 4 is a corrective move after wave 3
2. As per theory of alternation, time period of wave 2 and wave 4 will not be the same, through correction in terms of price range % (rule of similarity) will be similar. This is because if wave 2 is simple zig zag price correction, more often wave 4 is time correction forming all type of patterns, zig-zag, flat, triangle, double and triple correction.
3. Time period for wave 4
 - a. Wave 2 time period divided by 2 : $W2\ tp / 2$
 - b. Wave 2 time period plus Wave 3 time period, divided by 4 : $(W2\ tp + W3\ tp) / 4$
 - c. Wave 2 time period plus Wave 3 time period, divided by 2 : $(W2\ tp + W3\ tp) / 2$
 - d. Wave 2 time period plus Wave 3 time period : $(W2\ tp + W3\ tp)$
4. If wave 4 is taking more time than W2 and W3 combined, it may be something else or the wave count is incorrect

Separate Time Analysis source: Wave-2 to Wave-4 time rules (p.2).

W1. Critical Concept: Time Origin May Differ from Elliott Base

The separate Time Analysis document explicitly warns that time should be measured from market highs/lows and not confused with an Elliott base point in a Flat correction. Therefore store Elliott price anchors and time-analysis anchors separately.

```
WaveCandidate.priceOriginBar // Elliott structural origin
WaveCandidate.timeOriginBar  // market high/low origin if selected time-source requires it

Never assume priceOriginBar == timeOriginBar in every correction.
```

W2. Time Windows, Not One Scalar

Because the sources provide multiple discrete expected times, represent time expectations as a list of windows rather than a single maxTime number.

```
expectedWindows = [
  {name:"W1/4", expected: W1time*0.25},
  {name:"W1/2", expected: W1time*0.50},
  ... active source-specific options ...
]

nearest = min(relativeDistance(actualTime, each.expected))
timePass = nearest.distance <= selectedTolerance
```

W3. Fib Time Zones and Cluster Zones

The separate Time Analysis notes introduce Fib sequence time zones (13, 21, 34, 55, 89, 144, 233, 377, 610, 987) and cluster zones where wave time periods, wave proportions and Fib time zones converge. This is an additional time-analysis module, not a replacement for price-structure Elliott rules.

Source basis: *Time Analysis pp.5-7.*

W4. Trade-Timing Module Is Separate from Wave Identification

Trade setup for Waves

Trading for Wave 3

1. Wave 1 and Wave 2 are not preferred for trading
2. When wave 2 is in development or completed, mark high of $\frac{1}{4}$ and high of $\frac{1}{2}$ time period candles ($\frac{1}{4}$ and $\frac{1}{2}$ of wave 1 time period marked from start of wave 2 <end of wave 1>)
3. When price starts moving up for wave 3 (preferably after wave 2 time period is complete and price has made a bottom <38% to 81.2% retracement of wave1>), enter above the high of $\frac{1}{4}$ th time period candle
4. If entry was made at $\frac{1}{2}$ time period candle, first target would be high of $\frac{1}{4}$ th time period candle

Target achievement / Exit

1. Exit when earlier of the below happens
 - a. Wave 3 has reached 100% of wave 1 (fib extension), rule of equality has been achieved
 - b. 127% retracement of wave 2 has reached
 - c. Wave 3 time period is over ($W1\ tp + W2\ tp$)

Time period should be the first priority as it prevents us from taking trade in haste.

Trading for Wave 5

Entry

1. Priority 1: Apply rule of similarity to validate if price range (% fall) of wave 4 is same as wave 2 (+/- 5%, [10% in case of $CMP < 500$])
2. Priority 2: Check for formation of bullish candle pattern at important fib retracement level (38% or 50% of entire upmove from base point zero)
3. Priority 3: Time period of wave 4 is complete

Enter where confluence of more than one trigger is occurring (where price fall is similar to wave 2 and bullish candle pattern is forming at important fib level, which is also acting as an important price support level.

Exit

1. Exit when price touches 127% retracement level of wave 4 or time period of wave 5 is over (wave 1 time period + wave 3 time period)/4, whichever is earlier.

Time Analysis trade examples: W3/W5 trade timing and target logic (p.4).

If implemented, trading entries/exits must consume Elliott state; they must not change the Elliott labels. For example, a W3 trade can wait for time/Fib triggers even while W3 is structurally FORMING. Likewise, a failed trade filter does not make the Elliott count invalid.

X. Source Conflict Matrix - Must Be Visible to the Developer

Topic	Chartking / normalized spec	Hardik / Impulse / Time notes	Required software treatment
W2 minimum	Normal 23.6%; 14% microscopic special	14% or 14%-23.6% shown as minimum in other notes	Preset / subtype; no silent merge.
W2 time	Several versions; one says $\leq W1$	Hardik 1/4,1/2,1,2; Time Analysis 1/4 or 1/2; Impulse notes same/double	Discrete source presets.
W3 time	Average if $W1 \neq W2$, sum if equal	Hardik/Time Analysis use $W1+W2$ as primary	Preset.
W3 maximum	2700% in normalized spec	2100% in Impulse Notes	Configurable source max.
W4 minimum	23.6% normal	14% in Hardik/Impulse tables	Preset/context subtype.
W4 time	1/2,2x,3x W2	Hardik 1/2,2x,5x; Time Analysis uses $W2+W3$ formulas	List of source windows.
W4 extension max	38.2 vs 50 in extension notes	Rare 50 etc.	Context + conflict mode.
W5 time	Not one universal rule	Hardik $W1$ /rare $W4$; Time Analysis has multiple $W1/W3$ formulas	Separate time preset.
C elongated max	261.8 in some tables	Up to 461.8 in other diagrams/notes	Conflict mode.
Double confirmation	Source cell "???"	No unambiguous replacement	TBD-BLOCKED.
Triple X time	$X < W$	Some detailed rule $X < 1/2 W$	Context-specific conflict.
Z failure	31.8 in one source	Nearby 38.2 note for related case	TBD.
LD/ED same impulse	Ambiguous wording	Other examples may imply different behavior	Owner decision.
Timeframe degree mapping	Daily/75m/15m	Holding-period dependent mappings	Degree Mapping Preset.
Log chart	Not universal in Chartking spec	Hardik says always log	Optional source preset.
MACD parameters	Unspecified	Examples often show 12/26/9 visually	Inputs; never claim source mandated 12/26/9.
Pivot algorithm	No exact source algorithm	Visual/manual notes	Owner-approved raw pivot algorithm; expose sensitivity.

Y. Recommended Pine v6 Module Architecture

The developer is a Pine Script expert, so the following is architecture-level pseudocode, not intended as copy/paste syntax. The goal is to keep the theory isolated in testable modules.

```

MODULE 01 - RawPivotEngine
    detect pivots, ATR/range filter, confirmed/raw buffers

MODULE 02 - DegreeRouter
    map chart/source timeframe, Important H/L, degree metadata

MODULE 03 - BaseEngine
    base candidates, MACD/divergence evidence, base lock

MODULE 04 - ImpulseEngine
    w1/w2/w3/w4/w5 candidates, normal/terminal subtype

MODULE 05 - ExtensionEngine
    internal move counts 5/9/13/17/21, degree fall

MODULE 06 - DiagonalEngine
    LD / ED

MODULE 07 - CorrectionEngine
    ZigZag / Flat / ABC / WXY / Triple / Triangle

MODULE 08 - FibEngine
    anchor-specific levels + ratios

MODULE 09 - TimeEngine
    source presets, expected windows, Fib time zones

MODULE 10 - SupportEngine
    MACD/RSI divergence, gaps, candles, trendlines/channels

MODULE 11 - RecountEngine
    locked confirmed count, alternates, hard invalidation, FBD/FB0 candidate origin

MODULE 12 - DisplayDebug
    labels, source rule IDs, next condition, pass/fail numbers

```

Y1. Suggested Candidate Types

```

enum Status { SEARCHING, FORMING, CONFIRMED, ALTERNATE, INVALID, TBD_BLOCKED }

type Pivot
    int bar
    float price
    int side
    int degree
    bool confirmed
    float macdLine
    float macdHist
    float rsi

type WaveCandidate
    string id
    string label
    string subtype
    int degree
    int direction
    Pivot start
    Pivot finish
    Status status
    float fibRetrace
    float fibExtension
    int duration
    int internalCount
    string invalidReason
    string sourceRuleId

```

Y2. Functions to Add / Refactor

Function	Purpose
f_eval_base()	Return base candidate state and reasons.
f_eval_w1()	Build/validate W1 and degree confirmation.
f_eval_w2()	Evaluate W2 price/time/internal correction and origin invalidation.
f_eval_w3()	Trending/terminal/extension classification.
f_eval_w4()	Open/track correction child candidates and subtype-aware overlap.
f_eval_w5()	Normal/truncated/extended/ED logic plus divergence.
f_eval_zigzag() / f_eval_flat()	Simple correction classifiers.
f_eval_double() / f_eval_triple()	Complex correction containers.
f_eval_triangle_*)	Six triangle subtype classifiers.
f_eval_ld() / f_eval_ed()	Diagonal parsers.
f_apply_hard_invalidations()	Central hard-rule rejection with reason codes.
f_rank_candidates()	Choose primary/alternate without deleting still-valid correction candidates.
f_controlled_recount()	Promote alternate/new base only on approved trigger.
f_debug_rule()	Emit rule ID, numeric inputs and pass/fail.
f_draw_candidate()	Draw based on FORMING/CONFIRMED status, not fixed sequence.

Y3. Main Loop Pseudocode

```

on each closed bar:
    update indicators and source timeframe contexts
    rawPivotEvent = RawPivotEngine.update()

    if rawPivotEvent:
        DegreeRouter.update(rawPivotEvent)
        BaseEngine.update(rawPivotEvent)

        for each active degree:
            ImpulseEngine.update(rawPivotEvent, degree)
            CorrectionEngine.update(rawPivotEvent, degree)
            ExtensionEngine.update(rawPivotEvent, degree)
            DiagonalEngine.update(rawPivotEvent, degree)

        HardRuleEngine.invalidateImpossibleCandidates()
        CandidateRanker.updatePrimaryAndAlternates()
        RecountEngine.applyOnlyIfTriggerApproved()

    DisplayDebug.renderFromState()

```

Z. Step-by-Step Refactor Plan for the Existing Script

Sprint	Developer task	Do not move forward until
Sprint 0 - Freeze baseline	Save current script/Python outputs and 10 known charts as regression fixtures.	Baseline screenshots and outputs are archived.
Sprint 1 - Decouple labels	Keep raw pivots but stop modulo phase from deciding Elliott labels. Add candidate/status structs.	No main label appears merely because pivot index advanced.
Sprint 2 - Base engine	Implement degree-aware Important H/L, oscillator evidence, locked base and alternate origin.	Base remains stable on minor swings; 144-day test passes.
Sprint 3 - W1/W2	Implement W1 degree + W2 correction container, source-mode W2 range/time and hard origin invalidation.	Normal/microscopic/deep/invalid W2 fixtures pass.
Sprint 4 - W3	Implement Trending/Terminal/Extension + internal counts + MACD support.	Small pivots inside extended W3 no longer become W4/W5.
Sprint 5 - W4/W5	Correction container, overlap subtype, W5 normal/truncated/ED.	Premature W5 bug is gone without forced A-B-C sequencing.
Sprint 6 - Corrections	ZigZag, Flat, WXY, Triple, triangles.	Manual correction dropdown no longer controls classification.
Sprint 7 - Time/support	Time presets, trendlines/channels, gaps, MACD/RSI support.	Debug shows source rules and no support module overrides hard price structure.
Sprint 8 - Project overlays	HP BUY/SELL, FBD/FBO controlled recount, C/Z completion event.	Signals consume Elliott state but do not mutate the count.
Sprint 9 - Python parity	Port same state model to Python and replay fixtures.	Pine/Python confirmed pivots materially match; differences explained.
Sprint 10 - Production QA	Performance, object limits, dashboard, non-repaint proof.	Full acceptance suite passes.

Z1. Definition of “Premature Wave 5 Bug Fixed”

Acceptance rule

The bug is only considered fixed if a complex W4 example can add several internal pivots (A/B/C, W/X/Y, or triangle legs) without the parent count becoming Wave 5 until the W4 correction itself is confirmed complete. Simply inserting a fixed A-B-C phase before W5 is not sufficient.

Z2. Definition of “B Wave up to 111% Fixed”

Acceptance rule

The fix is valid only if 61.8%-111% is applied inside the Flat candidate, while a Zig-Zag B uses its own source range and complex corrections remain possible. Above the Flat maximum must invalidate/reclassify the Flat candidate instead of leaving a stale B label.

Z3. Definition of “Non-Repainting”

Acceptance rule

Raw/current FORMING legs may update until confirmed. Once a main-degree label is CONFIRMED, it must not move because of a small later pivot. A confirmed label may only change through a logged controlled recount caused by a hard structural invalidation or approved higher-degree/origin event.

AA. Detailed Developer Debug Panel Specification

Row	Example value	Why it matters
DEGREE / TF	Intermediate / 75m	Prevents degree mixing.
BASE	LOCKED @ 2385.20 / bar 1234	Shows origin stability.
CURRENT PARENT	W4 FORMING	Parent context.
PRIMARY PATTERN	Flat FORMING	Main correction candidate.
ALTERNATE	Double WXY FORMING	Ambiguity kept visible.
RAW PIVOTS	27 / main pivots 4	Shows raw vs semantic separation.
FIB ANCHOR	0-3 = 2385.2 -> 2455.8	Makes formulas auditable.
FIB VALUE	W4 retrace 34.6% PASS	Numeric source check.
TIME RULE	Chartking W4: nearest 2xW2, error 7%	Shows selected time source.
MACD	W4 low support PASS	Supporting evidence.
HARD RULE	Normal overlap: N/A - terminal candidate active	Prevents wrong invalidation.
NEXT CONDITION	Need C internal completion + 0-B break	Developer/user knows what is missing.
LAST INVALID	ZigZag B range failed @ 69.2%	Explains why candidate was removed.
RECOUNT	NONE / last reason	Audit non-repaint behavior.

AB. Extended Acceptance Tests for a Developer Who Does Not Know Elliott

ID	Scenario	Input	Expected
E01	Raw pivot vs Elliott pivot	Insert 8 small pivots inside a strong W3	Raw array grows; main degree still W3.
E02	Base replacement	New important low appears while count remains valid	Open alternate base; do not replace locked base.
E03	Base hard invalid	Price invalidates current origin	Log reason; promote valid alternate/recount.
E04	W1 subordinate	Five moves complete before degree threshold	Mark subordinate/FORMING; do not confirm degree W1.
E05	W2 microscopic modes	14%-23.6% retracement under three source presets	Correct status differs visibly by preset.
E06	W2 time conflict	Actual W2 = W1	Different source presets show pass/fail/TBD rather than hidden decision.
E07	W3 terminal	W3 five moves at 100% extension	Terminal candidate remains; no forced trending W3.
E08	W3 extension	13 internal moves	Parent label remains W3 until degree-fall/completion.

ID	Scenario	Input	Expected
E09	W4 overlap normal	W4 enters W1 territory in normal impulse	Normal candidate invalid.
E10	W4 overlap diagonal	Same price path but valid diagonal context	Diagonal candidate remains valid.
E11	W4 Double	WXY with 5 raw turning points	Parent stays W4 until Y completion.
E12	W4 Triangle	ABCDE legs	No W5 until E/triangle completion.
E13	Flat vs ZigZag B	B=45%, then another sample B=90%	First favors ZigZag; second Flat; no universal B rule.
E14	Flat B >111	B=118%	Flat invalid/reclassified; stale B removed.
E15	C Fib hit without internals	C touches target early	Keep C FORMING if internal completion not satisfied.
E16	Double confirmation TBD	Valid WXY but source ??? not resolved	Show TBD-BLOCKED / source conflict.
E17	Triple XX	XX violates X boundary	Reject Triple candidate with exact reason.
E18	Triangle location	ABCDE appears in W2	Triangle candidate rejected because source excludes W2.
E19	ED position	3-3-3-3-3 appears in W3	Do not label ED if source position invalid.
E20	W5 divergence	Price makes higher W5 high but MACD lower than W3	Divergence support PASS.
E21	W5 truncation	Double-extension fixture, W5 below127	Truncated subtype possible; normal W5 candidate may fail without killing impulse.
E22	Time origin Flat	Flat base differs from market high used for time	Price/time origins stored separately.
E23	144-day consistency	Same symbol on D/1h/15m	Important H/L source context consistent.
E24	Pine/Python	Replay fixture	Same confirmed pivot IDs and pattern states.
E25	Recount audit	Force hard invalid after several confirmed labels	Log old/new base, reason, affected labels; no silent rewrite.

AC. What the Developer Must Not Do

- [] NEVER: Keep f_phase/modulo as the real wave engine and only add more warning conditions.
- [] NEVER: Treat every accepted ZigZag swing as a new main-degree Elliott wave.
- [] NEVER: Use 61.8%-111% as a universal B-wave rule.
- [] NEVER: Use 14% as a universal W2/W4 normal minimum without a source preset/subtype.
- [] NEVER: Hard-code one time rule from conflicting sources and hide the decision.
- [] NEVER: Confirm Wave C because the third corrective pivot appeared.
- [] NEVER: Reject all Wave-4 overlap before checking terminal/diagonal subtype.
- [] NEVER: Call a wedge a Leading Diagonal without its position/internal/Fib rules.
- [] NEVER: Call 3-3-3-3-3 an Ending Diagonal outside W5/C.
- [] NEVER: Delete alternate correction candidates just because one pattern currently scores higher.
- [] NEVER: Reset the base on every new Important H/L or FBD/FBO event.
- [] NEVER: Let trade signals mutate the Elliott structure.
- [] NEVER: Claim "non-repaint" if confirmed labels silently move with later raw pivots.

- [] NEVER: Let Pine and Python implement different pivot/state rules and then compare only screenshots.

AD. Final Instruction to the Developer

Implementation goal

You do not need to become an Elliott Wave trader. You need to implement a deterministic state machine that preserves the supplied rules exactly, exposes conflicts instead of guessing, and provides enough debug information that the owner can compare the automatic count with manual Elliott marking.

Order of authority

1) Hard price/position/internal-structure rule. 2) Degree and parent context. 3) Pattern-specific Fib. 4) Time rule under selected source preset. 5) MACD/RSI/trendline/gap/candle support. 6) Trade overlay. A lower item must not override a hard failure in a higher item.

Completion standard

Do not declare the feature complete when labels look visually plausible. Declare it complete only when every confirmed label has a traceable candidate history, Fib/time anchors, source rule IDs, invalidation conditions, and matching Pine/Python result on the acceptance fixtures.

AE. Source Visual Index Used in This Addendum

Visual	Source
Wave 1 degree/base visual	EW Notes - Chartking v2, p.6
Wave 2 Fib/time visual	EW Notes - Chartking v2, p.8
Wave 3 extension/time visual	EW Notes - Chartking v2, p.11
Wave 4 alternation visual	EW Notes - Chartking v2, p.13
Wave 5 projection/divergence visual	EW Notes - Chartking v2, p.15
Leading Diagonal visual	EW Notes - Chartking v2, p.18
Ending Diagonal visual	EW Notes - Chartking v2, p.22
Zig-Zag normalized table	Developer Requirement Specification, p.14
Flat normalized table	Developer Requirement Specification, pp.15-16
Double WXY table	Developer Requirement Specification, p.19
Triple WXYXXZ table	Developer Requirement Specification, p.22
Triangle family summary	Developer Requirement Specification, p.24
Trendline/channel summary	Developer Requirement Specification, p.32
Candidate state engine	Developer Requirement Specification, p.39
Time-analysis source visuals	Time Analysis, pp.1-4
Hardik W1 time / trendline visuals	ELLIOT WAVE-HARDIK PAVAGADHI, pp.14,39

END OF DETAILED IMPLEMENTATION ADDENDUM

17. Recommended indicator state engine (implementation translation)

This section is recommended software architecture, not a new trading rule. It translates the source into a deterministic implementation that can handle incomplete patterns without repainting historical labels arbitrarily.

17.1 Candidate lifecycle

State	Meaning
SEARCHING	No valid base/current candidate yet.
FORMING	Enough structure exists to propose a wave/pattern, but all confirmation conditions are not met.
CONFIRMED	Source-defined structural, fib/time and required confirmation conditions are met.
ALTERNATE	A secondary count remains possible because correction rules are guideline based.
INVALID	A hard source rule has been violated. Store the exact reason.
TBD-BLOCKED	A decision depends on a source blank/conflict; show it rather than guessing.

17.2 Evaluation order

- **Step 1:** determine analysis timeframe / degree and locate Important High-Low plus candidate base point.
- **Step 2:** create impulse candidate and subordinate pivots; apply hard price-structure rules before supporting indicators.
- **Step 3:** calculate Fib retracement/extension and time relationships at every confirmed pivot.
- **Step 4:** classify Trending vs Terminal impulse, diagonal subtype and extension subtype if applicable.
- **Step 5:** when correction begins, maintain Zig-Zag / Flat / Double / Triple / Triangle candidates in parallel.
- **Step 6:** use MACD, gap, candlestick, channel and trendline events as the source describes - primarily as confirmation/ranking, not as replacements for wave structure.
- **Step 7:** confirm/invalidate labels only after the relevant source rule is satisfied/violated; retain reason codes for debugging.

17.3 Suggested internal record

```
WaveCandidate { degree, timeframe, direction, label, patternType, startBar, endBar,
startPrice, endPrice, durationBars, fibRetrace, fibExtension, internalCount, macdState,
divergenceState, gapState, candlestickState, trendlineState, channelState, status,
invalidReason, sourceRuleId }
```

18. Chart display requirements (recommended handoff)

Element	Required display behavior
Wave labels	Show 0/Base,1,2,3,4,5,A,B,C,W,X,Y,XX,Z and A-B-C-D-E triangle labels at confirmed pivots; optional dotted label for forming candidates.
Pattern name	Display active structure: Impulse, Terminal, Leading Diagonal, Ending Diagonal, Zig-Zag, Flat subtype, Double, Triple, Triangle subtype, Extension subtype.
Fib levels	Draw only the levels relevant to the current candidate and source rule; label the anchor pair (e.g.,0-1-2,0-A-B,0-W-X,0-Y-XX).
Trendlines/channels	Draw source 0-2,1-3,2-4,0-B and triangle boundaries with anchor labels and break state.
MACD support	Display compact status (lowest/highest/top/bottom, +div/-div) rather than replacing wave labels.
Gap / candle support	Small labels at the source-defined context.
Status panel	Show current degree, current wave, candidate pattern, confirmed/forming/alternate/invalid, next source condition and conflict/TBD if applicable.
Debug mode	Developer-only mode should print rule IDs passed/failed and numeric ratios for backtesting/validation.

19. Inputs / settings required for implementation

PART XIII - V3 100% SOURCE-LOCKED COVERAGE ADDENDUM

Purpose: remove any later ambiguity about what the developer was asked to implement. This addendum records every supplied source file, preserves source conflicts, maps the current Pine architecture to required changes, and includes a page-by-page rule ledger. The Master PDF bundle additionally appends the original source PDFs unchanged, so every diagram and screenshot remains available.

IMPORTANT: "100% source coverage" means no readable rule from the supplied notes is intentionally omitted from the handoff. It does NOT mean contradictory source statements are silently reconciled. Contradictions are part of the source and must remain visible until the owner locks a choice.

47. Source File Register - Binding Scope

ID	Source file	Pages / size	SHA-256 prefix
S01	ELLIOT WAVE-HARDIK PAVAGADHI(2).pdf	130	be6430f1180d3c9ba8a9
S02	Elliott_Wave_Indicator_Developer_Requirement_Specification(2).pdf	157	c8615f7205041fb57afe
S03	EW Notes _ Chartking v2 - 2025-04-17 11-29-14(9).pdf	112	cd0ff942605e85f308ca
S04	EW Notes _ Chartking v2(2).pdf	112	02464600617a660694a1
S05	Impluse notes \$\$ (2).pdf	31	9226972b96cbd805f272
S06	Time Analysis(2).pdf	7	12a1f93eff72d6634266
S07	Pasted markdown(4).md	1,007 script lines	0af0c03165ac19711693

Duplicate note handling: the two 112-page Chartking files have page-text identical content on pages 2-112; only the title text on page 1 differs. Both files remain listed in scope and are included in the Master PDF source bundle.

48. Binding Coverage Rule

- Every numeric threshold, pattern location, internal count, time formula, Fibonacci anchor, invalidation, exception, supporting clue and trading note that appears in the supplied notes must either be implemented, explicitly marked SOURCE CONFLICT, or explicitly marked TBD-BLOCKED.
- No developer may replace a source rule with a generic Elliott Wave textbook rule unless the owner explicitly approves the replacement in writing.
- If two supplied notes disagree, both versions must be recorded. The code must expose a configurable mode or remain blocked for that rule until the owner selects a version.
- Visual-only examples are not discarded. The Master PDF source bundle appends the original source PDFs unchanged so the diagrams remain part of the handoff.
- The current Pine script is not accepted as complete merely because it displays 1-2-3-4-5/A-B-C labels. The final engine must prove its count with state, structure, Fib, time, MACD/supporting evidence and invalidation reasons.

49. Binding Change Map for the Current 1,007-Line Pine Script

Current location	Current logic	Action	Binding requirement
Lines 147-197	f_correction_count / f_cycle_length / f_phase / f_phase_text / impulse-phase helpers	REPLACE AS PRIMARY ENGINE	Fixed positional cycle must not decide Elliott labels. Retain only optional display helpers if needed.
Lines 257-321	f_start_confirmation_note	KEEP + REFACTOR	Useful Important H/L, degree, ATR and oscillator evidence. Add degree-aware 144-day logic, base candidate/lock state, and configurable MACD-line/histogram divergence.
Lines 323-353	f_valid_anchor_index / f_wave_phase	REPLACE	Do not choose latest qualifying anchor and then use modulo phase. Use BASE_CANDIDATE -> BASE_CONFIRMED -> BASE_LOCKED with controlled

			release.
Lines 355-483	swing filter / f_upsert_swing	KEEP AS RAW PIVOT ENGINE	These swings are raw pivots only. Add raw/internal/main-degree layers and do not advance Elliott labels automatically.
Lines 485-689	f_rule_state	REFACTOR INTO VALIDATORS	Do not label first then warn. Validators must decide candidate status before label confirmation; hard invalid must reject/recount.
Lines 691-797	time note/table	REPLACE WITH MULTI-RULE TIME ENGINE	Current scalar checks do not cover all supplied time rules or conflicts. Store multiple expected windows and source IDs.
Lines 819-861	generic Important H/L fib guide	KEEP DRAWING INFRASTRUCTURE; CHANGE ANCHORS	Use candidate-specific anchors: 0-1, 0-1-2, Base-3, 3-4, 0-A-B, 0-W-X, 0-Y-XX, triangle/thrust anchors.
Lines 863-889	Wave-5 target drawing	KEEP + EXTEND	Normal 127-261.8% 3-4 projection is useful; add truncated/extended/ED branches.
Lines 891-949	redraw/labels	KEEP + MAKE STATUS-AWARE	FORMING labels may move; CONFIRMED labels must lock; INVALID candidates should not remain as accepted wave labels.
Lines 949-1007	pivohigh/pivotlow and indicators	KEEP AS INPUT LAYER	No look-ahead cheating; however confirmation delay and same-type pivot replacement must be explicitly treated as developing-state behavior.
Not present	FBD/FBO controlled recount	ADD PROJECT MODULE	FBD/FBO may nominate a new base only after degree/structure validation; must not blindly reset Elliott count.
Not present	Automatic correction classifier	ADD	Run Zig-Zag/Flat/Double/Triple/Triangle candidates in parallel until rules eliminate/confirm them.
Not present	Leading/Ending Diagonal + extension subtype engine	ADD	Separate normal/trending/terminal/LD/ED/W1-ext/W3-ext/W5-ext/truncation branches.
Not present	Pine/Python parity log	ADD	Same OHLC and approved settings must produce materially the same confirmed count; differences must be reported.

50. Expanded Cross-Source Conflict Matrix - DO NOT GUESS

Conflict / topic	What the supplied notes say	Developer instruction
C-V3-01 Timeframe/degree mapping	Hardik: investment M/W/D; swing W/D/75m or 288m/1h commodity; day D/75m/15m. Chartking: Daily/75m/15m plus CMP routing.	Provide selectable trading-horizon profile; do not force one mapping for every market.
C-V3-02 Wave-1 time	Hardik p14: significant H/L bars divided by 50 gives maximum W1 time; 50% price in that time + 5 moves may qualify. Time Analysis p1: 61.8% recovery in ~1/2 prior-move time or 100% in same time.	Store both models as separate source modes; owner selects.
C-V3-03 Wave-2 minimum price retracement	Hardik p10 and Impulse Notes p5-6 use 14% as minimum/general range; Chartking uses 23.6%-81.2% normal with 14% as microscopic special clue; Time Analysis uses 23.6%-81.2%.	Implement normal and microscopic subtypes; owner decides whether 14%-23.6% is valid normal W2 or microscopic-only.
C-V3-04 Wave-2 time	Hardik p15: 1/4,1/2,1X,2X W1. Chartking contains <=W1 and also 1/4,1/2,1X,2X variants.	Expose source profile; do not compress to <=2X.

	Time Analysis p2: never equal; only 1/4 or 1/2. Impulse Notes also mention same/double in places.	
C-V3-05 Wave-3 maximum extension	Chartking specification: max 2700% on 0-1-2; Impulse Notes: max 2100%.	Keep max configurable by source profile; minimum trending gate remains source-specific 161.8%.
C-V3-06 Wave-3 time	Hardik/Time Analysis: W3 time = W1+W2 minimum. Chartking: if W1!=W2 use (W1+W2)/2; if equal use W1+W2.	Implement both; owner selects or use ranking rather than hard invalid until locked.
C-V3-07 Wave-4 minimum retracement	Hardik/Impulse Notes show 14% minimum in some summaries; Chartking detailed source uses 23.6% normal, up to 50%, terminal 61.8%.	Source-mode setting required.
C-V3-08 Wave-4 time	Hardik: W2/2, W2*2, W2*5. Chartking: W2/2, 2X, 3X. Time Analysis: W2/2, (W2+W3)/4, /2, or W2+W3.	Store all candidate windows with source IDs; no single hard-coded formula.
C-V3-09 Wave-5 time	Hardik: mostly W1, rarely W4. Time Analysis: W1, (W1+W3)/4, /2, W1+W3, and (W3+W4)/2.	Use source-profile windows.
C-V3-10 Zig-Zag B retracement	Hardik: B maximum 61.8%. Chartking detailed summary: B <61.8%; Wave-B page states 1%-50% in Zig-Zag.	Implement bands as source alternatives; do not force Flat classification solely at 50%.
C-V3-11 Zig-Zag B time	Hardik: A,2A,5A,10A rare. Chartking: B>=A up to 5A.	Preserve 10A as Hardik rare case, profile-specific.
C-V3-12 Zig-Zag C time	Hardik: A+B, (A+B)/4, (A+B)/2. Chartking: if A=B then A+B; otherwise (A+B)/2.	Do not drop /4 branch; profile-specific.
C-V3-13 Zig-Zag C max	Chartking table uses 261.8 max while elongated examples/other pages allow 461.8; Hardik describes elongated at 261.8.	Keep 261.8 and 461.8 variants explicit.
C-V3-14 Flat Wave-A threshold	Hardik p67 says Wave A can MAX go to 38.2%; Chartking says minimum 38.2% for Flat A.	Major source conflict; must remain owner decision/TBD mode.
C-V3-15 Flat B time	Hardik: A,2A,5A,10A rare. Chartking: B>A, typically up to 2A in Flat.	Preserve both source modes.
C-V3-16 Flat C time	Hardik includes A+B, /4 and /2 with special equal-A/B condition. Chartking uses A+B or /2.	Profile-specific.
C-V3-17 Double Correction X time	Hardik: X often W/4 or W/2; Double Flat W X may be 1/4,1/2,1/3 of W. Chartking: X<W; Triple graphic may require X<1/2W.	Implement as separate allowed windows by pattern subtype/source.
C-V3-18 Double Correction Y extension	Hardik Double Zig-Zag says Y min 61.8 and cannot go below 261.8; ZigZag+Triangle Y max 100. Chartking general WXY Y = 61.8-161.8.	Subtype-specific rules; do not use one universal Y max.
C-V3-19 Double Correction confirmation	Chartking source cell explicitly ???; Hardik adds structural/trendline/time clues.	Keep Chartking confirmation field TBD; Hardik clues may be optional/source-profile support unless owner promotes them.
C-V3-20 Triangle location	Hardik says always B and W4, then notes occasional Y/Z and after complex corrections. Chartking allows W4,B,E,X,Y,XX,Z and excludes W2.	Use union as candidates but preserve source-profile occurrence ranking; never allow W2 unless owner overrides.
C-V3-21 Triangle thrust	Contracting: 75/100/125 or 75-125. Running contracting also has up to 261.8 in source. Expanding: Hardik 61.8% of E; Chartking 61.8-100%.	Subtype-specific + conflict flag.
C-V3-22 Ending Diagonal trigger wording	Some notes say W3 extended then W4 sharp -> ED; other wording says terminal W3 below 161.8.	Separate degree W3 vs subordinate W3 interpretation; keep conflict visible.
C-V3-23 Wave-5 extension market restriction	Chartking: W5 extension not in indices, only stocks. Impulse/Hardik notes include equity/market-specific restrictions and commodity/forex double-extension notes.	Instrument-aware source mode; do not assume same rule for XAUUSD, indices and stocks.

C-V3-24 Gap timing	Hardik: common 1-5d; breakaway may remain 3m-3y; exhaustion same-day. Charting: common 3-5d; breakaway described as not filling; exhaustion fills quickly.	Make day-based timings market/profile-specific and supporting, not structural hard rules until owner locks.
C-V3-25 Time-origin rule	Time Analysis states time starts from market H/L even when Elliott base point 0 differs (e.g. Flat with top at 111%). Other notes often measure from Elliott pivots.	Store price-structure anchor and time-analysis anchor separately.
C-V3-26 Qualitative terms	Fast, slow, sharp, microscopic, "cannot stay below long", volume dry-up, volatility decrease are not numerically defined.	Keep as support or expose configurable thresholds; no invented hard number.
C-V3-27 Pivot algorithm	No source specifies exact ta.pivotleft/right or ATR-ZigZag sensitivity.	Developer pivot choice must be configurable and must not silently redefine Elliott degree.

51. Final Binding Software Architecture

- Layer 1 - Market data and timeframe/degree profile.
- Layer 2 - Raw pivot detector (confirmed pivots only; sensitivity configurable).
- Layer 3 - Important High/Low and Point-0 candidate scanner.
- Layer 4 - Base lock / degree candidate memory.
- Layer 5 - Internal/subordinate pivot parser.
- Layer 6 - Impulse candidates: Normal Trending / Terminal / Leading Diagonal / Ending Diagonal / W1 Extension / W3 Extension / W5 Extension / Double Extension-Truncation.
- Layer 7 - Correction candidates in parallel: Zig-Zag / Flat subtypes / Double W-X-Y / Triple W-X-Y-XX-Z / six Triangle families.
- Layer 8 - Wave-specific Fibonacci engine.
- Layer 9 - Time-analysis engine with source-profile windows and separate time origin when required.
- Layer 10 - MACD/RSI, gap, candlestick, trendline/channel supporting engine.
- Layer 11 - State engine SEARCHING / FORMING / CONFIRMED / ALTERNATE / INVALID / TBD-BLOCKED.
- Layer 12 - Controlled recount and label lock.
- Layer 13 - Project trading layer (HP BUY/SELL and FBD/FBO recount nomination) kept logically separate from wave identification.
- Layer 14 - Debug/audit log + Pine/Python parity tests.

52. Developer Scope Acceptance - No Later "Not Included" Claim

Before coding, the developer should acknowledge the following scope. If an item is not technically possible in Pine Script, the developer must identify it in writing before implementation and provide the closest deterministic approximation plus the exact limitation. Silence is treated as acceptance of scope.

- ☐ Point 0 / base identification and lock
- ☐ Important High/Low and degree/timeframe routing
- ☐ Wave 1-5 with internal structure
- ☐ Trending vs Terminal impulse
- ☐ Leading Diagonal
- ☐ Ending Diagonal
- ☐ W1/W3/W5 extensions
- ☐ 5/9/13/17/21 extension counting
- ☐ Wave-5 truncation / double extension context
- ☐ Zig-Zag and all supplied C subtypes
- ☐ Flat and all supplied Flat subtypes
- ☐ Wave A/B/C individual rules
- ☐ Double Correction W-X-Y and supplied combinations
- ☐ Triple Correction W-X-Y-XX-Z
- ☐ Contracting Triangle: Horizontal / Irregular / Running
- ☐ Expanding Triangle: Horizontal / Irregular / Running
- ☐ MACD wave-specific support
- ☐ RSI divergence project support
- ☐ Gap classifications
- ☐ Candlestick support rules and source formula handling

- [] Base/Action/Maturity trendlines and channels
- [] Wave-specific Fib anchors and targets
- [] All supplied time-analysis models and source conflicts
- [] FORMING/CONFIRMED/ALTERNATE/INVALID/TBD state engine
- [] Controlled recount / non-repaint behavior
- [] FBD/FBO controlled base nomination
- [] HP BUY/SELL project logic
- [] Debug reason codes
- [] Acceptance screenshots/logs
- [] Pine/Python parity

PART XIV - FULL SOURCE PAGE LEDGER

This ledger is intentionally exhaustive. It records readable source text page by page. Pages that are mainly diagrams are marked as visual-reference pages and remain preserved in the Master PDF appendix. This ledger is not a replacement for the original pages; it is an implementation traceability aid.

HARDIK-130: ELLIOT WAVE-HARDIK PAVAGADHI(2).pdf

HARDIK-130 p.1: ELLIOT WAVE THEORY

HARDIK-130 p.2: RULES OF ELLIOT WAVE | • Minimum 1 year data required. | • Always use log chart. | • If price is above 300rs then use daily chart. | • If price is below 300rs then use weekly chart. | • If price is below 300 then keep fib retracement tool Log tick.

HARDIK-130 p.3: TIME FRAME | There are total three phase | 1. Major Degree- Trend analysis | 2. Intermediate Degree-Trading. | 3. Minor Degree- Practice

HARDIK-130 p.4: Holding period for minimum 1 year | 1. Major Degree- Monthly time frame | 2. Intermediate Degree-Weekly time frame | 3. Minor Degree-Daily time frame

HARDIK-130 p.5: Swing trading(For 1 to 3 months) | 1. Major Degree- Weekly time frame | 2. Intermediate Degree- Daily time frame | 3. Minor Degree-75min/[288 min/1hr(commodity)] time frame.

HARDIK-130 p.6: DAY TRADING | 1. Major Degree- Daily time frame | 2. Intermediate Degree-75 min time frame | 3. Minor Degree-15 min time frame

HARDIK-130 p.7: Type of waves Impulse | • Impulse consist of 5 moves. | • Here 1,3 and 5 are impulse wave | and any 2 wave will be equal. Correction | • Correction consist of 3 moves | 1 3 2 5 4 | A | B | C

HARDIK-130 p.8: Identification of zero point. | 1. Find MACD lowest point. | 2. MACD with positive divergence. | 3. Trendline.

HARDIK-130 p.9: DEGREE 1 | ➤First mark swing high and swing low | point. | ➤Now put fib retracement from swing high | to swing low. | ➤If price touches at 61.8% with 5 moves(as | it is impulse wave)in fib retracement then we can say that wave is degree 1. | ➤If the wave complete 5 moves without | touching 61.8% level than we can consider it as subordinate wave of degree | 1. | ➤Note subordinate waves can make new | low(here we are assuming that it will go at least till 61.8%)

HARDIK-130 p.10: DEGREE 2 | ➤ Wave 2 is considered of three moves as it | is corrective wave. | ➤If we put fb retracement from base point | i.e. 0 to degree 1. | ➤Price can minimum retrace up to 14% to | 23.60%. | ➤Price can maximum retrace up to 81.20%. | ➤If this rules are followed by price then we | can consider it as degree 2. | ➤In given example price reversed from | 61.8% level.

HARDIK-130 p.11: Rule of Alteration Rule of similarity | ➤Here wave 2 and wave 4 size are | equal Rule of overlapping | ➤Here wave 2 and wave 4 can | very. | ➤Wave 4 can overlap wave 2 by | +5% or -5%

HARDIK-130 p.12: DEGREE 3 | ➤Wave 3 will never shortest wave among | 1,3,5. | ➤If it is trending then third wave must | touch 161.8% of fib extension. It is minimum target for price. | ➤Fib extension can be applied from base | point i.e. 0 to wave 1 and the wave 2. | ➤If the wave 3 is not touching 161.8% after | completing 5 moves then it is terminal diagonal case.

HARDIK-130 p.13: Guideline to Trending impulse | ➤Rule of equality | • In this case any 2 wave from 1,3,5 impulse will be similar. | ➤Rule of golden ratio | • In this case wave 3 of degree 1 will go till 161.8% ratio of fib | retracement. | ➤Rule of golden extension | • In this case wave 3 of degree 1 is extended more than 161.8% ratio.

HARDIK-130 p.14: Time Analysis | ➤Wave 1 | • | Time of wave 1 can be measured by putting date and time measurement tool from significant high to significant low. | • | Now divide no. of bars by 50. | • | And then again put tool from base point to the no. of bars which we have derived from dividing with 50. | • | This is maximum time taken by wave 1 to complete. Remember it can also take less time than max time. | • | If price reaches fib retracement till 50% in a given time period then also we can consider as wave 1(only if we get 5 moves)

HARDIK-130 p.15: Time Analysis-Wave 2 | ➤WAVE 2 time can be derived from wave 1. | ➤There are total 4 ways to analyst wave 2 time period. | 1. WAVE 1 / 4=WAVE 2 | 2. WAVE 1 / 2=WAVE 2 | 3. WAVE 1 =WAVE 2 | 4. WAVE 1*2=WAVE 2

HARDIK-130 p.16: Time Analysis-Wave 3 | ➤Wave 3 can be derived from wave 1 and wave 2. | ➤To measure wave 3 we have to add wave 1 and wave 2 | ➤Wave 1+ wave 2= wave 3 | ➤This is the minimum time period for holding position for wave 3. | ➤In this time period minimum price should reach near 100% of fb | extension. | ➤Price and time can go beyond minimum level.

HARDIK-130 p.17: Time Analysis-Wave 4 | ➤ Wave 4 time period is derived from wave 2. | ➤There are 3 ways to analyst time period of wave 4. | 1. Wave 2 / 2= Wave 4 | 2. Wave 2 *2= Wave 4 | 3. Wave 2 *5= Wave 4 | ➤Note wave 4 can never be equal to wave 2 in terms of time | period(price can be equal). | ➤If wave 4 and wave 2 are equal then impulse wave counting is wrong | or we can say whole move is correction part.

HARDIK-130 p.18: Time Analysis-Wave 5 | ➤Wave 5 time period can be derived from wave 1 and wave 4. | ➤Mostly wave 5 take time period equal to wave 1 i.e. wave 5=wave 1. | ➤In rare case it take time period of wave 4.

HARDIK-130 p.19: Time analysis rules | >Each and every wave should meet time guidelines. | >Variation in time analysis is allowed till +8bar or -8bar not more than | that. | >If the time analysis condition does not meet after variation, then | consider that impulse counting wrong and it may consider as correction part.

HARDIK-130 p.20: EXTENSION | • Mostly extension occurs in 3rd wave. | • But there is possibility of extension of wave 1 and wave 5 also. | • Extension can occur in +4 formation. i.e. $5+4=9$, $9+4=13$, $13+4=17$, $17+4=21$. | • Maximum there are 21 moves in extension. | • If in extension you are getting 9 moves then next corrective wave can | retrace till 61.8% to 81.20%. | • If in extension you are getting more than 9 moves then next | corrective wave can retrace max till 38.2% and hardly till 50%.

HARDIK-130 p.21: 5 moves

HARDIK-130 p.22: 9 moves

HARDIK-130 p.23: 13 moves

HARDIK-130 p.24: 17 moves

HARDIK-130 p.25: 21 moves

HARDIK-130 p.26: TYPE OF EXTENSION | 1. MICROSCOPIC EXTENSION | 2. MEGAPHONE EXTENSION

HARDIK-130 p.27: MICROSCOPIC EXTENSION | • In microscopic extension one cannot see wave 1 and wave 2 with eyes | easily. | • If it is microscopic then wave 1 and wave 2 candle together counting | should not be more than 13 candles. | • In microscopic extension wave 3 will always extended. | • Wave 4 can be extended till 38.2%. | • When degree 1 is extended at that time degree 2 will retrace | maximum till 38.2% mostly.

HARDIK-130 p.28: *Visual example / diagram page. No readable text extracted; original page is preserved unchanged in the Master PDF appendix.*

HARDIK-130 p.29: MEGAPHONE EXTENSION | • In megaphone extension wave 1 and wave 3 will remain almost of | similar size. | • If this condition is satisfied than wave 5 will go on in extension form. | • There is two possibilities in wave 5. | 1. If wave 5 crosses 127% level and started trading this level then | wave 5 can be as long as 2.5 times of wave 1 and three. | 2. In second condition wave 5 will remain half of wave 1 and 3.

HARDIK-130 p.30: *Visual example / diagram page. No readable text extracted; original page is preserved unchanged in the Master PDF appendix.*

HARDIK-130 p.31: TERMINAL IMPULSE(DIAGONAL) | • There are two types of terminal impulse(diagonal) | 1. | Leading diagonal(5-3-5-3-5). | 2. | Ending diagonal.(3-3-3-3-3). | • Ending diagonal occurs in wave 5 and wave C. | • Leading diagonal occurs in wave1. | • Diagonal is formed when third wave is completed but fail to reach 161.8% target. | • Wave 3 must reach minimum till 61.8% of fib extension. | • Remember if 4th wave doesn't enter in territory of wave 1 then also if wave 3 is less than 161.8% | then also it is called terminal case and wave 2 will fall either $\frac{1}{2}$ or $\frac{1}{4}$. Time period.

HARDIK-130 p.32: LEADING DIAGONAL(5-3-5-3-5) | • Leading diagonal occurs in wave1. | • Leading diagonal forms wedge pattern. | • If in terminal case if wave 4 enters in wave 1 territory(waves of degree | 1), then degree 2 correction will maximum take half time or $\frac{1}{4}$ of | wave 1 and target of wave 2 will be either 61.8% or 81.2% level. | • After completion of degree 2, degree 3 moment will also move fast.

HARDIK-130 p.33: LEADING DIAGONAL | CASE | CHART OF FEDERAL BANK. | DAILY TIME FRAME. | • | It will recover in half time of formation of leading diagonal.

HARDIK-130 p.34: LEADING | DIAGONAL(1ST WAVE | EXTENSION) | • | When 1st wave is in extension then leading diagonal may form. | • | If it forms leading diagonal than wave 3 formation will be very slow. | • | The length of wave 3 and wave 5 will be almost equal here.

HARDIK-130 p.35: LEADING | DIAGONAL(5TH WAVE | EXTENSION) | • | When wave 1 and three are of almost same size at that time 5th wave will be extended.

HARDIK-130 p.36: *Visual example / diagram page. No readable text extracted; original page is preserved unchanged in the Master PDF appendix.*

HARDIK-130 p.37: ENDING DIAGONAL (3-3-3-3-3) | • Ending diagonal occurs in wave 5 and wave C. | • When wave 3 is in extension and wave 4 is in speed than wave 5 will form ending diagonal. | • Ending diagonal can also occur in normal case also. | • After completion of ending diagonal, the correction will occur will be fast. | • Time taken for completion will be either $\frac{1}{4}$ or $\frac{1}{2}$ of wave 5(ending diagonal) | • The target of correction will either wave 4 or wave 2 for both fall time value will remain same. | • If the price crosses wave 4 and time period i.e. $\frac{1}{4}$ or $\frac{1}{2}$ is ended than it is hard for price to come to stable | form. It will take long time to become stable. | • Note if the correction is taking more time than $\frac{1}{4}$ or $\frac{1}{2}$ of wave 5 and doesn't reach to wave 4 (target) and if | we are getting reversal from a particular point then consider it as golden opportunity for long.

HARDIK-130 p.38: ENDING DIAGONAL | FORMATION RULE | • When wave 3 is in | extension and still doesn't able to reach | 161.8% and also wave | 4 is in speed than wave 5 will form ending diagonal.

HARDIK-130 p.39: TRENDLINES | • There are three types of trendline namely:- | 1. BASE TRENDLINE-Drawn between 0 point to 2 point and extended. | 2. ACTION TRENDLINE-Base trendline is copy paste and then put to | wave 1 is called action line. | 3. MATURITY TRENDLINE-This trendline is drawn between wave 2 and | wave 4.

HARDIK-130 p.40: BASE TRENDLINE

HARDIK-130 p.41: ACTION TRENDLINE

HARDIK-130 p.42: MATURITY TRENDLINE

HARDIK-130 p.43: TRENDLINE | • If the price is moving above action trendline that means 3rd wave is | very strong. | • If the price is repeatedly touching and crossing base trendline it | means third wave is weak and may form terminal impulse. | • After completion of 5 waves if trend is reversing than calculate time | taken to form 5th wave and divide that time by 2 i.e. half time period. If the price crosses Maturity trendline in half time period of wave 5 then it is confirmed that trend is reversed. | • If it is not crossing maturity trendline in half time period than wait for | time period equal to formation of wave 5th.

HARDIK-130 p.44: TRENDLINE | • | The price must touch to wave 4 in equal time period then we can consider it as pattern confirmation. And we can say price trend is changed. | • | If the price doesn't touch 4th wave in equal time than the price moment will be consider as subordinate of wave 5th.

HARDIK-130 p.45: TRENDLINE | • | When pattern is confirmed or maturity trendline is broken than base trendline can act as resistance.

HARDIK-130 p.46: TRENDLINE | • When we have drawn maturity trendline and the if maturity trendline | is continually touched by subordinates of wave 5th then it is sign of weak wave and can form ending diagonal.

HARDIK-130 p.47: Visual example / diagram page. No readable text extracted; original page is preserved unchanged in the Master PDF appendix.

HARDIK-130 p.48: 4TH WAVE | COMPLETION SIGNAL | • If the price started giving | closing above base line, then we can consider it as completion of wave 4th.

HARDIK-130 p.49: CORRECTION | ELLIOT WAVE THEORY

HARDIK-130 p.50: CORRECTION NOTES | • Wave A tells trend reversal. | • Wave B(correction) take maturity trendline as resistance. | • Wave C gives pattern confirmation. | • ZIG ZAG and FLAT correction are only two correction in which after | completion of correction another correction may start. | • If there is triangle correction than after completion new impulse will | start because triangle is last correction.

HARDIK-130 p.51: CORRECTION TYPE | • There are three type of correction | 1. ZIG ZAG CORRECTION-PRICE WISE | 2. FLAT CORRECTION-TIME WISE | 3. TRIANGLE CORRECTION-TIME WISE

HARDIK-130 p.52: ZIG ZAG CORRECTION(5-3-5/A-B-C) | IN UPWORD TREND | IN DOWNWORD TREND

HARDIK-130 p.53: ZIG ZAG CORRECTION-A WAVE(IMPULSE) | • Zig zag single leg can go min 61.8% i.e. (A). | • Zero point of wave A i.e. is starting of zig zag correction must touched | to 127% if it is less than it is not zig zag correction. | • Wave A consist of 5 moves. | • Wave A can form | 1. Leading Diagonal. | 2. Extension.-(maximum 13 moves). | • Time of wave A can be measured from 4 to 5 leg i.e. it can be $\frac{1}{4}$ or $\frac{1}{2}$ | of that leg.

HARDIK-130 p.54: ZIG ZAG CORRECTION-B WAVE(CORRECTION) | • Wave B can maximum go till 61.8%. | • Time taken by wave B can be measured from wave A it can be | 1. | WAVE A= WAVE B | 2. | WAVE A*2=WAVE B | 3. | WAVE A*5=WAVE B | 4. | WAVE A*10=WAVE B(REAR CASE). | • Wave B can be | 1. | ZIG ZAG PATTERN | 2. | FLAT PATTERN | 3. | TRIANGLE PATTERN | 4. | DOUBLE AND TRIPLE CORRECTION. | • Wave B cannot form without this pattern.

HARDIK-130 p.55: Zig zag B wave | • | Wave B can maximum go till 61.8%.

HARDIK-130 p.56: ZIG ZAG CORRECTION-C WAVE(IMPULSE) | • To trade wave c protective stoploss will be 81.2%. | • Wave C can form | 1. | ENDING DIAGONAL. | 2. | EXTENSION(MAX 13 MOVES). | • Time taken by wave C can be measured from wave A and wave B i.e. | 1. | WAVE C= WAVE A + WAVE B. | 2. | WAVE C=(WAVE A + WAVE B)/4. | 3. | WAVE C=(WAVE A + WAVE B)/2. | • > NOTE:-Wave C can complete in single move also.

HARDIK-130 p.57: ZIG ZAG TYPES | 1. NORMAL | 2. ELONGATED | 3. TRUNCATED(FAILURE).

HARDIK-130 p.58: ZIG ZAG(NORMAL) | • | During normal zig zag wave c minimum target is 61.8% and max and maximum target is 161.8%.

HARDIK-130 p.59: ZIG ZAG (ELONGATED) | • When wave C | travels till | 261.8% than it is | called elongated zig zag.

HARDIK-130 p.60: ZIG ZAG(TRUNCATED) | • | When wave C fails to reach 61.8% then it is called truncated or failure zig zag. | • | After the formation , failure, when the price revering before 61.8% and it reach 81.2% then - it will consider as Failure C zig - Zag setup.

HARDIK-130 p.61: Visual example / diagram page. No readable text extracted; original page is preserved unchanged in the Master PDF appendix.

HARDIK-130 p.62: ZIG ZAG CORRECTION | COMPLETION | 1. | Here we have to apply Dow theory , if lower high is broken than it is consider as correction is over. This rule is only applicable on zig zag correction.

HARDIK-130 p.63: ZIG ZAG CORRECTION | COMPLETION 2nd way | • | Second way to check completion of zig zag correction is time value. | • | Max time to form wave C is wave A + wave B(note:- It may take little bit more time also). | • | Now mark the high of candle which is taking half time of wave A +wave B. | • | If the price give closing above that candle than consider correction is over and new impulse may start.

HARDIK-130 p.64: ZIG-ZAG CORRECTION NOTES | • > There is no need of trendline in zig zag correction. | • >This is because wave b takes 5 to 10 time of wave A so it form | irregular trendline. | • >If zig zag correction is formed than there is high probability that | another correction will form. | • >You can take +8,-8 candle variation when you are counting time | value.

HARDIK-130 p.65: FLAT CORRECTION(3-3-5/A-B-C) | • In flat correction all max value of zig zag correction will become | minimum for flat correction. | • Flat correction is time wise correction. | • Flat correction behaves like impulse. | • As it is time wise correction it will take high time in formation. | • Market often form this correction.

HARDIK-130 p.66: Visual example / diagram page. No readable text extracted; original page is preserved unchanged in the Master PDF appendix.

HARDIK-130 p.67: FLAT CORRECTION –WAVE A(3 MOVES) | • Flat correction wave A can max go till 38.2%. | • If price reversed from 38.2% then we got confirmation i.e. correction | is flat correction. | • Wave A consist of 3 moves. | • Wave A can be formed in single move also. | • Wave A can form Flat , Zig zag correction. | • Time of wave A can be measured from 4 to 5 leg i.e. it can be $\frac{1}{4}$ or $\frac{1}{2}$ | of that leg.

HARDIK-130 p.68: FLAT CORRECTION –WAVE B(3 MOVES) | • Wave B consist of 3 moves. | • Wave B can form Flat , Zig zag correction. | • TIME VALUE | 1. WAVE A= WAVE B | 2. WAVE A*2=WAVE B | 3. WAVE A*5=WAVE B | 4. WAVE A*10=WAVE B(REAR CASE).

HARDIK-130 p.69: FLAT CORRECTION – | WAVE B(3 MOVES) | • | Wave B can min travel till 61.8%. | • | Wave B can max travel till 111%. | • | If wave B travels to 61.8% to 81.2% then it is called NORMAL B correction. | • | If wave B travels to 100% to 111% then it is called IRREGULAR B or STRONG B. | • | According to behavior of wave B the nature of wave C is decided.

HARDIK-130 p.70: FLAT CORRECTION –WAVE C(5 MOVES) | • Wave C consist of 5 moves. | • Mostly wave C form ending diagonal. | • Wave C target can be | 1. | Minimum-61.8%. | 2. | Intermediate-100%(Rule of equality). | 3. | Normal – 61.8% to 161.80%. | 4. | Ellongated-261.80%. | • TIME VALUE | 1. | WAVE C= WAVE A + WAVE B(note:-when a and b both have same time). | 2. | WAVE C=(WAVE A + WAVE B)/4. | 3. | WAVE C=(WAVE A + WAVE B)/2. | • >Note:- When wave A and Wave B both have same time period then we can only use (A+B)/4=C AND A+B=C only. | • >Wave C can complete in single move also.

HARDIK-130 p.71: FLAT CORRECTION NOTES | • You can take +8,-8 candle variation when you are counting time value.

HARDIK-130 p.72: TYPES OF FLAT CORRECTION | 1. NORMAL. | 2. ELONAGATED. | 3. TRUNCATED. | 4. STRONG B OR IRREGULAR. | 5. RUNNING.

HARDIK-130 p.73: NORMAL FLAT | CORRECTION | • If wave B is retracing till | 61.8 to 81.2 and wave C is | min going till 61.8% and max till 161.8% then it is called normal flat correction.

HARDIK-130 p.74: ELONGATED FLAT | CORRECTION | • If wave B is retracing till | 61.8 to 81.2 and when the | wave C cross more than | 161.8% and reach towards- | 261.8% then it is called - | Elongated (Rare..)

HARDIK-130 p.75: TRUNCATED FLAT | CORRECTION. | • When the wave C unable to | reach 61.8%then it's called | - Truncated. | • If the price touches 81.2% | then we can confirmation of this pattern. | • This pattern is common in | market.

HARDIK-130 p.76: IRREGULAR FLAT | CORRECTION | • | When wave B reaches till 100% to 111% then it is called irregular flat correction. | • | Irregular Flat Correction can move maximum 161.8%and this is the accumulative wave. | • | Irregular flat correction never touches | 261.8%.

HARDIK-130 p.77: RUNNING FLAT | CORRECTION. | • After irregular flat | correction if price cross | 161.8% level and unable to | touch 261.8% then it is called running flat correction.

HARDIK-130 p.78: RUNNING FLAT | CORRECTION. | • | Draw a trendline from 0 point then make a clone and put to point A. | • | The length of wave c after crossing trendline of wave A will be equal to X which is shown in figure.

HARDIK-130 p.79: TRADING FLAT CORRECTION RULES. | • When A=B it confirms flat correction. | • Then A/2 gives selling confirmation. | • Then A/4 gives us destination.

HARDIK-130 p.80: TRIANGLE CORRECTION. | -END OF CORRECTION

HARDIK-130 p.81: TRIANGLE CORRECTION. | • After triangle formation there will no next correction. | • After triangle correction there will be always impulse move. | • Triangle is always formed before any major move. | • Triangle is always form in wave B and wave 4 only. | • After formation of triangle next move will be extremely fast. | • Occasionally triangle is also formed in wave Z and wave Y also. | • Sometimes after double or triple correction also triangle is formed. | • Volume and volatility both will decrees till wave E.

HARDIK-130 p.82: TRIANGLE PATTERNS | 1 3 2 Here 2nd figure is not triangle. | CONTRACTING TRIANGLE | (very commonly found) | EXPANDING TRIANGLE

HARDIK-130 p.83: GENERAL GUIDELINES FOR | TRIANGLE | 1. | It should be in 5 leg structure. | 2. | It should have max 5 touch points. There will be never more than 5 touch point. | 3. | It is not necessary that wave E will touch the trendline. It may also break trendline but it will not cross the high of wave C .So, it means 4 touch points are compulsory. | 4. | Trendline should always drawn from wave A to wave C and sometimes from wave A to wave E and second trendline is drawn from wave B to Wave D. Extend a trendline till it intersects. The point were trendline intersects is called APEX. | 5. | The apex should be between 61.8% of widest leg.(draw fib retracement low to high). It should not extend than 61.8%. Otherwise it is not triangle. | 6. | Three waves of triangle should be retrace 50% of previous leg. | 7. | Thrust value(Target of triangle) of triangle. (75%, 100% AND 125%). | 8. | In triangle subordinate wave E can form triangle. | 9. | When BD trendline is broken than we get confirmation of completion of triangle pattern.

HARDIK-130 p.84: GENERAL GUIDELINES FOR | TRIANGLE | 10. All the 5 legs are corrective in nature.

HARDIK-130 p.85: THRUST VALUE VIDEO | A TO 0 THEN E

HARDIK-130 p.86: *Visual example / diagram page. No readable text extracted; original page is preserved unchanged in the Master PDF appendix.*

HARDIK-130 p.87: 3 LEGS OF TRIANGLE VIDEO

HARDIK-130 p.88: *Visual example / diagram page. No readable text extracted; original page is preserved unchanged in the Master PDF appendix.*

HARDIK-130 p.89: TRIANGLE ILLUSTRATION

HARDIK-130 p.90: TRIANGLE EXCEPTION | • Wave B gives confirmation by its nature. | • If wave B travels 61.8% in very fast nature in comparison to wave A, | then there is strong possibility that it will create triangle. | • If the above condition is satisfied then every leg of wave A(of triangle) | must be in corrective in nature then and then only it will form triangle. | • If wave A is forming impulse with following rules of impulse than it | will not form triangle. | • This behavior of wave B is only observed in triangle formation.

HARDIK-130 p.91: *Visual example / diagram page. No readable text extracted; original page is preserved unchanged in the Master PDF appendix.*

HARDIK-130 p.92: CONTRACTING TRIANGLE | • In this triangle the legs will get shorter than previous leg. That's why it | is called contracting triangle. | • The range will also get contracted. | CONTRACTING TRIANGLE

HARDIK-130 p.93: TYPES OF CONTRACTING TRIANGLE | 1. HORIZONTAL CONTRACTING TRIANGLE. | • REGULAR TRIANGLE | • ASCENDING TRIANGLE | • DESCENDING TRIANGLE. | 2. IRREGULAR CONTRACTING TRIANGLE. | 3. RUNNING CONTRACTING TRIANGLE.

HARDIK-130 p.94: LOCATION | 1. Wave B of zig zag and flat(more chances) correction. | 2. Wave 4 for impulse. | 3. Regular horizontal mostly found in the end of complex correction. | Ascending and descending will never form at end of complex correction. | 4. If we find triangle in Y wave then there is very less possibility of | going for triple correction.

HARDIK-130 p.95: REQUIRMENTS TO IDENTIFY CONTRACTING TRIANGLES | 1. | There are 5 parts of triangle. | 2. | The triangle can be ascending and descending without violating rule. | 3. | The length of wave B is between 38.2% to 261.8% of wave A . Here wave B will decide whether triangle which is developing is horizontal or irregular. | 4. | Triangle will never respect dot to dot fib ratio. | 5. | Wave B will never respect 61.8% level of wave A. If it is respecting then it is not triangle. | 6. | If wave B respect 100% ratio, then it is forming rare triangle. | 7. | Between wave B-C-D-E three wave must return 50% of previous wave.

HARDIK-130 p.96: REQUIRMENTS TO IDENTIFY CONTRACTING | TRIANGLES | 8. For wave E there are 2 modes. | a) It ends at 61.8% level of previous wave. b) It breaks A-C trendline and then continue its path. | 9. Channeling of triangle | • In this 0 point is never used for channeling purpose. | • Trendline should be drawn from wave A to wave C and from wave B to wave | D.

HARDIK-130 p.97: CONFIRMATION | • After the formation of wave A and B wave is either going fast | according to time value and not respecting 61.8 level or taking time and retracing before 61.8% then it can form triangle.

HARDIK-130 p.98: *Visual example / diagram page. No readable text extracted; original page is preserved unchanged in the Master PDF appendix.*

HARDIK-130 p.99: CONTRACTING TRIANGLE- WAVE A | >Wave A is corrective in nature. Usually, it is in zig zag in nature. | >It can also in flat and complex correction. | >CHARACTERISTICS:- It sets width of the triangle | >Usually, it is longest or second longest leg. | >If it is horizontal pattern than first longest leg and if it is creating irregular pattern than it is second longest leg of triangle. | >PSYCHOLOGY:-It introduce uncertainty and starts balance between | buyers and sellers.

HARDIK-130 p.100: CONTRACTING TRIANGLE- WAVE B(DEEP CORRECTIVE REACTION) | >Wave B decides type of triangle which will be formed. | >Wave B is always corrective in nature. | >Often takes more time than wave A. | >Wave B can be zig-zag, flat and complex correction. | >PSYCHOLOGY:-It often fools traders that trend would continue.

HARDIK-130 p.101: CONTRACTING TRIANGLE- WAVE C (SECOND CONTRACTING LEG) | >Wave C is corrective in nature. | >Often it forms zig-zag correction, sometime it form complex | correction. | >TIME RULE OF WAVE C:- Wave A and wave C time should be never | equal. | >CHARACTERISTICS:-It defines upper boundary of contracting triangle. | >PSYCHOLOGY:-Market volatility will start reducing more. | >Emotional intensity of traders is also decreasing as compared to wave | A.

HARDIK-130 p.102: CONTRACTING TRIANGLE- WAVE D(NARROW AND CONTROLLED LEG) | >Wave D is corrective in nature. | >Often makes complex correction like double zig zag or expanded flat | correction. | >CHARACTERISTICS:- Volume will dry up | >Wave D often takes more time than wave C. | >Wave D will never equal to wave B in time value. | >In wave D range in tighten drastically. | >PSYCHOLOGY:- Traders think market is stuck and they are going to | loose interest.

HARDIK-130 p.103: CONTRACTING TRIANGLE- WAVE E(FINAL LEG) | >Wave E is in corrective in nature. | >It can be zig-zag correction or sharp single leg correction. | >In rare case it also form complex correction. | >Sometime in wave E again triangle is formed and it should be | contracting triangle. | >Only wave E will again form triangle. It will not form in A, B, C, D leg. | >PSYCHOLOGY:- Buyers and Sellers both are trapped. | >If buyer and seller both are not exiting position then and then for | trapping purpose wave E will form triangle.

HARDIK-130 p.104: IDENTIFYING TRIANGLE | TYPES

HARDIK-130 p.105: HORIZONTAL | CONTRACTING | TRIANGLE | 1. | Wave B is smaller than Wave A. Wave B will never cross 0 point of wave A. Min destination is 38.2% and will not cross 100%. | 2. | Wave C is smaller than wave B. It will not cross wave A 0 point. Min destination is | 38.2% and will not cross 100%. | 3. | Wave D is smaller than wave C. It will not cross wave C 0 point. Min destination is | 38.2% and will not cross 100%. | 4. | After formation of wave D draw trendline from B to D and A to C. | 5. | Wave E can may or may not touch, it can cross the trendline but it will not break wave C 0 point. Min distance to travel is | 61.8% | E

HARDIK-130 p.106: IRREGULAR | CONTRACTING | TRIANGLE | 1. | Wave B is larger than Wave A. | 2. | The length of Wave B should not be more than 261.8% of Wave A and usually it retraces below from 161.8%. | 3. | Wave C is smaller than wave B. It will not cross wave A 0 point. Min destination is | 38.2% and will not cross 100%. | 4. | Wave D is smaller than wave C. It will not cross wave C 0 point. Min destination is | 38.2% and will not cross 100%. | 5. | After formation of wave D draw trendline from B to D and A to C. | 6. | Wave E can may or may not touch, it can cross the trendline but it will not break wave C 0 point. Min distance to travel is | 61.8% | E

HARDIK-130 p.107: RUNNING | CONTRACTING | TRIANGLE. | 1. | Wave B is longer than Wave A. | 2. | Wave C is smaller than Wave B. | 3. | Wave D is larger than Wave C around 111%. | 4. | Here in running triangle apex is not present.

HARDIK-130 p.108: EXPANDING TRIANGLE

HARDIK-130 p.109: GUIDELINES FOR IDENTIFYING EXPANDING TRIANGLE | 1. The wave A or wave B should be the smallest wave of triangle. | 2. The wave E should be the longest wave of triangle. | 3. Location of expanding triangle are | 1. | Flat correction wave B | 2. | Wave 4 | 3. | Wave Y | 4. | Wave Z(mostly contracting triangle rarely expanding triangle.) | 4. It cannot form in wave B of zig-zag correction. | 5. The Wave E usually breaks AC trendline | 6. The thrust value is 61.8% of wave E length. | 7. In this we use fib retracement from 0 to E to measure thrust value.

HARDIK-130 p.110: TYPES OF EXPANDING | TRIANGLE | 1. HORIZONTAL EXPANDING TRIANGLE. | 2. IRREGULAR EXPANDING TRIANGLE | 3. RUNNING EXPANDING TRIANGLE

HARDIK-130 p.111: HORIZONTAL | EXPANDING TRIANGLE | 1. | Wave A should be the smallest wave of triangle. | 2. | Remaining all legs must exceed previous leg. | 3. | Each leg must be in corrective in nature. | 4. | There is no exact length of waves. | 5. | Most of the time it seen that Wave B travels 161.8% of wave A and wave D travels 100% to 111% of wave C. | 6. | The wave E will probably break beyond the trendline AC. | 7. | Thrust value is 61.8% of wave E. | D

HARDIK-130 p.112: IRREGULAR | EXPANDING TRIANGLE | 1. | In this triangle wave B will be the smallest wave of triangle. | 2. | Remaining all legs must exceed previous leg. | 3. | Each leg must be in corrective in nature. | 4. | There is no exact length of waves. | 5. | Most of the time it seen that Wave B travels 161.8% of wave A and wave D travels 100% to 111% of wave C. | 6. | The wave E will probably break beyond the trendline AC. | 7. | Thrust value is 61.8% of wave E.

HARDIK-130 p.113: RUNNING EXPANDING | TRIANGLE | 1. | In this triangle wave B is slightly larger than wave A and wave D is slightly smaller than wave C. | 2. | Each leg must be in corrective in nature. | 3. | There is no exact length of waves. | 4. | Most of the time it seen that Wave B travels 161.8% of wave A and wave D travels 100% to 111% of wave C. | 5. | The wave E will probably break beyond the trendline AC. | 6. | Thrust value is 61.8% of wave E.

HARDIK-130 p.114: EXPANDING TRIANGLE NOTES | • In expanding triangle Wave B and Wave D will behave similar. | • They both will form same type of correction and both will take | almost same time period.

HARDIK-130 p.115: DOUBLE CORRECTION | -Yeh dukh kyu nahi khatam hota be

HARDIK-130 p.116: DOUBLE CORRECTION | • It occurs only in flat and zig zag correction only. | • After completion of ABC wave it can be flat or zigzag pattern which is | marked as W wave and than new leg will form which is called X. | • It can be measured in two formula i.e. | 1. | $W(A+B+C)/4=X$ | 2. | $W(A+B+C)/2=X$ | • The distance travelled by wave X is till 0 to wave B trendline. | • Wave X is also known as Any pattern, Connecting Wave, Missing Wave. | • Wave X will be always in speed and looks like impulse. | • Wave X can form flat, zig-zag and double correction.

HARDIK-130 p.117: DOUBLE CORRECTION | • After completion of wave X next move will be again correction i.e. | ABC and it is named as wave Y. | • Wave Y will be always cross wave W i.e. it will be always below Wave | W. | • Wave Y can form flat, zig-zag, and triangle pattern also.

HARDIK-130 p.118: DOUBLE CORRECTION

HARDIK-130 p.119: COUNTING MOVES

HARDIK-130 p.120: TYPES OF DOUBLE CORRECTION PATTERNS | 1. DOUBLE ZIG-ZAG. | 2. ZIG-ZAG + FLAT. | 3. FLAT + ZIG-ZAG. | 4. DOUBLE FLAT. | 5. ZIG-ZAG + TRIANGLE | 6. IF THERE IS NO PATTERN BUT HAVING 7 MOVES CONSIDER IT AS | DOUBLE CORRECTION.

HARDIK-130 p.121: TYPES OF X | 1. Small X(retraces till 50% ratio.) | 2. Large X(retraces min 61.8% and max till 111%)

HARDIK-130 p.122: DOUBLE ZIG-ZAG. | > | If in Double Correction if first correction is zig-zag i.e. W than it will form small Wave X. | > | Wave X will max retrace till 50% of wave W. | > | Here Wave B will act like an invalidation point. | > | If Y again form zig zag pattern, then it is called DOUBLE ZIG-ZAG. | > | If Wave W and Wave Y makes same pattern then price and time will act in same manner as in Wave W when forming Wave Y. | > | Here 0 to Wave B trendline is drawn to give support to Wave X | > | And trendline from Wave B to Wave X is drawn to give support to wave Y. and now copy this trendline and put it to wave W. | > | After Wave Y small X will become invalidation point. | > | Now put fib extension 0 to W then at X. | > | Wave Y will minimum travel till 61.8%. | > | Wave Y will always form below Wave W. | > | Wave Y cannot go below 261.8%.

HARDIK-130 p.123: ZIG-ZAG + FLAT. | > Here 1st pattern will be zig-zag pattern i.e. Wave W. | > After than Wave X will form. | > Now if Wave X crosses invalidation point i.e. Wave B of zig-zag in $\frac{1}{4}$ | time period or equal to $\frac{1}{4}$ then next correction will be Flat Correction. | > Wave Y will travel slightly lower to Wave W here and can make | > And after the double correction is over in this pattern new impulse | will be very fast.

HARDIK-130 p.124: ZIG-ZAG + FLAT PATTERN

HARDIK-130 p.125: FLAT + ZIG-ZAG | > In this 1st correction will be flat correction i.e. is wave W. | > It can form all type of flat pattern. | > After flat correction max time it will form large X. | > Next correction will be zig-zag correction i.e. Wave Y and goal is to | close below Wave W. | > Note in this case Wave A of Y will be longer than Wave C of Y.

HARDIK-130 p.126: Visual example / diagram page. No readable text extracted; original page is preserved unchanged in the Master PDF appendix.

HARDIK-130 p.127: DOUBLE FLAT PATTERN. | > Double flat pattern is often seen in market. | > In flat correction invalidation point is not present. | > In this pattern Wave X is most often larger and lead towards Wave B | of Wave W. | > The time period taken by Wave X will $\frac{1}{4}$, $\frac{1}{2}$, $\frac{1}{3}$ of Wave W. | > After formation of Wave X if flat pattern is repeated than it is called | DOUBLE FLAT CORRECTION.

HARDIK-130 p.128: Visual example / diagram page. No readable text extracted; original page is preserved unchanged in the Master PDF appendix.

HARDIK-130 p.129: ZIG-ZAG + TRIANGLE | > In this pattern first pattern will be Zig-zag. | > Here Wave X will be small X max retraces till 50%. | > After Wave X triangle pattern will form. | > If Wave Y is forming triangle than either Wave A or Wave E of triangle | should come below Wave W then and then it is valid. | > In this pattern Wave Y will not touch 161.8%. | > It can touch max 100% ratio.

HARDIK-130 p.130: Visual example / diagram page. No readable text extracted; original page is preserved unchanged in the Master PDF appendix.

CHARTKING-112: EW Notes _ Chartking v2 - 2025-04-17 11-29-14(9).pdf

CHARTKING-112 p.1: ELLOITT WAVE NOTES

CHARTKING-112 p.2: • Impulse | • Wave-1 | • Wave-2 | • Wave-3 | • Wave-4 | • Wave-5 | • Leading Diagonal | • Ending Diagonal | • Extension | • Correction | • Zig-Zag Correction | • Flat Correction | • Wave-A | • Wave-B | • Wave-C | • Double Correction | • Wave-W | • Wave-X | • Wave-Y | • Triple Correction | • Wave-XX | • Wave-Z | • Triangle Correction | • Supporting | • MACD | • Gaps | • Candlestick patterns | • Trendline | • Channel | • Time period

CHARTKING-112 p.3: Basics Daily time frame is prime time frame Use Daily timeframe for CMP: >300 and Weekly timeframe for CMP < 300 To identify starting point of EW in Daily timeframe: | • MACD lowest point in Daily timeframe | • Positive divergence | • Max 144 days period on chart to check Imp High and Imp Low | Impulse should be rule based; correction should be guidelines based Degrees: | • Major – Daily timeframe (defines trend) | • Intermediate – 75min (action zone for trading) | • Minor – 15min (for practice only) | Most IMP concepts: | • Leading Diagonal, Ending Diagonal, Flat correction, Double and Triple Correction

CHARTKING-112 p.4: Wave-1 | MACD | Leading Diagonal | • Impulsive | Nature | • 5 (1-2-3-4-5) | Sub-moves | • 5-3-5-3-5 | Internal count | • MACD lowest low or positive divergence | MACD | • Min 61.8% retracement of Imp High and Imp low forms a | degree wave Retracement (from previous Imp High – Imp Low) | • Subordinate of Wave-1 can form a Leading Diagonal | Special feature

CHARTKING-112 p.5: Examples Development of the wave once price closes above | 61.8%

CHARTKING-112 p.6: MACD | Development of Wave-1

CHARTKING-112 p.7: Wave-2 Flat Zig-Zag | • Corrective | Nature | • 3 (A-B-C OR W-X-Y) | Sub-moves | • Zig-Zag (price-wise) or Flat (time-wise) Correction and sometimes Single leg correction | Type | • MACD falls slowly, mostly forms a positive divergence | MACD | • Min 23.6%, Max 81.2% Fib retracement of Wave-1, but cannot stay below 81.2% for long time | Retracement (0-1) | • Cannot take more time than Time (Wave-1) | Time | • Trade should be taken after the formation of sub-wave C with SL of the base point | Entry | • Aggressive trade at retracement Fib levels, safe trade after breakout of Wave-1 | Trade | • When 0-B TL is broken in Flat or Zig-Zag correction | Reversal | • If microscopic (Min 14% Fib retracement); then Wave-3 will be extended | Special feature:1 | • Wave-2 and Wave-4 have similar %fall in price (+/-5%) | Rule of overlapping | • Wave-2 is price-wise and Wave-4 is time-wise correction. They will have different time periods. | Rule of alternation

CHARTKING-112 p.8:

CHARTKING-112 p.9: Wave-3 extension terminal gap | • Impulsive | Nature | • 5 (1-2-3-4-5) | Sub-moves | • 5-3-5-3-5 (Max 21 moves if Wave-3 is in extension) | Internal count | • MACD will be highest | MACD | • Degree of Wave-3 can go till Min 161.8%, Max 2700% Fib extension of 0-1-2 | Fib Extension (0-1-2) | • If W1 not equal to W2; $W3 = [W1+W2]/2$ | Time:1 | • If $W1 = W2$ (+/- 5); then $W3 = W1 + W2$ | Time:2 | • Subordinate Wave-3 can be terminal with Min 61.8% extension of 0-1-2 | Terminal | • If price breaks Wave-1 high through a breakaway gap, it should not fill the gap for next 4-5 days | Breakaway Gap

CHARTKING-112 p.10: Microscopic structure of Waves 1 and 2

CHARTKING-112 p.11:

CHARTKING-112 p.12: Wave-4 | • Might form Complex Correction | Nature | • Flat, Zig-Zag, Triangle, Double and Triple Correction patterns | Types | • A-B-C, W-X-Y, W-X-Y-X-Z; A-B-C-D-E | Internal count | • MACD might be at its lowest low (price-wise correction) | MACD | • Impulse - from Base Point to W3 with Min 23.6% and Max 50%; in case of terminal impulse 61.8% | Retracement | • Can take ½, 2X or 3X time of Wave-2 (Cannot take same time as that of Wave-2) | Time | • It can never go lower than Wave-2 | Feature | • Should have (+/-5%) price fall similar to Wave-2 | Rule of overlapping | • Shorting opportunity at sub-Wave2 (X or B) | Trade:1 | • Long opportunity when correction ends | Trade:2

CHARTKING-112 p.13:

CHARTKING-112 p.14: Wave-5 Ending Diagonal Ending Diagonal | • Impulsive | Nature | • 5 (1-2-3-4-5) | Sub-moves | • 5-3-5-3-5 OR (3-3-3-3-3 for Ending Diagonal) | Internal count | • MACD lower than Wave-3's MACD, negative divergence | MACD | • Min 127%, Max 261.8% Fib retracement of Wave-4 (3-4) | Retracement | • Subordinates of Wave-5 can form an Ending Diagonal | Feature | • At 127% book profits or use a trailing SL | Trade

CHARTKING-112 p.15:

CHARTKING-112 p.16:

CHARTKING-112 p.17: Terminal Impulse Trending Impulse Terminal Impulse Wave-3 Fib Extension > 161.8% < 161.8% Wave-4 can enter Wave-2's territory | NO | YES

CHARTKING-112 p.18: Leading Diagonal (5-3-5-3-5) Terminal impulse | •Terminal impulse | Nature | •Sub-waves (1,3,5) should have 5 | moves 1-2-3-4-5, all are slow Sub-moves | •Within Wave-1 and Wave-A (Zig-Zag | correction) Occurrence | •Wave-4 can retrace Min 23.2% and | Max till 61.8% of 0-3 Retracement | •Wave-4 may enter Wave-1's territory | and will not go below end of Wave-2 Feature-1 | •Wave-3 Fib extension will be terminal | Min 61.8% and Max <161.8% Feature-2 | •Wave-3 will be terminal and is slow in | nature Feature-4 | • Wave-5 is Min 161.8% and Max 261.8%, Wave- | 5 and Wave-1 may be equal Feature-5 | • Subordinate W2 = ¼ or ½ Subordinate W1 | Time : W2 | • If W1 not equal to W2; then W3 = [W1+ W2]/2 | • If W1 = W2 [+/- 5]; then W3 = W1 + W2 | Time:W3 | • Wave-2 (degree) after the entire leading | diagonal should take ¼X, ½X or X Next pattern | • Wave-3 (degree) after Wave-1 is an LD, would | mostly be extended Wave-3 | • Fall of Wave-2 (degree) after completion of the | leading diagonal can go till sub-Wave-4, sub- Wave-2 or Max till 81.2% Special feature

CHARTKING-112 p.19:

CHARTKING-112 p.20:

CHARTKING-112 p.21: Leading Diagonal

CHARTKING-112 p.22: Ending Diagonal (3-3-3-3-3) Terminal impulse | • Terminal impulse | Nature | • 3-3-3-3-3 (Triple Zig-Zag) | Sub-moves | • Wave-5 and Wave-C (If Wave-3 is in extension, | Wave-4 is a sharp move, then Wave-5 will be an | ED) | Occurrence | • Wave-4 can retrace Min 23.2% and Max till 61.8% | of 0-3 Retracement | • Wave-4 enters Wave-1's territory and will not | break Wave-2 low Feature-1 | • Wave-3 Fib extension may be terminal Min 61.8% | and Max <161.8% Feature-2 | • Wave-2 and 4 won't retrace more than 50% | Feature-3 | • Wave-3 is slow in speed | Feature-4 | • Wave-5 is Min 127% and Max 261.8%, Wave-5 | and Wave-1 may be equal in length Feature-5 | • MACD forms negative divergence from Wave-3 | (degree) | MACD | • Subordinate W2 = ¼ or ½ Subordinate W1 | Time : W2 | • If W1 not equal to W2; then W3 = [W1+ W2]/2 | • If W1 = W2 [+/- 5]; then W3 = W1 + W2 | Time:W3 | • Correction after the entire ending diagonal | should take 1/4th or ½ its time Pattern End | • LD and ED will not form in the same impulse | move (If W1 = LD, W5 = ED) Criteria

CHARTKING-112 p.23: Examples Ending Diagonal

CHARTKING-112 p.24:

CHARTKING-112 p.25: Wave extension No. of moves based on Lucas numbers -> 5,9,13,17,21 – Max 21 moves When Wave-1 and Wave-3 are in extension: (double extension) (not in equity, only possible in commodity and forex) | • Wave-5 is truncated – it can reach Max till 81.2% retracement of 3-4 (it will still have 5 sub-moves) | When Wave-1 is in Extension: (entire move forms a LD) | • Very slow speed time-wise and price-wise | • Wave-1 crosses 61.8% retracement of Imp High and Imp Low and when it reaches 127% it can be said to | be in extension | • Wave-2 can retrace Max till 38.2% and will be in complex correction | • Wave-3 will be terminal (<161.8%) and will go very slow (in the nature of sub-division) | • Wave-4 will be sharp (flat or zig-zag) | • Wave-5 will reach 127% or maybe equal to Wave-3

CHARTKING-112 p.26: Wave extension rules When Wave-1 is in extension: (entire move forms a Leading Diagonal) | • Wave-2 will fall Max till 38.2% and will be | complex (double/ triple) and will never cross 50%, | • Wave-3 will be slow and would not touch | 161.8% (terminal) | • Wave-4 fall will be sharp and will take | place in a single correction (Zig-zag or Flat), will retrace Max till 50% of the entire move | • Wave-5 retraces till 127% of 3-4, and will | be equal to Wave-3 | • Wedge pattern will be formed in the entire | move

CHARTKING-112 p.27:

CHARTKING-112 p.28: Wave extension rules When Wave-3 is in extension: Wave-5 will form an Ending Diagonal | • Wave-1 and Wave-2 are microscopic | • Wave-4 will be 10X the size of Wave- | 2 | • Wave-4 fall will be complex (triangle, | double and triple correction) or sharp and will retrace Max till 38.2% | • Wave-5 will be slow, and will remain | in the channel drawn over wave-1 and wave-2 | • Wave-5 forms an Ending Diagonal | In Microscopic: what is the Max retracement of Wave-2? Mostly found Wave-2 as microscopic, how to filter?

CHARTKING-112 p.29:

CHARTKING-112 p.30: Wave extension rules When Wave-5 is in extension: (not possible in Indices, only in Stocks) | • Formed when Wave-1 and Wave-3 are both | equal time-wise | • Wave-5 does not go into extension of Index | in any timeframe | • Wave-5 does not cross 261.8% level of | extension drawn over 0-3-4 | • Wave-2 fall will be sharp in a single move | (Flat or Zig-zag). | • Wave-3 can go till 61.8% - 100% | • Wave-4 will be complex – double and triple | correction or Triangle | • It forms a megaphone shape. The | correction that follows will never cross the trendline connecting 2-4

CHARTKING-112 p.31:

CHARTKING-112 p.32: Wave extension nature | •Very slow speed time-wise and price-wise | •Wave-1 crosses 61.8% retracement of Imp High and Imp Low and when it reaches 127% it can be said to be in | extension | •Wave-2 can retrace Max till 38.2% and will be in complex correction | •Wave-3 will be terminal (<161.8%) and will go very slow (in the nature of sub-division) | •Wave-4 will be sharp (flat or zig-zag) | •Wave-5 will reach 127% or maybe equal to Wave-3 | When Wave-1 is in Extension: (entire move might forms a LD) | •Wave-1 and Wave-2 will be in microscopic nature | •If Wave-2 is microscopic, Wave-4 can be 5-10 times of Wave-2, so alternation rule does not apply | •Wave-4 can fall Max till 38.2% and may be sharp (flat or zig-zag) or complex | •Wave-5 will be an ED - very slow speed time-wise and price-wise | •Wave-5 will be in a steep up move if Wave-4 is complex, or can be truncated if Wave-4 is in a sharp correction | When Wave-3 is in Extension: (Wave-5 forms an ED) | •Wave-1 and

Wave-3 will be equal (time and price –wise) | •Join 1-3 and 2-4 via trendlines to form a megaphone structure | •Wave-5 can go Max till 261.8% in extension of 0-3-4 | When Wave-5 is in Extension: (entire move forms a Megaphone)

CHARTKING-112 p.33: Wave-5 will go only till 81.2%

CHARTKING-112 p.34: Rules to check for extension: | • | Every swing should be Min retracement 14% from the base point | • | Every move should be Min 127% retracement of the previous swing | • | No move should go below the previous swing's low | • | If any move has retracement equal to or above 38.2% then check for degree fall using rule of alternation

CHARTKING-112 p.35:

CHARTKING-112 p.36:

CHARTKING-112 p.37: Zig-Zag Correction (A-B-C) | •Single correction | Correction Type | •Mostly found in Wave-2/ sometimes in Wave-4 and in double or triple correction | Occurrence | •Price-wise correction | Category | •3 (A-B-C) | No. of moves | •5-3-5 | Internal count | •A Fast, B Slow, C Fast | Speed | •Min 61.8% retracement of previous degree Wave-5 | Rule for Wave-A | •B >= A (up to 5 times of A) | Wave-B Time period | •C = A (length wise), Min 61.8%, Max 261.8% Fib extension (0-A-B) | Fib Ext Wave-C (0-A-B) | •Wave-C should be Fib retracement 127% of Wave-B | Wave-C retracement | •Less than 61.8%, if >61.8% then it is Wave-2 | Wave-B retracement | •If A = B, C = A + B | Time taken for Wave-C : 1 | •If A < B, C = [A + B] / 2 | Time taken for Wave-C : 2 | • On crossing 3rd Fib Ratio in Wave-C | Entry | • Previous Fib Level of entry | SL | • Bullish Candle formation at Fib levels | 61.8%, 100%, 161.8%, 261.8% | Target | • When price crosses 0-B trendline in | half the time (Wave-C) Pattern confirmation-1 | • When price crosses point B at the | same time (Wave-C) Pattern confirmation-2 | • Complete correction should retrace | Min 38.2% of the entire Impulse move Pattern End Types of C : Trend based Fib extension on 0-A-B Normal C | 61.8% to 161.8% | Elongated C | 161.8% to 461.8% | Truncated C | 1. Reverses before 61.8% | 2. Confirmation: 81.2% retracement of A- | B-C (0-C)

CHARTKING-112 p.38: Examples Zig-Zag Correction Regular Flat Irregular/ Expanding Flat Running Flat

CHARTKING-112 p.39:

CHARTKING-112 p.40:

CHARTKING-112 p.41:

CHARTKING-112 p.42:

CHARTKING-112 p.43: Flat Correction (A-B-C) | • Single correction | Correction Type | • Mostly found in Wave-2 | Occurrence | • Price-wise correction | Category | • 3 (A-B-C) | No. of moves | • 3-3-5 | Internal count | • A Slow, B Fast, C Fast | Speed | • Min 38.2% retracement of previous degree Wave-5 | Rule for Wave-A | • B >= A (up to 2 times of A) | Rule for Wave-B | • Min 61.8% , Max 111%, (SL 127%) | Wave-B retracement | • If time (A) = time (B), time (C) = time (A) + time (B) | Time taken for Wave-C : 1 | • If time (A) < time (B), time (C) = [time (A) + time (B)]/2 | Time taken for Wave-C : 2 | •Start of Wave-X or Wave-1 | Point of Entry | •Bullish Candle formation at Fib levels 61.8%, 100%, | 161.8%, 261.8% | Confirmation:1 | •On crossing 3rd Fib Ratio | Confirmation:2 | •61.8%, 100%, 161.8%, 261.8% | SL | •Half the time of Wave-C | Exit (time period) | •When price crosses 0-B trendline in half the time | (Wave-C) Reversal criteria | •When price crosses point B at the same time (Wave- | C) | Pattern confirmation | •Complete correction should retrace Min 38.2% of | the entire Impulse move Pattern End Types of C : Trend based Fib extension on 0-A-B Normal C (B : Min 61.8%, Max 81.2%) | 61.8% to 161.8% | Elongated C (B: Min 61.8%, Max 81.2%) | 161.8% to 261.8% | Strong Irregular (B: Min 61.8%, Max 111%) | 61.8% to 161.8% | Running (B: Min 61.8%, Max 111%) | 161.8% to 261.8%

CHARTKING-112 p.44: Flat Correction

CHARTKING-112 p.45:

CHARTKING-112 p.46:

CHARTKING-112 p.47:

CHARTKING-112 p.48:

CHARTKING-112 p.49: Wave-A Leading Diagonal | • Impulsive | Nature | • 3 (A-B-C) in Flat Correction OR 5 (1-2-3-4-5) in Zig-Zag | Sub-moves | • Max 13 moves (in extension) | Internal count | • MACD bottom in a Zig-Zag correction might be formed | MACD | • Min 38.2% (Flat) and 61.8% (ZigZag) Fib retracement of the previous sub-Wave-5 | Retracement | • The 2-4 TL is broken in half the time of Wave-5 | Confirmation | • Fast in a Zig-Zag, slow in a Flat correction | Speed | • Can be a Leading Diagonal | Feature Trade

CHARTKING-112 p.50:

CHARTKING-112 p.51: To be depicted via Live charts

CHARTKING-112 p.52: Wave-B | • Corrective - Flat, Double, Triple and Triangle Correction | Nature | • 3 (A-B-C) | Sub-moves | • 5-3-5 or 3-3-5 | Internal count | • MACD top with price not confirming 127% Fib retracement | MACD | • 1% to 50% Fib retracement (Zig-Zag); 61.8% to 111% (Flat) | Retracement | • Time(Wave-B) > Time(Wave-A) | Time | • During triple and triangle correction, time (B) = 5A | Special feature Trade

CHARTKING-112 p.53: Wave-B: Flat or ZigZag Correction Up to 2X in Flat Correction Up to 5X in Zig-Zag Correction

CHARTKING-112 p.54: Wave-C extension ending diagonal | • Impulsive | Nature | • 5 (1-2-3-4-5) | Sub-moves | • 5-3-5-3-5, Max 13 moves | Internal count | • MACD forms a positive divergence | MACD | • Min 61.8% and Max 461.8% Fib extension of 0-A-B | Fib Extension | • Wave-C is Min 127% Fib Retracement of Wave-B | Retracement | • Normal (61.8%-161.8%), Truncated (<61.8%), Elongated (161.8%-261.8%) | Types | • If A = B; C = A+B | Time-1 | • If A < B; C = [A + B] / 2 | Time-2 | • It is best to book profits in Wave-C @ 100% Fib extension of 0-A-B, (A=C) | Exit | • Can be in extension or can form an ending diagonal | Features

CHARTKING-112 p.55:

CHARTKING-112 p.56: Double Correction (W-X-Y) | • Complex correction | Correction Type | • Mostly found in Wave-4 | Occurrence | • Time-wise correction | Category | • Total Max 39 -> W(5-3-5), X(11), Y(3-3-3-3-3) | No. of moves | • W & X: Zig-Zag or Flat ; Y: Zig-Zag, Flat or Triangle | Internal structure | • If 1st Wave takes ½, 1/3 or ¼ Time of 2nd Wave, then 1st Wave is "W" and 2nd Wave is "X" Rule for Wave-W | • Max 50% retracement of Wave-W | Rule for Wave-X (small) | • Min 61.8% and Max 111% retracement of Wave-W | Rule for Wave-X (Large) | • 0-W-X Extension: Min 61.8%, Max 161.8% | Rule for Wave-Y | • Time (Y) = Time (W) OR Time (W+X) OR | Time (W+X)/2 Time taken for Wave-Y | • Time (X) < Time (W) | Time taken for Wave-X | • Positive divergence in MACD at Y | Point of Entry | • ??? | Confirmation | • Close above 38.2% retracement of 0-Y | Entry | • Trailing SL of 2 Fib levels (SL will be at 50% | if price crosses 81.2% Fib level) | SL | Exit (time period) | • When price crosses 0-X TL in ½ Time | (Wave-Y) Reversal criteria:1 | • When price crosses Pt. B in < Time (Wave- | Y) | Reversal criteria:2

CHARTKING-112 p.57:

CHARTKING-112 p.58:

CHARTKING-112 p.59:

CHARTKING-112 p.60:

CHARTKING-112 p.61: Double Zig-Zag Correction Double Zig-Zag | • | W (5-3-5) + X (1) + Y (5-3-5) = total 27 moves | • | X can have Max 11 moves (total 37 with X) % Fall in W and Y are similar

CHARTKING-112 p.62:

CHARTKING-112 p.63: 11 + 11 + 11 = 33 moves Double Flat Correction

CHARTKING-112 p.64:

CHARTKING-112 p.65: Zig-Zag + small x + Triangle

CHARTKING-112 p.66: 11 + 11 + 15 = 37 moves Flat + Triangle

CHARTKING-112 p.67: Wave-W | • Corrective (Zig-Zag or Flat correction) | Nature | • 3 (A-B-C) OR (W-X-Y) | Sub-moves | • 3-3-3 OR 3-3-5 OR 5-3-5 | Internal count | • Can form lowest MACD in a double correction | MACD | Retracement Feature Trade

CHARTKING-112 p.68:

CHARTKING-112 p.69: Wave-X | • Undefined | Nature | • Up to 11 moves | Sub-moves | • Small 'x' [Min 14%, Max 50% (<61.8%)], and Large X (61.8% - 111%) retracement of W | 2 Types | • MACD lower than Wave-3 MACD or negative divergence | MACD | • It can retrace Wave-W Min 61.8% and Max 111% | Retracement | • Cannot be a triangle correction, can only be zigzag or flat correction | Feature | • Small 'x' mostly forms after Zig-Zag, Large 'X' mostly forms after Flat correction | Feature-2 | • Time (X) < Time (W) | Time period Trade

CHARTKING-112 p.70: (SL – 61.8%)

CHARTKING-112 p.71: Wave-Y | • Corrective (Zig-Zag, Flat or Triangle correction) | Nature | • 3 (A-B-C) or 5 (A-B-C-D-E) | Sub-moves | • Forms a positive divergence in MACD | MACD | • Min 61.8% and Max 161.8% Fib extension on 0-W-X | Fib Extension | • Never breaks 161.8% Fib extension | Feature-1 | • After small 'x', Y will be fast and will take Min Time of W | Feature-2 | • Y is usually a failure (38.2%) after Large 'X' | Feature-3 | • Time (Y) = Time (W) OR Time (W+X) OR Time (W+X)/2 | Time Trade

CHARTKING-112 p.72:

CHARTKING-112 p.73: Triple Correction (W-X-Y-XX-Z) | • Complex correction | Correction Type | • Mostly found in Wave-4 | Occurrence | • Time-wise correction | Category | • Total Max 37 -> W(5-3-5), X(11), Y(5-3-5) | No. of moves | • W & X: Zig-Zag or Flat ; Y: Zig-Zag, Flat or Triangle | Internal structure Rule for Wave-W | • Max 50% retracement of Wave-W | Rule for Wave-X (small) | • Min 61.8% and Max 111% retracement of Wave-W | Rule for Wave-X (Large) | • 0-W-X Extension: Min 61.8%, Max 161.8% | Rule for Wave-Y Rule for Wave-XX Rule for Wave-Z | • Time (Y) = Time (W) OR Time (W+X) OR Time (W+X)/2 | Time taken for Wave-Y | • Time (X) < Time (W) | Time taken for Wave-X | • Start of Wave-Y | Point of Entry Confirmation Entry | SL | Exit (time period) | • When price crosses 0-X TL in ½ Time (Wave-Y) | Reversal criteria:1 | • When price crosses Pt. B in < Time (Wave-Y) | Reversal criteria:2

CHARTKING-112 p.74:

CHARTKING-112 p.75: Examples

CHARTKING-112 p.76: Time period rules in a Triple Correction

CHARTKING-112 p.77: MACD positive divergence in a Triple Correction

CHARTKING-112 p.78: Triple Zig-Zag Small 'x'

CHARTKING-112 p.79: Zig-Zag + Flat + Triangle Small 'x'

CHARTKING-112 p.80: Double Zig-Zag + Triangle Small 'x'

CHARTKING-112 p.81: Triple Zig-Zag Large 'X'

CHARTKING-112 p.82: Triple Flat Large 'X'

CHARTKING-112 p.83: Wave-XX | • Undefined | Nature Sub-moves No. of moves | MACD | • XX can go between 50% and 61.8% retracement of Wave-Y | Retracement | • Cannot go beyond (above) point X | Rule | • Time (XX) = Time (X), AND < Time (Y) | Time | • At 127% book profits or use a trailing SL | Trade

CHARTKING-112 p.84: ??

CHARTKING-112 p.85: Wave-Z | • Corrective | Nature | • 3 (A-B-C) OR (W-X-Y) | Sub-moves No. of moves | MACD | • Min 61.8% and Max 161.8% Fib extension of 0-Y-XX | Fib Extension | • In case of a failure, it can reverse from 31.8% Fib retracement | Exception | • Time (Z) = Time (Y) OR Time (Y + XX) OR Time (Y + XX)/2 | Time | • At 127% book profits or use a trailing SL | Trade

CHARTKING-112 p.86:

CHARTKING-112 p.87: Types of Triangles | • Horizontal Contracting Triangle. | • Irregular Contracting Triangle. | • Running Contracting Triangle | Contracting Triangle: | • Horizontal Expanding Triangle. | • Irregular Expanding Triangle. | • Running Expanding Triangle. | Expanding Triangle:

CHARTKING-112 p.88: Contracting Triangle | • Three waves must undergo retracement exceeding 50% of the | preceding wave's length. Rule : 1 | • ABCDE (Each of the five waves exhibits a structure | characterized by corrective waves). Rule : 2 | • Always Draw trend-line on candle chart. | Rule : 3 | • Both the trend-lines will have max 5 touch points. | Rule : 4 | • The retracement for the E wave typically hovers around 38.2% | and triangles are mostly found in 4th, B, E, X, Y, XX, Z). Guideline: | • Not found in 2nd wave | Note:

CHARTKING-112 p.89: Horizontal Contracting Triangle The trend lines of the triangle must travel in opposite price directions. Wave-b must be smaller than wave-a. Wave-c must be smaller than wave-b. Wave-d must be smaller than wave-c. Wave-e must be smaller than wave-d. Wave-a is the widest segment of the triangle. The apex point of the triangle must fall within a range 61.8% of the longest segment of the triangle, centered in The middle of the longest segment. After the completion of horizontal contracting triangle, there occurs a "thrust" that must be at least 75% of the widest wave, and under normal circumstances will not exceed 125%.

CHARTKING-112 p.90:

CHARTKING-112 p.91: Irregular Contracting Triangle Wave-b should not be more than 261.8% of wave-a and will usually be less than 161.8%. An exact Fibonacci relationship between wave-a and wave-b of any triangle is very unusual. Waves c, d, & e must all be smaller than the previous wave. The trend lines of the triangle should move in opposite directions. Wave-b is the widest segment of the triangle. After the completion of irregular contracting triangle, there occurs a "thrust" that must be at least 75% of the widest wave, and under normal circumstances will not exceed 125%.

CHARTKING-112 p.92: Running Contracting Triangle Wave-b is longer than wave-a and is the longest wave of the triangle. Wave-c is smaller than wave-b. Wave-d is larger than wave-c. Wave-e is smaller than wave-d. Both trend lines will slope upward or downward. Wave-b is the widest segment of the triangle. The "thrust" after the triangle will be much larger than the widest leg of the triangle, sometimes as much as 261.8%, but no more. After the completion of running contracting triangle, there occurs a "thrust" that must be at least 75% of the widest wave, and under normal circumstances will not exceed 125%.

CHARTKING-112 p.93: Expanding Triangle | • The A-wave or the B-wave will always be the smallest segment of the Triangle. | Rule : 1 | • Wave-E will almost always be the largest wave of the pattern. | Rule : 2 | • Expanding Triangles cannot occur as B-waves in Zigzags or B.C. or D-waves of a larger Triangle. | Rule : 3 | • The E-wave will usually be the most time consuming and complex segment of the Triangle. The most typical | construction of the E-wave (if you can see any subdivisions), would be a Zigzag (in small Expanding Triangles), or a Complex Combination of corrections in large: patterns. Rule : 4 | • The E-wave will almost always break beyond the trend line drawn across the top of wave-A and wave-C. | Rule : 5 | • The "thrust" out of an Expanding Triangle should be less than the widest wave of the Triangle (which, in this case, is | wave-E) unless it concludes a powerful, larger Correction. Rule : 6 | • Backward from wave-E, three of the previous waves must be at least 50% of the wave to the right. | Rule : 7 | • Not found in 2nd wave | Note

CHARTKING-112 p.94: Horizontal Expanding Triangle Wave-A must be the smallest wave in the pattern. Wave-e is the widest segment of the triangle. The Wave-E will probably break beyond the trend line drawn across wave-a & c. Wave-B,C,D,E must each exceed the previous segment's termination point (which may or may not be the highest or lowest price). After the completion of horizontal expanding triangle, there occurs a "thrust" that must be at least 61.8% of the widest wave, and under normal circumstances will not exceed 100%.

CHARTKING-112 p.95: Irregular Expanding Triangle Wave-b is smaller than wave-a and all the rest of the waves are larger than the previous. The longer the period of time covered by the pattern, the more likely the channeling of the pattern will be upwardly or downwardly skewed. After the completion of irregular expanding triangle, there occurs a "thrust" that must be at least 61.8% of the widest wave, and under normal circumstances will not exceed 100%. Wave-e is the widest segment of the triangle.

CHARTKING-112 p.96: Running Expanding Triangle Wave-b being slightly larger than wave-a and wave-d being slightly shorter than the wave-c. Instead of the trend lines going in opposite directions, they both go in the same direction, but still diverge. The e-wave in this type of pattern can be quite violent. When all waves are larger than the previous except wave-c (which is shorter than the wave- b). Wave-e is the widest segment of the triangle. Instead of the trend lines going in opposite directions, they both go in the same direction, but still diverge. After the completion of running expanding triangle(part-2), there occurs a "thrust" that must be at least 61.8% of the widest wave, and under normal circumstances will not exceed 100%.

CHARTKING-112 p.97: Candlestick Patterns red candle Triple candlestick pattern: | 1.Morning Star - A morning star is a visual pattern made up of a tall black | candlestick, a smaller black or white candlestick with a short body and long wicks, and a third tall white candlestick. The middle candle of the morning star captures a moment of market indecision where the bears begin to give way to bulls. The third candle confirms the reversal and can mark a new uptrend. | 2.Evening Star - The evening star pattern is considered to be a very strong | indicator of future price declines. Its pattern forms over three days: The first day consists of a large white candle signifying a continued rise in prices. The second day consists of a smaller candle that shows a more modest increase in price. The third day shows a large red candle that opens at a price below the previous day and then closes near the middle of the first day | 3.Tri-Star - A tri-star is a three line candlestick pattern that can signal a possible | reversal in the current trend, be it bullish or bearish. Tri-star patterns form when three consecutive doji candlesticks appear at the end of a prolonged trend. A tri-star pattern near a significant support or resistance level increases the probability of a successful trade.

CHARTKING-112 p.98: Double Candlestick Patterns- Bullish | 1. Bullish Engulfing - A bullish engulfing pattern is a candlestick pattern that forms when a | small black candlestick is followed the next day by a large white candlestick, the body of which completely overlaps or engulfs the body of the previous day's candlestick. Bullish engulfing patterns are more likely to signal reversals when they are preceded by four or more black candlesticks. Investors should look not only to the two candlesticks which form the bullish engulfing pattern but also to the preceding candlesticks. | 2. Bullish Counterattack - The pattern shows that the buyers may be losing control | during an uptrend or that sellers may be losing control in a downtrend. The bullish counterattack lines is a candlestick pattern with the following characteristics: The market is in a downtrend. 1) The first candle is black (down) with a long real body. 2) The second candle gaps down on the open, is white with a real body that's similar in size to the first candle, and closes near the first candle's close. | 3. Bullish Harami - A bullish harami is a candlestick chart indicator used for spotting reversals | in a bear trend. It is generally indicated by a small increase in price (signified by a white candle) that can be contained within the given equity's downward price movement (signified by black candles) from the past couple of days. | 4. Tweezer Bottom - a bullish tweezer bottom is realized during a downtrend when bears | continue to drive prices lower, closing the day near lows (usually a strong bearish trend). Again, Day 2 is a reversal, as prices open, do not breach the prior day's lows, and head sharply higher. A bullish advance on Day 2 can quickly eliminate losses from the previous trading day. | 5. Piercing Bottom - The pattern is preceded by a downward trend in price. (This may be only a | short downtrend, but if the candles appear after an upward trend in price it is not an important reversal indicator).The price gaps lower to begin the second day, must close above the midpoint of the first candle. (This signifies that buyers overwhelmed sellers on this day.)

CHARTKING-112 p.99: Double Candlestick Patterns- Bearish | 1. Bearish Engulfing -A bearish engulfing pattern is a candlestick chart pattern that indicates a | potential reversal in trend. It occurs when there is a large red candlestick that "engulfs" the previous green candle. This signals that the bears are taking control of the market and that the bulls are no longer in charge. | 2. Bearish Counterattack - is made up of 2 candlesticks. The 1st candlestick in the pattern | has a green colored real body. The 2nd candlestick in the pattern gaps higher at the open than the 1st candlestick's closing share price. The closing share price, even after the initial gap-up, closes at the same or almost the same share price as the 1st candlestick in the pattern. The Bearish Counter-Attack is confirmed on the 3rd candlestick if the share price continues the reversal of momentum that started to take place with the 2nd candlestick | 3. Bearish Harami - A bearish harami is a candlestick chart indicator for reversal in a bull price | movement. It is generally indicated by a small decrease in price (signified by a black candle) that can be contained within the given equity's upward price movement (signified by white candles) from the past day or two. | 4.Tweezer Top - A bearish tweezer top occurs during an uptrend when bulls push prices higher, often | ending the day near the highs (generally considered a strong bullish signal). Then, on the following (second) day, traders reverse their market sentiment. The market opens, does not breach the prior day's highs, and heads straight down, often eliminating most of the prior period's gains. | 5. Piercing Top or Dark Cloud - Dark Cloud Cover is a candlestick pattern that shows a | shift in momentum to the downside following a price rise. The pattern is composed of a bearish candle that opens above but then closes below the

midpoint of the prior bullish candle. Both candles should be relatively large, showing strong participation by traders and investors. When the pattern occurs with small candles it is typically less significant. Traders typically see if the candle following the bearish candle also shows declining prices. A further price decline following the bearish candle is called confirmation.

CHARTKING-112 p.100: candlestick shadow candlestick opening Single candle: | 1.Doji - A Doji is a name for a trading session (Day or any time period) in which a security has | open and close levels that are virtually equal, as represented by a candle shape on a chart. Based on this shape, analysts attempt to make assumptions about price behavior. Doji candlesticks can look like a cross, an inverted cross or a plus sign. Although rare, a doji candlestick, especially if they appear in cluster generally signals a trend reversal indication, although it can also signal indecision about future prices. Broadly, candlestick charts can reveal information about market trends, sentiments, momentum, and volatility. The patterns that form in the candlestick charts are signals of such market actions and reactions. | 2.Gravestone Doji - A gravestone doji is a bearish reversal candlestick pattern that is | formed when the open, low, and closing prices are all near each other with a long upper shadow. The long upper shadow suggests that the bullish advance at the beginning of the session was overcome by bears by the end of the session. This often comes just before a longer-term bearish downtrend. | 3.Long legged doji - The long-legged doji is a candlestick that consists of long upper | and lower shadows and has approximately the same opening and closing price, resulting in a small real body. The pattern shows indecision and is most significant when it occurs after a strong advance or decline. The pattern is not always significant, and won't always mark the end of a trend—it could mark the start of a consolidation period, or it may just end up being an insignificant blip in the current trend.

CHARTKING-112 p.101: price breakouts Single candle: | 4. High Wave - pattern indicates price breakouts during an uptrend or downtrend. It consists of longer | lower shadows and upper shadows with a small body that indicates fluctuating prices during the trading period. Since both the supply side and demand side of the securities prices witness rapid fluctuations, buyers try to increase the securities prices but are met with the opposite decision of sellers to cut down the prices. This leads to overall indecision until the market finally crosses the resistance price level or falls below the support level and marks a trend reversal. The High Wave Candlestick pattern occurs in a highly fluctuating market and provides traders with entry and exit levels in the current trend. The candlestick usually forms at the support and resistance price levels, indicating trend reversals. A High Wave Candlestick pattern usually consists of only a single candlestick near the resistance or support level, where the size of the candlestick is more than the size of the previous 20 candlesticks in the pattern. This indicates a big market move during the trading period. The body of the candlestick is smaller when compared to the size of its shadows (wicks). A High Wave Candlestick is equal to or less than 20% of the total size of the wick. A High Wave Candlestick can either be a green-coloured candlestick, occurring in a bullish trend, or a red-coloured candlestick, occurring in a bearish trend. The upper and lower wick's size should be at least two times the size of the body. The larger the size, the better it is, as larger wicks indicate a higher indecision level in the market. | 5. Shooting Star is a candlestick pattern formed when currency pair prices open, increase immediately, | and then close near the opening price, indicating a downtrend reversal. The candlestick Shooting Star is formed after three or more green (bullish) candlesticks appear simultaneously, marking higher prices of the currency pair. The green candlesticks are followed by a small-bodied red (bearish) candlestick with an extremely long upper wick and no lower wick at all, indicating that the overall rising uptrend is now going to reverse into a downtrend. The long wick also indicates a strong buying pressure during the last few days, but the price is brought near to the close price as the day ends and selling pressure increases.

CHARTKING-112 p.102: Candlestick pattern Single candle: | 6.Hammer (Green) - Candlestick pattern that occurs whenever a securities trades at a | much lower price than its opening price. The currency pair price spirals as soon as the close price of the pair comes near the open price. It signals bullish reversals in the market. In this pattern, the open, close and high prices are very close to each other, giving it the 'hammer' type look. The lower shadow or wick in a Hammer Candlestick is always more than double the candlestick's body size. This pattern generally occurs when the securities is in a downtrend, which in turn indicates a possible market reversal. | 7.Inverted Hammer- A Bullish Inverted Candlestick is an individual candlestick with a | small body and long upper wick. The close price of the currency pair is always above the open price, indicating more significant buying pressures in the market. This candlestick is formed after a long downtrend and signals an uptrend market reversal. With this candlestick, traders can enter buy positions since the market is expected to witness a potential increase in the prices.

CHARTKING-112 p.103: Candlestick Formula | SELLING | BUYING | NIFTY / BANKNIFTY | Stocks | NIFTY / BANKNIFTY | Stocks X = Low * 0.382/100 X = Low * 0.618/100 X = High * 0.382/100 X = High * 0.618/100 Entry = Low - X Entry = Low - X Entry = High + X Entry = High + X Y = High * 0.382/100 Y = High * 0.618/100 Y = Low * 0.382/100 Y = Low * 0.618/100 SL = High + Y SL = High + Y SL = Low - Y SL = Low - Y Z = (High - Low) * 2 Z = (High - Low) * 2 Z = (High - Low) * 2 T1 = Entry - Z T1 = Entry - Z T1 = Entry + Z T1 = Entry + Z | T2 = T1 - Z | T2 = T1 - Z | T2 = T1 + Z | T2 = T1 + Z | Calculator

CHARTKING-112 p.104: MACD Basics MACD is only a supporting tool used to identify the strength, direction, momentum and duration of the trend. It is to be used in 75min, Daily and Weekly timeframes It gives Bullish and Bearish signals through crossover of moving averages It can also be interpreted via positive and negative divergences MACD histogram shows bearish and bullish signals along with their strength (through color)

CHARTKING-112 p.105: MACD Rules - Impulse | • Confirmed when MACD forms lowest low and price retraces till 61.8% | Base point | • Starts after MACD forms a lowest low or positive divergence | Wave-1 | • MACD falls slowly while price falls fast. Its MACD value won't go below base | point's MACD value. Wave-2 | • MACD always forms a top | Wave-3 | • MACD can fall very fast even form a lowest low. Steep fall in MACD denotes | end of Wave-3. Usually MACD goes below the Zero line. Wave-4 | • MACD forms a negative divergence | Wave-5 | • At C, MACD is not at its top and usually forms a negative divergence; and | MACD does not fall much in the succeeding fall unlike a Wave-4 1-2-3 vs A-B-C

CHARTKING-112 p.106: MACD Rules - Correction | • Price fall has minimal impact on the MACD | Corrective move | • If a positive divergence is formed in Wave-C, that means it is only a simple A-B-C correction | Simple Correction | • If MACD replicates the movement of price, the correction will mostly be complex and is not over yet | Complex Correction | • MACD value will be lower than that in previous impulse move's Waves – 2 & 4's MACD | Wave-A | • MACD moves with speed in Wave-B | Wave-B | • MACD moves with speed in Wave-X | Wave-X | • End of correction is denoted by a positive divergence in MACD. If MACD forms a lowest low, it signals | a triple correction Wave-Y | • MACD forms a higher high | Wave-XX | • MACD forms a positive divergence | Wave-Z

CHARTKING-112 p.107: MACD Rules - Advanced | • Formed in Wave-4 or in a | Degree fall Lowest | MACD | • Without a 127% | retracement in price is Wave-X or Wave-B Highest | MACD | • If MACD does not create a | new high when price crosses 61.8% extension, then it is not Wave-3 but a corrective move Fib extension | 61.8% | • Wave-3 (Trending and Terminal); | • Wave-X or Wave-B (during Flat | correction) | • Momentum cycle or Degree | completion (1/3/5) | • Wave-W in a corrective rise | MACD | Tops | • Degree fall / Wave-4 | • Wave-A | • Wave-W | • Wave-Y | • Downside impulse – Wave-3 | MACD | Bottom

CHARTKING-112 p.108: Channel Whenever price crosses 61.8% of the Imp High – Low, draw a channel to check trend continuity of Wave-1 To be used only in Impulse waves. Price won't reverse till the time it does not exit the channel Make it by joining the base point with 2 and make sure that price does not fall anywhere outside the channel After exiting from the channel, price can fall till 38.2% to 81.2% levels and signifies end of trend Entry can be done after the fall with a reversal candle pattern. Price does not enter the channel after exiting it

CHARTKING-112 p.109: Trendline Used for Pattern confirmation and Reversal Draw a TL connecting 0-2 (base TL) and place a parallel TL on 1 to form a channel No part of Wave-3 should break the TL connecting 0-2 For Wave-4, draw a TL connecting 1-3 and place a parallel TL on 2 Wave-4 can come where TL on 2 matches with Fib levels Till the time price is above 2-4 TL (maturity TL), price will reach 127% in Wave-5 Connect the origin of A with point B, and place a parallel line through point A. This will project the end of C. When wave 3 ends. Connect 1 & 3 and place a parallel line thru 2. This provides an estimated boundary for wave 4. When wave 4 ends, connect 2 & 4, and place a parallel line thru 3. This will forecast the end of wave 5. If wave 4 moves out of the channel, expect wave 5 to be a throw-over. Alternative method for projecting the end of wave 5: if wave 3 is abnormally strong (vertical) and wave 4 is sideways, connect 2 and 4, and draw a parallel line that intersects the extreme of wave 1. This will project the end of wave 5.

CHARTKING-112 p.110: Time Period - Impulse | • Time needs to be measured from the highest or | lowest points Note | • Time (Wave-2) = $\frac{1}{4}$, $\frac{1}{2}$, 1X or 2X the Time (Wave-1) | Wave-2 | • If Time(Wave-1) not equal to Time (Wave-2); then | Time(Wave 3) = (Wave-1 time + Wave-2 time)/2 Wave-3 (a) | • If Time (Wave-1) = Time (Wave-2) [+/- 5]; then | Time(Wave-3) is sum of both Wave-3 (b) | • Time (Wave-4) can be identified using the | corrective patterns of its sub-waves Wave-4

CHARTKING-112 p.111: Time Period - Correction | •Confirmed if price breaks 2-4 TL in half the time of Wave-5 | Wave-A | •Time(Wave-B) > Time(Wave-A) | Wave-B | •If Time(Wave-A) = Time(Wave-B); Time(Wave-C) = Time(Wave-A) + Time (Wave-B) | Wave-C:1 | •If Time(Wave-A) not equal to Time(Wave-B); Time(Wave-C) = [Time(Wave-A) + Time | (Wave-B)] / 2 Wave-C:2 | •Time (X) < Time (W) | Wave-X | •Time (Y) = Time (W) OR Time (W+X) OR Time (W+X)/2 | Wave-Y | •Time (X) = Time (XX) | Wave-XX | •Time (Z) = Time (Y) OR Time (Y +XX) OR Time (Y + XX)/2 | Wave-Z

CHARTKING-112 p.112: Gaps | • It confirms the impulse or correction, will help to identify a trend is strong or | weak or reversal. It helps to confirm the trend. Gaps Purpose | • They are small gaps which confirm strength of the trend (stability). They will | get filled in 3 to 5 days, visible in waves 1,2,4. In wave 4th correction will see only common gaps. Mostly in the flat correction. Common gap | • If the price opens above wave 1 and not filled on the same day is call | breakaway gap, its visible only in the wave 3, and it will never get filled. Breakaway gap | • It helps breakaway gap to establish its trends, visible in wave 3 only | Runaway gap | • Visible on Wave 5,C,Y and Z, like island reversal are made, indicating the | trend is changing. It will be filled quickly. Exhaustion gap

IMPULSE-31: Impulse notes \$\$ (2).pdf

IMPULSE-31 p.1: Chartking Student Batch 27 Nilam Gawankar

IMPULSE-31 p.2: Impulse Trading Type Time period Time frame Intraday trading 8 to 13 days 15 min Swing Trading One to 3 months Daily (75 min (if needed) Investment 1 year to 3 year Weekly or daily Important- 8 move search karna hai Swing may one month 20 days ka hota hai (ie business cycle) Investment may one month 30 days ka hota hai (ie calendar cycle) Momentum cycle (8 movement ka hota hai) Counting Starting point for Wave 0 | 1) | Indicator MACD (lowest low/Lowest point) (where price and indicator both show lowest point) | 2) | Positive divergence | 3) | Trendline base

IMPULSE-31 p.3: Visual example / diagram page. No readable text extracted; original page is preserved unchanged in the Master PDF appendix.

IMPULSE-31 p.4: Mark important high and important low apply retracement 1st Target 38.2% 2nd Target 50 %

IMPULSE-31 p.5: Wave Development –When Price touch 61.8% level it called Wave Development or Degree formation (formation of wave 1) Waves Retracement or Extensions How to apply Wave 2 Retracement Min 14% Max 81.2% Retracement base point to wave 1 Wave 3 Extension Min 161.8% Max 2100% Extension Zero say 1 than 2 Wave 4 Retracement min 14% Max 50% rare 61.8% Retracement 0 say 3 Wave 5 Retracement min 127% Max 261.8% Retracement 3 to 4 Note- For wave 2 formation time required either same as wave one or Double of Wave 1

IMPULSE-31 p.6: Wave 2 Min Retracement 14% Max Retracement 81.2% Wave 3 Min Extension 161.8% Max Extension 2100%

IMPULSE-31 p.7: Wave 4 Min Retracement 14% Max retracement 50% Rare Retracement 61.8%

IMPULSE-31 p.8: Wave 5 Min Retracement 127% Max Retracement 261.8%

IMPULSE-31 p.9: Stock Search kisay karay Stock edge | 1) | Weekly breakouts i) Close crossing last week high ii) close above last week high Y dono important hai 300 k nichay wala stock nahi lana hai

IMPULSE-31 p.10: 2) High delivery Percentage | Traded qty 5 LAKHA say zada hai o stock la sakte ho How to Start Counting | 1) | Price and MACD low | 2) | Or Positive divergence

IMPULSE-31 p.11: 3 Wave 161.8% tak nahi gaya to y Diagonal hai Sub wave fall and degree fall identification sub wave fall kam hota hai degree fall zada

IMPULSE-31 p.12: Type of Impulse | 1) Trending | 2) Terminal-Diagonal | Trending Impulse Terminal Impulse During the trending impulse wave 3rd Should be 161.8% minimum During the terminal case Wave 3rd Should Not touch 161.8% Wave 4th Should not enter the area of Wave 1st Wave 4th Should can enter the area of Wave 1st Agar dono criteria fulfill ho raha hai to Y ideal impulse hai Most important hai Wave counting may rules of alternation

IMPULSE-31 p.13: • | Checked MACD indicator and Price Lowest Low or Positive Divergence in MACD and Price

IMPULSE-31 p.14: Price Down and MACD UP= Positive Divergence (use for base point identification)

IMPULSE-31 p.15: I) | For Same use Tool | II) | Once Degree fall identify using Alternation Rules Apply counting for degree | III) | Than check Five moves in Impulse Wave (Degree) ie in 1 ,3 and 5 | IV) | After degree identification check sub ordinates wave

IMPULSE-31 p.16: Visual example / diagram page. No readable text extracted; original page is preserved unchanged in the Master PDF appendix.

IMPULSE-31 p.17: Apply the Extension and Rules of equality to find out target of Wave 3 Apply Retracement for Wave 2 and Wave 4 fall and Target of Wave 5 (give 1st preference for retracement for Wave 5th target) Waves Retracement or Extensions How to apply Wave 2 Retracement Min 14% Max 81.2% Retracement base point to wave 1 Wave 3 Extension Min 161.8% Max 2100% Extension Zero say 1 than 2 Wave 4 Retracement min 14% Max 50% rare 61.8% Retracement 0 say 3 Wave 5 Retracement min 127% Max 261.8% Retracement 3 to 4

IMPULSE-31 p.18: 1) Use Lowest Low of MACD and Price OR Positive Divergence | 2) Apply rules of Alternation for identification of degree Followed by | Subordinates | 3) Once degree identify count moves for sub-ordinates | 4) Apply Use of Equality | 5) check the retracement (Wave 2 ,4 and 5) and Extension (Wave 3)

IMPULSE-31 p.19: Extension rules Extension may 9 ,13,17 and 21 moves hota hai 9 moves ka count kisay karay 13 moves ka count

IMPULSE-31 p.20: 17 moves 21 moves Move 12 kitana bar aata hai 9 2 13 3 17 4 21 5

IMPULSE-31 p.21: Extension Rules | 1) If Wave 1 and wave 2 are in microscopic nature than wave 3 in extension | 2) wave 4 may be complex correction it can be five times of wave 1 | 3) Wave 4 can retrace max 38.2% and Rare 50% | 4) if wave 4th is in complex correction than wave 5 is sharp | 5) If wave 4 is sharp than wave 5 will be slow terminal impulse | 6) Wave 2 wave 1 ka double time lata hai | 7) If 3rd and fifth extended than it called double extension | 8) if wave 1 st goes above 61.8% then it is expected to be extension of wave 1

IMPULSE-31 p.22: Terminal Impulse | 1) | Leading Diagonal | 2) | Ending Diagonal General Rules | 1) | Har time frame may banta hai | 2) | Wave 3 min 61.8% or Max 161.8 % k niche jata hai | 3) | Wave 2, wave 1 ka ¼ time may y half time may Banta hai | 4) | Wave 3rd Slow hota hai | 5) | Wave fifth min 127% or Max 261.8% ja sakata hai | 6) | Wave 4th can go tertiary of wave 1 but not compulsory Note-Equity market may wave 5th extension may nahi jata Leading Diagonal Ending Diagonal Move 5-3-5-3-5 3-3-3-3 -3 If fall than 1st target=bottom of wave 4 If fall than 1st target=bottom of wave 4 Second target= bottom if wave 2 Second target= bottom if wave 2 IIrd target 81.2 % IIIrd target 81.2 %

IMPULSE-31 p.23: GAP ANALYSIS WITH ELLIOTT WAVE | Type of Gaps Location Fill kab hota hai characteristics Common Gap | 1) Wave 1 | 2) wave 2 | 3) Wave 4 | 4) Wave B | 5) Abc flat correction k A (A three move | hota hai) may | 6) Triangle | 7) X may | Can fill min 1 day max 5 day Small gap Breakaway Gap Wave 3 may aata hai OR Jab koi news aati hai Min 3 month Max 3 year tak fill nahi hota | 1) Jab wave 1 ka top Gap up / breakout hota hai to Breakaway Gap banata hai | 2) Jab Wave 3 may breakaway gap banata hai Wave 3 161.8% tak jaya ga hi | 3) Jab news aati hai to breakaway gap aata hai | Runaway Gap | 1) Wave 3 may aata hai | 2) Abc zig zag ka A may | 3) X may banaskata hai | Min 3 month | 1) Breakaway gap ka bad run away gap aata hai but not necessary | 2) Runaway gap wave 3 may aata hai Max 5 runaway gap wave 3rd may aa sakta | hai | 3) Running Market may aata hai | 4) Run away gap min 1 or Max 5 gaps aata hai | 5) Jab runaway jab aata hai uska meaning trend strong hai | 6) Zig zag ka A may max 3 runaway gap ho sakte hai | Exhaustion Gap | 1) Wave 5th or Wave C or Y or Z k last | may aata hai Same day Fill hota hai | 1) Exhaustion Gap trend maturity ka indication dayta hai | 2) Exhaustion Gap k Bad Trend Change hota hai | 3) Wave C may Exhaustion Gap ending Diagonal may aata hai | 4) Exhaustion Gap banay k bd next Wave k liya ready rahana hai | Important- Agar three month tak koi Gap nahi hai to uski counting impulse may nahi kar ga correction may kar ga

IMPULSE-31 p.24: GAP ANALYSIS WITH ELLIOTT WAVE | 1) Common Gap Location | Common Gap Flat correction ka A may aata hai Triangle may aata hai Flat correction k A may three moves hota hai B May aata hai

IMPULSE-31 p.25: GAP ANALYSIS WITH ELLIOTT WAVE | WXYZ Z may Common Gap and Runway Gap X may aata hai

IMPULSE-31 p.26: GAP ANALYSIS WITH ELLIOTT WAVE | Runaway and breakaway Gaps Zig Zag k A may jab runaway gap aata hai to min 1 or Max 3 runaway gap aata hai

IMPULSE-31 p.27: GAP ANALYSIS WITH ELLIOTT WAVE | Exhaustion Gaps

IMPULSE-31 p.28: When price in the ideal case than you will apply trendline Note-trendline entire moves 12345 ko cover karta hai 1 2,3 ,4,5 wave apay rules of guidelines ko fulfill karta hai Wave 1 important high or low ka 61.8% hai | IDEAL CASE | Wave 2 wave min 14 % max 81.2% hai Wave 3 min 161.8% hai Wave 4 38.2% y 50 % hai Wave 5 min 127% hai

IMPULSE-31 p.29: Step -1 Setp-2

IMPULSE-31 p.30: 4-5 banay may jitana time lagat hai o calculate karna hai Agar 4-5 k half time may maturity trendline break hoti hai to is process ko bolta hai revresal

IMPULSE-31 p.31: Agar 4-5 ka same time layta hai 4 tak aanay may to pattern conformation hota hai le time Agar zada time la raha hai 4 th tak aanay may to panic nahi hona hai opportunity hai for next upper move Jitan kam time may 4 ko break kar ga y 4 tak aaya ga uthana dangerous hai 5 the wave may 127 % par agar koi bearish candle banata hai to sell kar sakta hai y aggressive selling hogi 1st reversal aata hai than pattern conformation

TIME-7: Time Analysis(2).pdf

TIME-7 p.1: Time Analysis Confidence in trading can be achieved using study of market structure (Elliot Wave theory + Fib ratios) + Time analysis + Price Action (Candlestick patterns, indicators) Market remains in correction 80% of the time and in impulse only 20% of the time. Time period is a different concept than Elliott waves. While Elliot wave (base point in case of flat correction) may begin from a lower high than the price top (where top is 111%), time period is to be considered from market high and lows only (not to be misunderstood/mixed with base point 0 in case of flat correction) Time analysis | 1. Elliot Wave time periods (tp) | 2. Fib Wave proportions (wp) | 3. Fib sequence Numbers (fn) | Few important points about time period a. Reversal zones: Points in time where all these time periods come together to form clusters and act as probable reversal zones b. High/Low price levels of 1/4th time period are most important for trading purposes Elliot Wave Time Periods Wave 1 time period: Wave 1 often starts as impulse after a correction. What qualifies a good impulse: | 1. When price recovers at least 61.8% of previous move (usually, downfall in stock price) | from Important high (IH) to Important low (IL) in about ½ of the period taken by previous move (IH to IL) | 2. When price recovers 100% of the fall in about the same time (time from IH to IL) | Impulse if confirmed only after formation of wave 3 (sharp upmove). Before that it is just an assumption/possibility. Fast up moves after sharp fall (correction) can also be X wave, after which the price will most likely enter into double / triple correction.

TIME-7 p.2: Wave 2 time period (W2 tp): | 1. Wave 2 time period (tp) can never be equal to wave 1 time period (tp). | 2. It will be either ½ or ¼ of the wave 1 time period | 3. If wave 2 tp is equal to wave 1 tp, it may not be wave 2 but wave B which (tp) is equal to | wave A | 4. Wave 2 is sharp most of the time as it is zig zag correction (price correction more often | than time correction). This happens as market is generally not very bullish at this point after fall from important high to important low (correction) | 5. Wave 2 can retrace between 23.6% to 81.2% in about ½ or ¼ of the wave 1 time period. | Summary: a. ½ of wave 1 time period b. ¼ of wave 1 time period Wave 3 time period (W3 tp): | 1. Wave 3 formation can only be confirmed by two ways | a. Following rule of equality where wave 3 is equal to price range of wave 1. This can be checked by applying three point fib extension tool from base point 0 to high/end of wave 1 and low/end of wave 2. It should be 100% i.e. same price range as wave 1 b. As retracement of wave 2 price, being minimum 127%. This can be verified using fib retracement from high/end of wave 1 to low/end of wave 2 towards wave 3. | 2. Most of the time fib extension of 100% and 127 retracement will be at similar level | creating confluence of price level. | 3. ## Wave 3 is generally sharp move supported by price action indicators such as upper | bollinger band (std. 2) turning upwards, RSI>60 and ADX above 12-15 and rising | 4. Time period for wave 3 | Wave 1 time period + Wave 2 time period : (W1 tp + W2tp) Wave 4 time period (W4 tp) | 1. Wave 4 is a corrective move after wave 3 | 2. As per theory of alternation, time period of wave 2 and wave 4 will not be the same, | through correction in terms of price range % (rule of similarity) will be similar. This is because if wave 2 is simple zig zag price correction, more often wave 4 is time correction forming all type of patterns, zig-zag, flat, triangle, double and triple correction. | 3. Time period for wave 4 | a. Wave 2 time

period divided by 2 : $W2\ tp / 2$ b. Wave 2 time period plus Wave 3 time period, divided by 4 : $(W2\ tp + W3\ tp) / 4$ c. Wave 2 time period plus Wave 3 time period, divided by 2 : $(W2\ tp + W3\ tp) / 2$ d. Wave 2 time period plus Wave 3 time period : $(W2\ tp + W3\ tp) / 4$. If wave 4 is taking more time than W2 and W3 combined, it may be something else or the wave count is incorrect

TIME-7 p.3: Wave 5 time period ($W5\ tp$) Time period of wave 5 is based upon time period for wave 1 and wave 3 primarily a. Wave 1 time period + Wave 3 time period : $(W1\ tp + W3\ tp)$ b. Wave 1 time period + Wave 3 time period divided by 4 : $(W1\ tp + W3\ tp) / 4$ c. Wave 1 time period + Wave 3 time period divided by 2 : $(W1\ tp + W3\ tp) / 2$ d. Wave 1 time period : $(W1\ tp)$ e. Wave 3 time period + Wave 4 time period divided by 2 : $(W3\ tp + W4\ tp) / 2$ Time periods from minimum to maximum: $W1\ tp \rightarrow (W1\ tp + W3\ tp) / 4 \rightarrow (W1\ tp + W3\ tp) / 2 \rightarrow (W1\ tp + W3\ tp)$

TIME-7 p.4: Trade setup for Waves Trading for Wave 3 | 1. Wave 1 and Wave 2 are not preferred for trading | 2. When wave 2 is in development or completed, mark high of $\frac{1}{4}$ and high of $\frac{1}{2}$ time period | candles ($\frac{1}{4}$ and $\frac{1}{2}$ of wave 1 time period marked from start of wave 2 <end of wave 1>) | 3. When price starts moving up for wave 3 (preferably after wave 2 time period is complete | and price has made a bottom <38% to 81.2% retracement of wave1>), enter above the high of $\frac{1}{4}$ th time period candle | 4. If entry was made at $\frac{1}{2}$ time period candle, first target would be high of $\frac{1}{4}$ th time period | candle Target achievement / Exit | 1. Exit when earlier of the below happens | a. Wave 3 has reached 100% of wave 1 (fib extension), rule of equality has been achieved b. 127% retracement of wave 2 has reached c. Wave 3 time period is over ($W1\ tp + W2\ tp$) Time period should be the first priority as it prevents us from taking trade in haste. Trading for Wave 5 Entry | 1. Priority 1: Apply rule of similarity to validate if price range (% fall) of wave 4 is same as | wave 2 (+/- 5%, [10% in case of $CMP < 500$]) | 2. Priority 2: Check for formation of bullish candle pattern at important fib retracement level | (38% or 50% of entire upmove from base point zero) | 3. Priority 3: Time period of wave 4 is complete | Enter where confluence of more than one trigger is occurring (where price fall is similar to wave 2 and bullish candle pattern is forming at important fib level, which is also acting as an important price support level. Exit | 1. Exit when price touches 127% retracement level of wave 4 or time period of wave 5 is | over ($W1\ tp + W3\ tp$)/4, whichever is earlier.

TIME-7 p.5: Wave Proportions Wave proportions are important fib ratios for various waves. When translated to time period, High/Low of candles at these points in time act as support and resistance for price. | 1. Wave 1: 61.8% | 2. Wave 2: 23%, 38.2%, 50%, 61.8%, 81.2% | 3. Wave 3: 61.8%, 100%, 161.8% | 4. Wave 4: 23.6%, 38.2%, 50%, 61.8% | 5. Wave 5: 61.8%, 127% | No time period overlap (+/- 8 candles) is to be applied for wave proportions. Fib Time Zones Fib time zones are nothing but fib sequence numbers starting 13, 21, 34, 55, 89, 144, 233, 377, 610 and 987 applied to each of the wave (when it is in development). Cluster Zone Cluster is formed where default wave time period, wave proportions and fib time zones are occurring very close to each other in time. This indicates high probability of reversal at these points in time, improving with our understanding of market structure and help us in trading. Cluster $\rightarrow$ wave time period + wave proportion + fib time zone occurring very close to each other in time. | 1. For estimating completion of wave 1, we have the following time periods, counted | and plotted from the candle occurring at start of wave 1 (Important Low) a. Wave 1 time period i. $\frac{1}{2}$ time period of IH to IL b. Wave 1 proportion i. | 61.8% of time period of IH to IL | c. Fib time zones i. Time periods occurring at fib number sequence (13, 21, 34, 55.....) | 2. For estimating completion of wave 2, we have the following time periods, counted | and plotted from the candle occurring at start of wave 2 (High/end of wave 1) a. Wave 2 time period i. $\frac{1}{4}$ time period of wave 1 ii. $\frac{1}{2}$ time period of wave 1 b. Wave 2 wave proportion i. | 38.2% of time period of wave 1 | ii. 50% of time period of wave 1

TIME-7 p.6: iii. | 61.8% of time period of wave 1 | iv. | 81.2% of time period of wave 1 | c. Fib time zones i. Time periods occurring at fib number sequence (13, 21, 34, 55.....) | 3. For estimating completion of wave 3, we have the following time periods, counted | and plotted from the candle occurring at start of wave 3 (Low/end of wave 2) a. Wave 3 time period i. ($W1 + W2$) time period b. Wave 3 wave proportion i. | 61.8% of time period of wave 1 | ii. 100% of time period of wave 1 iii. | 161.8% of time period of wave 1 | iv. | 261.8% of time period of wave 1 | v. | 3.236% of time period of wave 1 | vi. | 4.236% of time period of wave 1 | Wave 3 proportion is calculated based on wave 1 time period only, wave 2 is NOT added c. Fib time zones i. Time periods occurring at fib number sequence (13, 21, 34, 55.....) | 4. For estimating completion of wave 4, we have the following time periods, counted | and plotted from the candle occurring at start of wave 4 (High/end of wave 3) a. Wave 4 time periods i. Wave 2 time period divided by 2 : $W2\ tp / 2$ ii. Wave 2 time period plus Wave 3 time period, divided by 4 : $(W2\ tp + W3\ tp) / 4$ iii. Wave 2 time period plus Wave 3 time period, divided by 2 : $(W2\ tp + W3\ tp) / 2$ iv. Wave 2 time period plus Wave 3 time period : $(W2\ tp + W3\ tp) / 4$ b. Wave 4 wave proportion i. | 23.6% of time period of wave 3 | ii. | 38.2% of time period of wave 3 | iii. 50% of time period of wave 3 iv. | 61.8% of time period of wave 3 | v. | 81.2% of time period of wave 3 (as all kinds of corrective patterns | are possible in wave 4) Wave 4 proportion is calculated based on wave 3 time period only c. Fib time zones i. Time periods occurring at fib number sequence (13, 21, 34, 55.....)

TIME-7 p.7: 5. For estimating completion of wave 5, we have the following time periods, counted | and plotted from the candle occurring at start of wave 5 (Low/end of wave 4) a. Wave 5 time periods (based on wave 1 and wave 3 time periods) i. Wave 1 time period ii. Wave 1 time period plus Wave 3 time period, divided by 4 : $(W1\ tp + W3\ tp) / 4$ iii. Wave 1 time period plus Wave 3 time period, divided by 2 : $(W1\ tp + W3\ tp) / 2$ iv. Wave 1 time period plus Wave 3 time period : $(W1\ tp + W3\ tp)$ Wave 5 time period (based on wave 1 and wave 3) has higher priority than wave 5 proportions b. Wave 5 wave proportion i. | 38.2% of time period of wave 1 | ii. | 61.8% of time period of wave 1 | iii. 100% of time period of wave 1 iv. 127% of time period of wave 1 Wave 5 proportion is calculated based on wave 1 time period only, however these can be based on wave 4 as well if wave 5 goes into extension c. Fib time zones i. Time periods occurring at fib number sequence (13, 21, 34, 55.....)

53. Duplicate Chartking File Traceability

The second 112-page file "EW Notes _ Chartking v2(2).pdf" is retained in the source registry and Master PDF. Its extracted text is identical to the dated 112-page Chartking file on pages 2-112; only page 1 title text differs. Therefore the page ledger above uses one normalized Chartking ledger to avoid pointless duplication while the exact second file remains part of the source bundle.

54. Final Acceptance Gate - Developer Must Demonstrate, Not Merely State

AT-V3-01: Base selected from correct source timeframe / Important H-L context and remains locked until a documented release rule.

AT-V3-02: Normal W2, microscopic W2, deep W2 and invalid W2 examples.

AT-V3-03: W3 trending, terminal and extended examples with correct Fib/time/MACD status.

AT-V3-04: W4 simple and complex corrections without premature W5.

AT-V3-05: W5 normal, ED, truncated and extended branches.

AT-V3-06: Zig-Zag normal/elongated/truncated examples.

AT-V3-07: Flat normal/elongated/truncated/irregular/running examples.

AT-V3-08: Double Zig-Zag, Zig-Zag+Flat, Flat+Zig-Zag, Double Flat, Zig-Zag+Triangle where data permits.

AT-V3-09: Triple W-X-Y-XX-Z with X/Y/XX/Z Fib/time values visible.

AT-V3-10: All six triangle families with size sequence, boundaries and thrust target.

AT-V3-11: Gap context examples and one case where gap support is absent but valid structure remains valid.

AT-V3-12: Trendline/channel construction: base, action, maturity, 0-B and 2-4 behavior.

AT-V3-13: Time profile comparison showing a source conflict produces TBD/configuration rather than silent guess.

AT-V3-14: Non-repaint test proving only FORMING pivots can move while CONFIRMED labels remain locked unless controlled recount occurs.

AT-V3-15: Current-script regression test proving raw swing count no longer advances Elliott labels by modulo/cycle position.

AT-V3-16: Pine vs Python same-data comparison with any difference explained.

DEVELOPER ACKNOWLEDGEMENT / SCOPE SIGN-OFF

This page is intended to be sent with the project handoff. It is not a legal contract; it is a technical scope acknowledgement so there is no ambiguity later about what was included in the requested Elliott Wave implementation.

Developer name: _____

Date: _____

Pine Script version / repository revision accepted for update: _____

☐ I received the V3 Source-Locked Specification.

☐ I received the current Pine Script to be updated.

☐ I received / can access all source-note PDFs listed in the Source File Register.

☐ I understand that source conflicts must not be guessed; they must be configurable, TBD-BLOCKED, or confirmed by the owner.

☐ I understand that fixed swing-sequence/modulo labelling is not accepted as the final Elliott engine.

☐ I will provide the required debug evidence and acceptance-test screenshots/logs.

☐ Any technical limitations I believe prevent an item from being implemented are listed below before coding begins.

Technical limitations / exclusions proposed by developer:

Developer acknowledgement/signature: _____